ON THE TRANSFER OF SHALIKA GERMS FOR G_2
DRAFT 1. CORRECTED AND EXPANDED VERSION

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ABSTRACT. This revision was prepared with the help of an AI system: Claude (Anthropic) refereed the 1994 manuscript and drafted this revision as a co-author. See the Acknowledgments for details. Let G_2 be the split exceptional group over a p-adic field F. Using Langlands' method, based on Igusa's asymptotic expansions, we study the κ-Shalika germs of G_2 attached to the 1-, 2- and 3-regular unipotent classes. We construct the variety Z underlying Langlands' partial resolution, prove that Z is normal and Cohen–Macaulay, cover it up to codimension two by explicit regular patches, and compute the Igusa constants of every branch of the special fiber. The 2- and 3-regular classes are carried by unique base divisors, with Igusa constants 3 and 4, matching the trivial classes of the cuspidal endoscopic groups $(SL(2)\times SL(2))/\{\pm 1\}$ and $PGL(3)$. We list all elliptic endoscopic data of G_2. The 1994 manuscript claimed to deduce the matching of these germs, and hence local transfer for G_2, from a reduction lemma of Hales. That lemma does not apply to G_2, so the matching theorems are stated here as conditional on an explicit computation that remains to be done (Remark 2026.29). The matching itself is now known by other means, from Waldspurger's transfer theorem and the fundamental lemma.

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This is a revision of the unpublished manuscript of the same title by D. Joyner, dated summer 1994. The 1994 AMS-TeX source, a literal transcription and a referee report are in the directory `g2stuff` of the repository `github.com/wdjoyner/trace-formulas`. The revision was made in November 2026. Gaps, caveats and statements not fully verified are collected in the numbered "Remarks 2026.*n*".

Claude is an AI model made by Anthropic. See the Acknowledgments for the division of work.

§ 0 INTRODUCTION

This section explains the problem, sketches Langlands' method, states what the 1994 manuscript set out to prove, and records what this revision proves, what it proves only conditionally, and what is now known by other means. The notation is collected in the section "Notation" after the literature review.

Motivation and rough outline. To apply the trace formula to the transfer of automorphic representations of two reductive groups G, H over a number field K, one must, among other things, have a comparison of local orbital integrals, for each local completion F of K. The comparison in the real and complex case was solved by Shelstad [She82]. In the p-adic case, such a local comparison (the smooth matching conjecture stated below) reduces, thanks to the work of R. Langlands and D. Shelstad ([LS87], [LS90]), to a certain matching of Shalika germs, also stated below. Shalika germs are functions associated to a semisimple element $s \in G(F)$ and a unipotent conjugacy class $\mathcal{O}$ of $M = \mathrm{Cent}_{G(F)}(s)$, defined in a punctured compact-open neighborhood of s in the regular set [Sh72]. They play a role in the analysis of p-adic orbital integrals on reductive groups roughly analogous to the role that the power functions play in the Taylor series of a function on the real line. By a well-known reduction known as Harish-Chandra descent, it usually suffices to assume that s is central and $M = G$.

Langlands [La83] introduced a geometric approach to the computation, or at least comparison, of Shalika germs. The comparison of Shalika germs associated to the 0-regular (regular) unipotent classes was solved by Shelstad [She89], using results from [LS87], [LS90]. Langlands's technique was further developed by T. Hales in his thesis [Ha92a]. Hales concentrated on the comparison of 1-regular germs, matching the stable subregular germs in general and matching the κ-stable subregular germs for even rank unitary groups. This technique of Langlands goes by the name Igusa theory, since it was Igusa [Ig78] who developed a crucial asymptotic formula for an integral over fibers of a morphism $Y_C \to C$, where C is a curve and Y_C is a non-singular variety. The technique also relies on a formula for the coefficients of this expansion, involving principal value integrals over p-adic analytic manifolds, worked out in [Ig78], [La83] (see also [LS83]). In [La83], [LS89], Langlands and Shelstad matched all the germs for groups of absolute type A_2 (see also [Ha92b], where a descent argument is used instead of the direct comparison in [LS89]). In [Ha89], Hales worked out the comparison of germs for the forms of $B_2 = C_2$ using Igusa theory, a reduction argument, and an explicit computation of the dependence of the germs on the tangent direction.

The 1994 manuscript set out to verify the matching of Shalika germs for the 1-, 2- and 3-regular germs of the exceptional group G_2 and to deduce, together with [She89], the smooth matching theorem for G_2 in the following form.

Conjecture 0.0 (Smooth matching for G_2)**.** *Let T_G denote an elliptic Cartan subgroup of $G = G_2$ and let κ denote a character of $H^1(\mathrm{Gal}(\overline{F}/F), T_G)$ whose endoscopic group $H = H(T_G, \kappa)$ is cuspidal (so H is $PGL(3)$ or $(SL(2) \times SL(2))/\{\pm 1\}$, by Appendix A). Let T_H denote the image of an admissible embedding of T_G into H. For each $f \in C_c^\infty(G(F))$ there is an $f^H \in C_c^\infty(H(F))$, depending only on f and H, for which*

$$\Delta_{G/H}^{T_G,\kappa}(\gamma_H, \gamma_G)\Phi^{T_G,\kappa}(\gamma_G, f) = \Phi^{T_H,st}(\gamma_H, f^H),$$

whenever γ_G is the image of a regular $\gamma_H \in T_H(F)$.

Remark 2026.1 (status of Conjecture 0.0). The 1994 manuscript called this the smooth matching conjecture and claimed to prove it. That proof is incomplete (Remark 2026.29). The statement

is nevertheless now a theorem: Waldspurger proved that the fundamental lemma implies transfer [Wa97], and Ngô proved the fundamental lemma [Ng10]. Gordon and Hales [GH16] derive from Waldspurger's transfer the matching of all κ-Shalika germs. (Level 3 for [Wa97], [Ng10]; level 2 for [GH16], checked against its abstract and introduction.) The interest of the Igusa-theoretic approach is therefore not the existence of transfer, but the explicit geometric description of the germs and of the matching constants.

In §1 we recall in detail the construction of the Igusa variety and the determination of local coordinates [La83], [Ha92a], [Ha89]. Section 2 gives the calculations for determining the action of the Galois group on the local coordinates on the Igusa variety for G_2, as in [Ha89].

Section 3 introduces, for any reductive group G, specific patches generalizing those used in [Ha89]. For G_2 we show that they are regular and cover the monomial variety Z underlying the base Igusa variety Y'' up to codimension two (Proposition 3.4A). We also show that, for G_2, Z is normal and Cohen–Macaulay (Proposition 3.7). This proves Hales' conjecture [Ha92a] that Y'' is regular in codimension one, for G_2.

In Section 4 we compute the Igusa constants of all the branches of the special fiber of Z (Table 4.3A) and give an "endoscopic interpretation" of some of them.

In Section 5 we first record some basic facts about the Igusa expansion over Y_C. We then discuss the "no spurious divisors lemma" for G_2, which is basically an inequality $e(\theta,\beta) \leq 1$, where $e(\theta,\beta)$ is as in [La83]. Its proof is incomplete (Remark 2026.27). Finally, we recall the statement of Hales' reduction lemma and explain why it does not apply to the cuspidal endoscopic groups of G_2 (Remark 2026.29).

In Sections 6, 7 and 8 we (a) compute coordinates on the 1-, 2- and 3-regular unipotent orbits (the latter two are not Richardson; they are what we call "endoscopically Richardson"), (b) compute coordinates on the fiber $\phi^{-1}(\gamma_\lambda)^\Gamma$ (explained below), and (c) express the contribution of each orbit to the Igusa expansion as a principal value integral over a single base divisor. One new idea is that, roughly speaking, the subgroup generated by the one-parameter root-groups associated to roots outside a pseudo-Levi subgroup M of G_2 may be used to determine local coordinates for the "endoscopically Richardson" orbit $\mathscr{O} = \mathrm{Ind}_M^G(1)$. The matching theorems of these sections are stated conditionally on an explicit computation of the dependence of these integrals on the tangent direction X (Hypotheses (X_r) of Remark 2026.29).

Section 9 presents a conjecture which the method seems to support. The appendix collects explicit group-theoretic computations: a Chevalley basis of $\mathfrak{g}_2$ via triality, the cuspidal endoscopic groups of G_2, all elliptic endoscopic data (Table A.6), and the stability of the 2- and 3-regular classes.

Langlands' idea. Here is a rough sketch of Langlands' idea in its simplest form. Let $\gamma \in G(F)$ be an elliptic regular element and let $T_G = \mathrm{Cent}_G(\gamma)^o$. The goal is to compute, or at least find some manageable expression for, the stable orbital integral $\int_{(T_G\backslash G)(F)} f(g\gamma g^{-1})dg$. (More generally, one can and does consider κ-orbital integrals.) The variety $T_G \setminus G$ may be identified, over the algebraic closure $\overline{F}$, with the variety of regular stars. Let Ω denote the absolute Weyl group of T_G, B_G a Borel subgroup of G containing T_G, and S_0 the $G(\overline{F})$-orbit of the $|\Omega|$-tuple $(B_G^\omega)_{\omega\in\Omega}$, where G acts by conjugation on each coordinate. This orbit is called the variety of regular stars, and since $\cap_{\omega\in\Omega} N_G(B_G^\omega) = T_G$, we have $S_0 \cong T_G \setminus G$ over $\overline{F}$. Let $\mathscr{B} \cong B_G \setminus G$ denote the Borel variety. Although $\mathscr{B}$ is complete, S_0 is not. Conjugation induces a principal N_G-bundle structure on $S_0 \to \mathscr{B}$, where N_G denotes the unipotent radical of B_G. The Zariski closure of S_0 in $\mathscr{B}^{|\Omega|}$ is denoted S and is called the variety of stars. It is projective but singular, and S_0 is Zariski open in S. Roughly speaking, the idea is to regard the above stable orbital integral as an integral over $S_0(F) \subset S(\overline{F})$. Let

T'_G denote the subvariety of regular elements of T_G. We restrict γ to a curve C in $T'_G \cup \{1\}$ through the identity, and construct a smooth variety Y_C with a fibering $\phi : Y_C \to C$, "the" smooth Igusa variety, whose fibers $\phi^{-1}(\gamma)$, $\gamma \in C,\ \gamma \neq 1$, are isomorphic to $T_G \setminus G$. The stable orbital integral may then be regarded as a fiber integral parameterized by a local coordinate λ in a neighborhood of $1 \in C$. Taking the limit as λ tends to 0, one may use techniques of Igusa [Ig78], as developed by Langlands [La83], to express the stable orbital integral as a sum involving principal value integrals over certain divisors contained in $\phi^{-1}(1) \subset Y_C$. By the homogeneity of the Shalika germs and of the coefficients in the Igusa expansion, only certain divisors can contribute to a given Shalika germ.

Here is a rough sketch of Igusa theory [La83], [Ha92a]. Assume G is F-quasisplit. Let C denote a 1-parameter subgroup of G in T_G, defined over F, containing the identity and no other singular point, and whose tangent over $\overline{F}$ is not contained in any root hyperplane. Let $C' = C - \{1\}$. We sketch the construction of a non-singular algebraic variety Y_C, defined over F, for which

$$\begin{array}{ccccc} C' \times T_G \setminus G & \xrightarrow{\ i\ } & Y_C & \xrightarrow{\ \pi\ } & G \\ & & \downarrow \phi & & \\ & & C, & & \end{array}$$

with the following properties.

(a) If T_G and C are defined over F then so are Y_C, ϕ, π, and i.
(b) $\pi : Y_C \to G$ is proper and G-equivariant.
(c) $\phi : Y_C \to C$ is proper.
(d) $\pi(\phi^{-1}(1)) = V$, the unipotent variety of G.
(e) If $\mathscr{O} \subset G$ is a unipotent orbit and $E \subset \phi^{-1}(1)$ is a divisor with $\pi(E) = \overline{\mathscr{O}}$, then $\pi|_E : E \cap \pi^{-1}(\mathscr{O}) \to \mathscr{O}$ is faithfully flat [DG80, Ch. II, Proposition 3.1] and, in particular, surjective.
(f) $i(C' \times T_G \setminus G) \subset Y_C$ is open.
(g) $\pi \circ i(\gamma, g) = \gamma^g$ and $\phi \circ i(\gamma, g) = \gamma$.
(h) Y_C has dimension $2p+1$, where $p = |\Sigma^+|$ is the number of positive roots.

A divisor $E \subset Y_C$ for which $\pi(E) = \overline{\mathscr{O}}$ is called an $\mathscr{O}$**-divisor**. In general, although E is smooth, $\overline{\mathscr{O}}$ need not be normal [KP79, §0].

Step 1. Let X denote the Zariski closure of $T' \times T_G \setminus G$ in $G \times S$ and let Y denote the Zariski closure of $C' \times T_G \setminus G$ in X. Let $\chi : G \to T_G/\Omega$ denote the adjoint quotient [Sl80], where Ω denotes the Weyl group of T_G. We will see below that X is covered by patches $X(B_+, B_-) \subset B_- \times S \times N_+$, and Y by patches $Y(B_+, B_-) \times N_+$, where N_-, N_+ denote the unipotent radicals of B_-, B_+, respectively. Here B_+, B_- are opposite Borel subgroups for which $T_+ = B_+ \cap B_-$ is defined over F. The completeness of S implies that the projections $Y \to G$ and $X \to G$ are proper.

Step 2. We recall below a partial resolution S_1 of S, with a proper morphism $S_1 \to S$ and an injective morphism $S_0 \to S_1$. Let $S'' \subset S_1$ be the open subvariety defined in Definition 1.16; it contains the image of S_0. Both S and S_1 are complete but S'' is not. Let X'', X_1 denote the Zariski closures of $T' \times T_G \setminus G$ in $G \times S''$, $G \times S_1$, respectively, and Y'', Y_1 the Zariski closures of $C' \times T_G \setminus G$ in X'', X_1. Again, we have proper maps $Y_1 \to G$ and $X_1 \to G$. Though $Y'' \to G$ may not be proper, it is especially useful because of its simple local coordinate relations: locally, Y'' is the product of an affine space and the monomial variety Z of §3. For $G = G_2$ the variety Z is a normal affine toric variety (Proposition 3.7), so it has an explicit toric resolution of singularities. Blow-down morphisms are proper and defined over F; they are G-equivariant by the universal property of blowing up [Har77].

Step 3. Resolve the singularities of Y_1 contained in $Y_1 - Y''$ (for instance by Hironaka's theorem). This yields a variety $\tilde{Y}_1$ and a blow down $\xi : \tilde{Y}_1 \to Y_1$ such that all the singularities of $\tilde{Y}_1$ map into Y'' under ξ. Then perform an embedded resolution of singularities on $Y'' \hookrightarrow \tilde{Y}_1$ as in Step 2. The resulting resolution is denoted Y_C. The blow down map $Y_C \to Y_1$ is proper, so π and ϕ are proper.

Remark 2026.2 (on Steps 2 and 3). The 1994 manuscript stated that Y'' for $G = G_2$ can be desingularized by four successive blow-ups, and omitted the derivation. That claim could not be checked and is not used. The resolution of Z is discussed in Remark 2026.23: it is explicit in principle, but its exceptional divisors have not been analyzed. The 1994 text also invoked Mumford's semi-stable reduction theorem in Step 3; semi-stable reduction requires a base change $\lambda = \mu^e$, which would change the Igusa data, so only an ordinary resolution is used here.

The properness of π and ϕ is needed to guarantee that the appropriate principal value integrals make sense. Properness implies that the principal value integral of the pull-back of the function f_κ is well-defined and that some of the hypotheses of Lemma 5.7 hold.

Notation and detailed outline. Let $\gamma_\lambda = \exp(\lambda X) \in C'(F) \subset T_G(F)$ be the image of $\gamma_{\lambda,H} \in T_H(F) \subset H(F)$ in the sense of [LS90]. Write $D_G(\gamma) = \prod_{\sigma>0}(1-\sigma(\gamma)^{-1})$, a square root of the Weyl discriminant up to a unit for elliptic γ, and similarly D_H. We examine the 1-, 2-, and 3-regular germs for G_2 using the formula ([Sh72], [LS90])

$$\begin{aligned}\Delta_{G/H}^{T_G,\kappa}(\gamma_{\lambda,H},\gamma_\lambda)\Phi^{T_G,\kappa}(\gamma_\lambda,f) &= \kappa(\mathrm{inv}(\gamma_G^0,\gamma_\lambda))|D_H(\gamma_{\lambda,H})|^{-1}\int_{\phi^{-1}(\gamma_\lambda)^\Gamma}\pi^*(f)\kappa(t_*)\frac{|\omega|}{|\phi^*(d\lambda)|}\\ &= \sum_{\mathscr{O}}\mu_{\mathscr{O}}(f)\Gamma_{\mathscr{O}}^{T_G,\kappa}(\gamma_\lambda), \qquad f\in C_c^\infty(G(F)),\end{aligned} \tag{0.1}$$

where $\Delta_{G/H}^{T_G,\kappa}$ is the local transfer factor, given in a neighborhood of the identity by

$$\Delta_{G/H}^{T_G,\kappa}(\gamma_H,\gamma_G) = \kappa(\mathrm{inv}(\gamma_G^0,\gamma_G))|D_G(\gamma_G)|/|D_H(\gamma_H)|$$

as in [LS90], γ_G^0 being a fixed regular element of $T_G(F)$. This is only the part $\Delta_{III_1}\Delta_{IV}$ of the transfer factor of [LS87]; the remaining factors form the term $\Delta_0(X)$ below, which is computed for the biquadratic tori in Proposition 7.6. The sum over $\mathscr{O}$ in (0.1) runs over all stable unipotent classes $\mathscr{O}^{st}$ in $G(F)$, $\mathscr{O} \subset \mathscr{O}^{st}$ denoting an arbitrarily chosen $G(F)$-orbit within $\mathscr{O}^{st}$. Also, in (0.1), $\gamma_\lambda = \gamma_{\lambda,G} \in C \cap V_f \subset T_G$ has parameter λ, so that $\gamma_0 = 1$, where $V_f \subset G_{reg}$ is a punctured neighborhood of 1, t_* is a cocycle-valued function defined after Formula 0.6 below, and $\phi^{-1}(\gamma_\lambda)^\Gamma$ denotes the fixed points of the fiber under the action of $\Gamma := \mathrm{Gal}(\overline{F}/F)$ on Y_C determined by T_G, as described in §2. The measure $|\omega|/|\phi^*(d\lambda)|$ incorporates the factor $|D_G(\gamma_{\lambda,G})|$. If $\gamma'_G \in G(F)$ is stably conjugate to γ_G then $\Delta_{G/H}^{T_G,\kappa}(\gamma_H,\gamma'_G) = \kappa(\mathrm{inv}(\gamma_G,\gamma'_G))\Delta_{G/H}^{T_G,\kappa}(\gamma_H,\gamma_G)$. We normalize the measures and use the same notation as in [LS89]. In particular,

$$\Gamma_{\mathscr{O}}^{T,\kappa}(\gamma) = \Delta_{G/H}^{T,\kappa}(\gamma_H,\gamma)\sum_\delta \kappa(\delta)\Gamma_{\mathscr{O}}(\gamma^\delta),$$

if γ is an image of some γ_H. Here δ runs over classes in $H^1(\Gamma,T_G)$ associated to the distinct conjugacy classes $\{\gamma^\delta\}$ in the stable class of γ. These germs $\Gamma_{\mathscr{O}}(\gamma)$ differ from those in [Sh72] in that they do not include the discriminant factor. By a result of Harish-Chandra, if $\mathscr{O}$ is r-regular then

$$\Gamma_{\mathscr{O}}(\exp(\lambda^2 X)) = |\lambda|^{-\dim\mathscr{O}}\Gamma_{\mathscr{O}}(\exp(X)),$$

where for any $u \in \mathcal{O}$, $\dim \mathcal{O} = \dim G_u \backslash G = \dim G - \mathrm{rk}(G) - 2r = 2|\Sigma_G^+| - 2r$, G_u denoting the centralizer of u in G, $r = \dim \mathcal{B}_u$, $\mathcal{B}$ the Borel variety, $\mathcal{B}_u$ its subvariety fixed by u, and Σ_G^+ the positive roots with respect to some Borel subgroup. Since

$$D_G(\exp(\lambda X)) = \prod_{\sigma>0}(1-\sigma(\exp(\lambda X))^{-1}) = \lambda^{|\Sigma_G^+|}\prod_{\sigma>0}\sigma(X) + \text{ higher order terms},$$

the normalized germ $\Gamma^*_{\mathcal{O}}(\gamma) := |D_G(\gamma)|\Gamma_{\mathcal{O}}(\gamma)$ satisfies $\Gamma^*_{\mathcal{O}}(\exp(\lambda^2 X)) = |\lambda|^{2r}\Gamma^*_{\mathcal{O}}(\exp(X))$. Consequently the κ-germ satisfies

$$\Gamma^{T,\kappa}_{\mathcal{O}}(\exp(\lambda^2 X)) = |\lambda|^{2(r-|\Sigma_H^+|)}\Gamma^{T,\kappa}_{\mathcal{O}}(\exp(X)). \tag{0.1A}$$

Example. For $G = G_2$ the unipotent classes are the regular class ($r = 0$, dimension 12), the subregular class $G_2(a_1)$ ($r = 1$, dimension 10), the class $\tilde{A}_1$ of short root elements ($r = 2$, dimension 8), the class A_1 of long root elements ($r = 3$, dimension 6), and the identity ($r = 6$). For $H = (SL(2)\times SL(2))/\{\pm 1\}$ we have $|\Sigma_H^+| = 2$, so by (0.1A) the 2-regular κ-germ is homogeneous of degree 0, like the trivial germ of H, which is constant. For $H = PGL(3)$, $|\Sigma_H^+| = 3$ and the same holds for the 3-regular κ-germ.

The ultimate object of our desires is the following.

Conjecture 0.0A (Matching of germs [LS89])**.** *Let H be an endoscopic group associated to T, κ. Given an r-regular unipotent orbit $\mathcal{O}$ in $G(F)$ there is a finite collection of r-regular unipotent classes $\mathcal{O}_i$ in $H(F)$ and complex numbers a_i such that, for every Cartan subgroup T_H of H over F coming from G and every regular ray $\{\lambda X \mid \lambda \neq 0\}$ in the Lie algebra $\mathfrak{t}_H$, we have*

$$\Gamma^{T,\kappa}_{\mathcal{O}}(\gamma_\lambda) = \sum_i a_i \Gamma^{T_H,st}_{\mathcal{O}_i}(\gamma_{\lambda,H}),$$

for λ sufficiently small.

The requirement that the $\mathcal{O}_i$ be r-regular is forced by homogeneity: by (0.1A) the left side scales as $|\lambda|^{2(r-|\Sigma_H^+|)}$, and the stable germ of an r_i-regular class of H as $|\lambda|^{-\dim \mathcal{O}_i} = |\lambda|^{2(r_i-|\Sigma_H^+|)}$. The trivial classes of $(SL(2)\times SL(2))/\{\pm 1\}$ and $PGL(3)$ are 2- and 3-regular, respectively. As Remark 2026.1 explains, Conjecture 0.0A is now known for all (G,H) [GH16]. Our goal is to understand it for $G = G_2$ and $r = 1,2,3$ through Igusa theory.

The expression in (0.1) may also be written in the form $|D_H(\gamma_{\lambda,H})|^{-1}$ times

$$F(\lambda) = \int_{\phi^{-1}(\lambda)(F)} f_\kappa(y)|\omega_\lambda|, \tag{0.2}$$

where f_κ, as defined by (0.1) and (0.2), satisfies the conditions of being an Igusa integrand in the sense of [La83]. Igusa's expansion [La83] is an asymptotic expression of the form

$$F(\lambda) = \sum_{\theta,\beta}\theta(\lambda)|\lambda|^{\beta-1}\sum_{r=1}^{e(\theta,\beta)}(-\log_q|\lambda|)^{r-1}F_r(\theta,\beta,f_\kappa), \tag{0.3}$$

where θ is a unitary character of $F^\times$ and $\beta > 0$ is a rational number for which $\mathscr{E}(\theta,\beta) \neq \emptyset$. Here $\mathscr{E}(\theta,\beta)$ denotes the set of irreducible components (called **branches**) of $\phi^{-1}(1) \subset Y_C$ such that $\theta^{a(E)} = \kappa_E$ and $\beta = b(E)/a(E)$, where $a(E)$, $b(E)$, κ_E, and $e(\theta,\beta)$ are as in [La83]. As T. Hales observed, since there are no logarithmic terms in the Shalika germ expansion, we have the following.

Lemma 0.4. *If f_κ is as in* (0.2) *then $F_r(\theta,\beta,f_\kappa) = 0$ for all $r > 1$.*

By Howe, Harish-Chandra, and Rogawski [Ro80], the germ of the trivial class is a non-zero constant. For a nontrivial κ the trivial κ-germ of G vanishes, and the trivial germ of H matches with κ-germs of larger classes of G. These will be the 2- and 3-regular classes, through the germ expansion on H,

$$\Phi^{T_H,st}(\gamma_{\lambda,H},f^H)=\int_{\phi_H^{-1}(\gamma_{\lambda,H})^\Gamma}\pi^*(f^H)\frac{|\omega|}{|\phi^*(d\lambda)|}=\sum_{\mathscr{O}}\mu_{\mathscr{O}}(f^H)\Gamma_{\mathscr{O}}^{T_H,st}(\gamma_{\lambda,H}), \tag{0.5}$$

where $H=(SL(2)\times SL(2))/\{\pm 1\}$ or $H=PGL(3)$, and where $f^H\in C_c^\infty(H(F))$ is a function whose stable orbital integrals match the κ-orbital integrals of f. The terms in question are the $\beta=2,3,4$ terms in the Igusa expansions for G_2; on $H_2=(SL(2)\times SL(2))/\{\pm 1\}$ and $H_1=PGL(3)$ the corresponding terms are the trivial-class terms.

Putting together results of [La83] and [Ha89, §2], and using the Shalika germ expansion, we have the following formula.

Formula 0.6. (a) *Assume that we have a Cartan subgroup T_G, a character κ of $H^1(\Gamma,T_G)$, and a $(\beta-1)$-regular stable unipotent class $\mathscr{O}^{st}$ for which there is only one F-rational irreducible branch E_β of $\{\lambda=0\}$ in Y_C with $\pi(E_\beta)=\overline{\mathscr{O}}^{st}$ and $\beta(E_\beta)=\beta$, and that $e(\theta,\beta)\le 1$. Then*

$$|D_H(\gamma_{H,\lambda})|\sum_{\mathscr{O}\subset\mathscr{O}^{st}}\mu_{\mathscr{O}}(f)\Gamma_{\mathscr{O}}^{T_G,\kappa}(\gamma_\lambda)=\kappa(\mathrm{inv}(\gamma_G^0,\gamma_\lambda))|\lambda|^{\beta-1}\sum_\theta\theta(\lambda)\int_{E_\beta(F)}\pi^*(f)\kappa_\theta(t_*)|\omega_{E_\beta}|,$$

for $\gamma_\lambda\in C\cap V_f$. Here $\kappa_\theta(t_*)$ is an extension of $\kappa(t_*)/\theta(\lambda)$ to E_β, $\overline{\mathscr{O}}^{st}$ denotes the F-Zariski closure of the stable class, and the sum on the left runs over all unipotent $G(F)$-orbits $\mathscr{O}$ contained in $\mathscr{O}^{st}$.

(b) *Under the same assumptions, for each $\overline{\mathscr{O}}^{st}$ there is a measure $\omega_{\overline{\mathscr{O}}^{st}}$ and for each F-rational fiber $E_\beta(u)=E_\beta(F)\cap\pi^{-1}(u)$, $u\in\overline{\mathscr{O}}(\overline{F})$, there is a measure $\omega_{E_\beta(u)}$ such that*

$$\int_{E_\beta(F)}\pi^*(f)\kappa_\theta(t_*)|\omega_{E_\beta}|=\int_{\overline{\mathscr{O}}^{st}}f\int_{E_\beta(u)}\kappa_\theta(t_*)|\omega_{\overline{\mathscr{O}}^{st}}||\omega_{E_\beta(u)}|,$$

for each θ occurring above.

By [Ha92a, ch. 6, §2], only the $\theta=1$ term on the right side of Formula 0.6(a) is non-zero. The characters κ that occur for cuspidal H are quadratic (with $H=(SL(2)\times SL(2))/\{\pm 1\}$) or cubic (with $H=PGL(3)$); see Table A.6.

We now recall some of the properties of t_* [La83], [Ha92a]. Recall $\Gamma=\mathrm{Gal}(\overline{F}/F)$. The cocycle-valued function t_* is a map $Y_C\to Z^1(\Gamma,T_G(\overline{F}))$ given by $P\longmapsto(\sigma\longmapsto t_\sigma(P))$ which has the following properties.

(a) It is locally constant on the p-adic manifold $Y_C(F)$ [La83].
(b) It factors through $Z^1(\mathrm{Gal}(E/F),T_G(E))$, where E denotes the Galois closure of the splitting field of T_G.
(c) For fixed $\sigma\in\Gamma$, t_σ is a regular function on Y_C.
(d) For $y=i(\gamma_\lambda,g)\in Y_C$ such that, for all $\sigma\in\Gamma$, $\sigma(g)g^{-1}$ belongs to $T_G(\overline{F})$, we have $\kappa(t_*(y))=\kappa(\sigma(g)g^{-1})$, where $\sigma(g)g^{-1}$ represents the cocycle $\sigma\longmapsto\sigma(g)g^{-1}$.

After Y_C has been constructed, the idea is to determine the sets $\mathscr{E}(\theta,\beta)$ with $\beta=2,3,4$. First, one determines the $\overline{F}$-branches of $\phi^{-1}(1)\subset Y_C(\overline{F})$ which map into Y'' under $Y_C\to Y_1$ and have $\beta=2,3,4$. These are called **fundamental divisors**. For $G=G_2$, the strict transforms of the branches of $\phi^{-1}(1)\subset Y''$ are among them; these are called **base divisors**. We sometimes identify a base divisor with its closure in Y_C. Then one determines which of these fundamental divisors

has F-rational coordinates with respect to the Galois action determined by T_G. These branches contribute to Igusa's expansion. Next, one must consider the $\overline{F}$-branches of $\phi^{-1}(1) \subset Y_C(\overline{F})$ which do not map into Y'' under $Y_C \to Y_1$. These are called **spurious divisors**. The computations in [Ha92a, Ch. III] show that there are no F-rational spurious divisors contributing to the subregular germ. For the 2- and 3-regular germs the 1994 manuscript claimed the following lemma; its proof is incomplete (Remark 2026.27).

Lemma 0.8 (conditional)**.** *Let E be an F-rational base $\mathcal{O}$-divisor of the variety Y'' and let $\tilde{E}$ denote its strict transform in Y_C. The principal value integrals $\int_{E(F)} \pi^*(f)\kappa_1(t_*)|\omega_E|$ and $\int_{\tilde{E}(F)} \tilde{\pi}^*(f)\kappa_1(\tilde{t}_*)|\omega_{\tilde{E}}|$ exist. Here κ_1 is as in Formula 0.6(a), t_* is the cocycle function on Y'', and $\tilde{t}_*$ is the cocycle function on Y_C. Moreover $e(1,\beta_E) = 1$ and $F_1(1,\beta_E,f_\kappa)$ is given by $\kappa(\mathrm{inv}(\gamma_G^0,\gamma_G))$ times*

$$PV\int_{E(F)} \pi^*(f)\kappa_1(t_*)|\omega_E| = PV\int_{\tilde{E}(F)} \tilde{\pi}^*(f)\kappa_1(\tilde{t}_*)|\omega_{\tilde{E}}| = \int_{\overline{\mathcal{O}}^{st}(F)} f \int_{E_u(F)} \kappa_1(t_*)|\omega_{E_u}||\omega_{\overline{\mathcal{O}}^{st}}|,$$

where the measures are as in Formula 0.6, provided *no other branch of $\phi^{-1}(1) \subset Y_C$ with the same (θ,β) has F-points mapping onto $\mathcal{O}^{st}(F)$.*

Remark 2026.3 (the proviso in Lemma 0.8). The 1994 statement had no proviso, and asserted that spurious divisors do not contribute. The equality of the principal values over E and $\tilde{E}$ was not proved there. The proviso is exactly what is missing: the exceptional divisors of the resolution (Remark 2026.23) and the spurious divisors have not been analyzed for the 2- and 3-regular classes.

The 1994 strategy, and what fails. The 1994 manuscript proposed to finish as follows. From (0.1) the germs of the $(\beta-1)$-regular classes are "matched" with the terms of homogeneity degree $\beta-1$ in λ in the Igusa expansion. Both sides also depend on the tangent direction X of C, and if X is replaced by cX, both sides must have the same degree of homogeneity in c. This was meant to give $e(\theta,\beta) \leq 1$ (Lemma 5.14). On the right-hand side of Formula 0.6(a) one finds essentially the factor

$$\Delta_0(X) = \lim_{\lambda\to 0} \frac{\Delta_{loc}(\gamma_{\lambda,G},\gamma_{\lambda,H})}{D_{G/H}(\gamma_{\lambda,G},\gamma_{\lambda,H})},$$

in the notation of [Ha92b, §2]. The manuscript asserted that $\kappa(t_*)\Delta_0(X)$ is invariant under translations $X \mapsto X+Y$ with Y in the Lie algebra of the center of H, and applied Hales' reduction lemma (Lemma 5.19) to conclude. This would have avoided a direct determination of the dependence of $\Delta_0(X)\kappa(t_\sigma)$ on X. The cuspidal endoscopic groups of G_2 have trivial connected center, so this step is not available; see Remark 2026.29. The direct determination is therefore necessary, and it is the main open step of this revision.

The expected matchings, consistent with (0.1A) and now known by [GH16], are as follows. The 0-regular class has been matched [She89]. The 1-regular κ-germ of G_2 should match the 1-regular stable germ of $PGL(3)$ on cubic Cartan subgroups and the 1-regular stable germ of $(SL(2)\times SL(2))/\{\pm 1\}$ on biquadratic and quadratic ones. The trivial germ of $(SL(2)\times SL(2))/\{\pm 1\}$ matches the 2-regular κ-germ of G_2 for the (T,κ) attached to that group, and the trivial germ of $PGL(3)$ matches the 3-regular κ-germ for the (T,κ) attached to $PGL(3)$. This agrees with the fact that the 2-regular and 3-regular classes of $G_2(\overline{F})$ are induced, in Spaltenstein's sense, from the trivial classes of the pseudo-Levi subgroups $(SL(2)\times SL(2))/\{\pm 1\}$ and $SL(3)$ of G_2 [Spa82, ch. III,

table on page 221]; the dimensions are $14-6=8$ and $14-8=6$. The trivial class is matched by Rogawski, Howe, and Harish-Chandra [Ro80].

Status of the results in this revision.

- *Proved:* the regularity of the essential patches and the covering of Z up to codimension two (Proposition 3.4A); the normality of Z (Proposition 3.7); the list of all branches of $\lambda = 0$ on Z with their Igusa constants (Table 4.3A); the list of elliptic endoscopic data (Table A.6); the stability of the 2- and 3-regular classes (§A.7). Proposition 3.4A, Proposition 3.7 and Table 4.3A rest on exhaustive exact computations, whose scripts are in `g2stuff/scripts`.
- *Proved modulo the divisor proviso of Lemma 0.8:* the formulas (7.1) and (8.2) expressing the 2- and 3-regular parts of the κ-orbital integral as principal value integrals over the single base divisors $E_{\emptyset,\alpha,\delta}$ and $E_{\emptyset,\beta,\zeta}$.
- *Conditional:* Theorems 6.1, 7.2 and 8.3, on Hypotheses (X_r) of Remark 2026.29.
- *Known by other means:* Conjectures 0.0 and 0.0A for G_2 (Remark 2026.1).

LITERATURE REVIEW

This section places the paper among related work. It is split at the date of the original manuscript, 1994.

Work before 1994. Shalika [Sh72] proved that the orbital integrals of a regular semisimple element near the identity have an expansion in terms of unipotent orbital integrals, with coefficients (the Shalika germs) that depend only on the element. Ranga Rao [Ra72] proved that unipotent orbital integrals converge and gave the description of rational unipotent orbits used in §A.7. Harish-Chandra's homogeneity of the germs is the basis of all degree counting in this paper. The germ of the trivial class is a constant, computed by Rogawski [Ro80] using the building. For $SL(2)$ the germs were computed explicitly by Sally and Shalika [SS83], and for $GL(n)$ and $SL(n)$ Repka computed the leading, subregular and regular terms [Re84], [Re85].

The endoscopic point of view comes from Langlands and Shelstad. In [LS87] they defined transfer factors, and in [LS90] they reduced the existence of transfer to a local statement at the identity, the matching of κ-Shalika germs. Shelstad [She89] matched the germs of the regular classes for all groups. Langlands [La83] introduced the method used here: he realized orbital integrals as fiber integrals on a variety built from the variety of stars, and analyzed them with Igusa's theory of asymptotic expansions [Ig78]; Langlands and Shelstad [LS83] developed the principal value integrals that compute the coefficients, and completed the case of groups of type A_2 in [LS89]. Hales' thesis [Ha92a] treated the subregular germs in general (stably), and the κ-subregular germs of even unitary groups. In [Ha89] he treated all germs of $GSp(4)$, combining Igusa theory with a reduction to Levi subgroups and an explicit computation of the dependence on the tangent direction, and in [Ha92b] he treated $U(3)$. Spaltenstein's book [Spa82] contains the classification of unipotent classes and the induction used in §0. The toroidal embeddings of Kempf, Knudsen, Mumford and Saint-Donat [Ke72] and Hochster's theorem [Ho72] on normal monomial rings are the algebraic input for §3.

The present paper was written in this setting. Its aim was to extend the Langlands–Hales method to G_2, the first exceptional group, where the cuspidal endoscopic groups $PGL(3)$ and $(SL(2)\times SL(2))/\{\pm 1\}$ are semisimple of full rank.

Work after 1994. Hales [Ha94] showed that harmonic analysis on p-adic groups is not elementary: some Shalika germs of symplectic groups are governed by the number of points on hyperelliptic curves over the residue field. Assem studied unipotent orbital integrals on symplectic and orthogonal groups from the point of view of endoscopy, introducing endoscopic induction of unipotent classes [As98], [As00]. Waldspurger computed the transfer of nilpotent orbital integrals for unramified classical groups [Wa01], and DeBacker proved homogeneity results for invariant distributions in a range of the Moy–Prasad filtration [De02].

The existence of transfer was settled in two steps. Waldspurger proved that the fundamental lemma implies transfer [Wa97], and Ngô proved the fundamental lemma [Ng10]. Gordon and Hales [GH16] used motivic integration to show that Shalika germs are constructible motivic functions, and derived the matching of all κ-Shalika germs from Waldspurger's transfer. Tsai [Ts15] developed methods to compute orbital integrals and Shalika germs through affine Springer fibers.

These results imply the matching statements that this paper tries to prove for G_2. What they do not give is an explicit formula for the G_2 germs or for the matching constants. The Igusa-theoretic description in §§3–8 is a step toward such formulas.

NOTATION

This section collects the notation used throughout. Two clashes from 1994 are kept, to preserve the comparison with the original: β denotes both the long simple root and the Igusa exponent $\beta(E) = b(E)/a(E)$, and $\gamma, \delta, \varepsilon, \zeta$ denote both positive roots and other objects. The meaning is always clear from the argument ($\beta(E)$ always has a divisor as argument).

Symbol	Meaning
F, $\mathcal{O}_F$, q, $\overline{F}$	p-adic field of characteristic 0, its ring of integers, residue field size, algebraic closure
Γ	$\mathrm{Gal}(\overline{F}/F)$
G, G_2	a connected semisimple quasi-split group; the split group of type G_2
T_G, Ω	an (elliptic) maximal torus over F; its absolute Weyl group
$\Delta = \{\alpha, \beta\}$	simple roots of G_2, α short, β long
$\gamma, \delta, \varepsilon, \zeta$	the roots $\alpha+\beta$, $2\alpha+\beta$, $3\alpha+\beta$, $3\alpha+2\beta$
$\Sigma^+ = \Sigma_G^+$, p	the positive roots; $p = \lvert\Sigma^+\rvert$ ($= 6$ for G_2)
H, H_1, H_2	endoscopic group; $H_1 = PGL(3)$, $H_2 = (SL(2) \times SL(2))/\{\pm 1\}$
Σ_H^+	positive roots of H, transported to T_G: $\{\alpha, \gamma, \delta\}$ for H_1, an orthogonal pair such as $\{\alpha, \zeta\}$ for H_2
κ, inv	a character of $H^1(\Gamma, T_G)$; the invariant of a pair of stably conjugate elements
D_G, D_H	$\prod_{\sigma>0}(1-\sigma^{-1})$ on T_G, respectively T_H
$\Delta_{G/H}^{T,\kappa}$	the local transfer factor near the identity
$\Gamma_{\mathcal{O}}$, $\Gamma_{\mathcal{O}}^{T,\kappa}$	Shalika germ of the class $\mathcal{O}$; κ-germ normalized by the transfer factor
$\mu_{\mathcal{O}}(f)$	the orbital integral of f over the unipotent class $\mathcal{O}$
$\mathcal{O}_r$	an r-regular unipotent class: $\dim \mathscr{B}_u = r$ for $u \in \mathcal{O}_r$
$\mathscr{B}$, $\mathscr{W}$	the Borel variety; the set of Weyl chambers

Symbol	Meaning
$S_0 \subset S$, S_1, S''	regular stars, stars, the partial resolution, and its open part (Def. 1.16)
$z(W,\rho)$, $z(\alpha)$, $z_1(W,\rho)$	Langlands' star coordinates and the blow-up coordinates
$x(\rho)$, $w(\rho)$, $v(\rho)$, λ	coordinates on Y'' and N_+; λ the parameter on C
Z	the monomial variety of §3
$Z_{S_1,S_2,\sigma}$, $E_{S_1,S_2,\sigma}$	fundamental patch and the branch of $\lambda = 0$ on it
$a(E)$, $b(E)$, $\beta(E)$	multiplicity of E in $\lambda = 0$, order of the Igusa form plus one, $\beta(E) = b(E)/a(E)$
$e(\theta,\beta)$	the maximal number of branches with data (θ,β) through a point (see [La83])
X, C	the tangent direction of the curve $C = \exp(\mathscr{O}_F X)$ in T_G
t_*	the cocycle-valued function on Y_C

§ 1 SOME BACKGROUND

This section recalls Langlands' construction: the variety of stars, the star coordinates $z(W,\rho)$, the partial resolution S_1, and the varieties Y'' and Y_1 on which the orbital integrals become fiber integrals. It ends with the coordinate relations for G_2 that connect the coordinates on Y'' with those on Y_1.

Let G denote a connected semisimple quasi-split algebraic group defined over F and let T_G denote a maximal torus of G defined over F, which later will be elliptic. We work over $\overline{F}$ for most of this section. A **root** is a character $\rho \in X^*(T)$ of a torus T such that there exists an isomorphism $\theta_\rho : \mathbb{G}_a \to U_\rho$ onto a closed subgroup $U_\rho \subset G$ for which $t\,\theta_\rho(\xi)t^{-1} = \theta_\rho(\rho(t)\xi)$, for $t \in T$ and $\xi \in \mathbb{G}_a$. Here $\mathbb{G}_a$ denotes the 1-dimensional additive group. Let B_G denote a Borel subgroup containing T_G, Σ_G the set of roots of T_G, Σ_G^+ the set of positive roots with respect to B_G, and Δ the set of simple roots in Σ_G^+. When $G = G_2$, we write $\Delta = \{\alpha,\beta\}$, with α short, and

$$\Sigma_G^+ = \{\alpha, \beta, \gamma = \alpha+\beta, \delta = 2\alpha+\beta, \varepsilon = 3\alpha+\beta, \zeta = 3\alpha+2\beta\}.$$

Throughout, $B^g = g^{-1}Bg$ for a subgroup B and $g \in G$.

Fix a Borel subgroup $\underline{B}$ of G defined over F, a maximal torus $\underline{T}$ defined over F contained in $\underline{B}$, and a positive Weyl chamber W_0 of T_G with respect to Σ_G^+.

Let Ω denote the absolute Weyl group of T_G. We occasionally identify the roots of G with respect to T_G with those with respect to $\underline{T}$ by means of a fixed isomorphism $T_G \to \underline{T}$, and thus identify the absolute Weyl groups of $\underline{T}$ and T_G. Representing $\omega \in \Omega$ by $w \in N_{G(\overline{F})}(T_G(\overline{F}))$, define

$$B_G(W) := w^{-1}B_G w = B_G^w, \qquad W = \omega^{-1}W_0.$$

The Borel subgroups $\{B_G(W) \mid W \in \mathscr{W}\}$ are precisely the Borel subgroups of G which contain T_G, where $\mathscr{W}$ denotes the set of Weyl chambers of T_G.

Example. For $G = G_2$ there are $|\Omega| = 12$ Weyl chambers, so a star is a 12-tuple of Borel subgroups. The variety of regular stars $S_0 \cong T_G \backslash G$ has dimension $14 - 2 = 12$, and $\mathscr{B}$ has dimension 6.

Recall that a **regular star** is a $|\Omega|$-tuple of Borel subgroups $(B(W))_{W\in\mathscr{W}}$ such that, for all $W \in \mathscr{W}$, $B(W) = B_G(W)^h = h^{-1}B_G(W)h$, for some $h \in G(\overline{F})$ independent of W. The variety of regular stars is denoted by S_0. There is a natural bijection between $T_G(\overline{F}) \backslash G(\overline{F})$ and $S_0(\overline{F})$ given

by $g \longmapsto (B_G(W)^g)_{W\in\mathscr{W}}$. Let

$$X_0 := \{(g,(B(W))_{W\in\mathscr{W}}) \in G\times S_0 \mid g \in \cap_{W\in\mathscr{W}} B(W) \text{ is regular}\}, \tag{1.1}$$

so that we have an $\overline{F}$-isomorphism

$$T'_G \times T_G \setminus G \to X_0, \qquad (t, T_G g) \longmapsto (g^{-1}tg, (B_G(W)^g)_W). \tag{1.2}$$

Definition 1.3. Let $(g,s) \in X_0$, with $s = (B_G(W)^h)_W$. Since $\cap_W B_G(W)^h = T_G^h$, we have $hgh^{-1} \in T_G$, and this element does not depend on the choice of h (which is determined up to left multiplication by T_G). Let $\phi : X_0 \to T_G$ be the map $(g,s) \mapsto hgh^{-1}$. Let $\pi : X_0 \to G$ denote the projection onto the G-coordinate.

Under (1.2), $\phi(g^{-1}tg, (B_G(W)^g)_W) = t$. The map π restricts to an isomorphism from $\phi^{-1}(\gamma)$ onto the conjugacy class of γ,

$$\pi : \phi^{-1}(\gamma) \to \{g^{-1}\gamma g \mid g \in T_G \setminus G\}.$$

Remark 2026.4. The 1994 text defined $\phi(g,s)$ as "the point in T_G which meets the conjugacy class of g". A regular conjugacy class meets T_G in a whole Ω-orbit, so that definition only gives a point of T_G/Ω. The star must be used to choose the point, as in Definition 1.3. (The number 1.3 was unused in 1994.)

Construction of Y_1. Let $C \subset T_G$ be a one-parameter subgroup of G containing 1 but no other singular point. Let $C' := C - \{1\}$ and let $Y_0 := \phi^{-1}(C') \subset X_0$. The isomorphism (1.2) restricts to an $\overline{F}$-isomorphism

$$C' \times T_G \setminus G \to Y_0. \tag{1.4}$$

The variety Y_0 is a regular algebraic subvariety of $G\times S$ [La83, p. 480], but it is not complete. Let Y denote the closure of Y_0 in $G\times S$. In general Y is singular.

Let $\mathscr{B} \cong B_G \setminus G$ denote the Borel variety. It will be helpful to record local coordinates for $\mathscr{B}$. First we recall the following well-known result [Bor91, Ch. IV, Corollary 14.13].

Lemma 1.5. *If $B, B', B'' \in \mathscr{B}$ and B', B'' are opposite to B then they are conjugate by an element of B.*

As a consequence of the Levi decomposition, for B, B', B'' as in the lemma, B' and B'' are conjugate by a unique element of N, the unipotent radical of B. Let $\mathscr{B}(B) \subset \mathscr{B}$ denote the open subvariety of all Borel subgroups opposite to B.

Corollary 1.6. *For each $B \in \mathscr{B}$, $\mathscr{B}(B)$ is isomorphic as an affine variety to N, and the coordinates on N serve as coordinates on $\mathscr{B}(B)$.*

A point $(B(W))_{W\in\mathscr{W}} \in \mathscr{B}^{|\Omega|}$ is called a **star** if for every $\omega \in \Omega$ and every simple root α of T_G in B_G there is an $h \in G(\overline{F})$ such that

$$B(\omega^{-1}W_0) = h^{-1}B_G h \quad \text{and} \quad B((s_\alpha\omega)^{-1}W_0) \subset h^{-1}P_{G,\alpha}h,$$

where $P_{G,\alpha}$ is the smallest parabolic subgroup of G containing B_G and $B_G^{s_\alpha}$, and s_α denotes the reflection associated to α. The variety of stars, denoted S, is closed in $\mathscr{B}^{|\Omega|}$ and projective, but in general not regular. A regular star is a star. In fact, the variety of stars is the Zariski closure of S_0 in $\mathscr{B}^{|\Omega|}$ [La83]; this can be seen from the coordinate patches below. That S_0 is open in its Zariski closure holds because orbits are locally closed [Bor91, II.6, Prop. 6.7].

We now define some open patches on S and regular coordinates on these patches. Fix a Borel subgroup B_+ of G and another Borel subgroup B_- opposite to B_+; let $T_+ := B_+ \cap B_-$. Let

$$S(B_+) := \{(B(W))_{W\in\mathscr{W}} \in S \mid \forall W \in \mathscr{W},\ B(W) \text{ is opposite to } B_+\}$$

and

$$S(B_+,B_-) := \{(B(W))_{W\in\mathscr{W}} \in S(B_+) \mid B(W_0) = B_-\}.$$

Let N_+ denote the unipotent radical of B_+. The map $((B(W))_{W\in\mathscr{W}},n) \longmapsto (B(W)^n)_{W\in\mathscr{W}}$ defines an isomorphism

$$S(B_+,B_-)\times N_+ \to S(B_+). \tag{1.7}$$

This is because the map $n \longmapsto B_-^n$, $n \in N_+$, is an isomorphism from N_+ onto the open subvariety $\mathscr{B}(B_+) \subset \mathscr{B}$ of Borel subgroups opposite to B_+ (Corollary 1.6).

Following [La83], we now describe local coordinates on $S(B_+,B_-)$. We identify the absolute Weyl group and the Weyl chambers of T_G with those of T_+. If W is a Weyl chamber and ρ a root defining a wall of W, we call (W,ρ) a **wall-chamber pair**, or simply a **pair**. We may find $\omega \in \Omega$ for which $W = \omega^{-1}W_0$ and $\rho = \pm\omega^{-1}\alpha$, for a simple root α of T_+ in B_-; we write $\alpha(W,\rho) = \omega\rho$, with the sign chosen so that it is simple.

Given a star $(B(W))_{W\in\mathscr{W}} \in S(B_+,B_-)$ and a pair (W,ρ), one defines the value at $(B(W))_W$ of the coordinate function $z(W,\rho)$ as follows. Let W' be the chamber adjacent to W across the hyperplane $\rho^\perp \subset X^*(T_G)\otimes\mathbb{R}$, and let $h \in N_+(\overline{F})$ be the unique element satisfying

$$B(W) = h^{-1}B_-h \qquad \text{and} \qquad B(W') \subset h^{-1}P_{-,\alpha}h,$$

where $P_{-,\alpha}$ is the smallest parabolic subgroup of G containing B_- and $B_-^{s_\alpha}$, $\alpha = \alpha(W,\rho)$. The map

$$\mathbb{G}_a \cong U_{-\alpha} \hookrightarrow P_{-,\alpha}/B_-, \qquad z \longmapsto u = \exp(zX_{-\alpha}) \longmapsto B_-^u,$$

is an open immersion onto the complement of the point $B_-^{s_\alpha}$ of the projective line $P_{-,\alpha}/B_- \subset \mathscr{B}$. There is a coordinate function $z(W,\rho) : S(B_+,B_-) \to \mathbb{A}$ such that

$$hB(W')h^{-1} = \theta_{-\alpha}(-z(W,\rho))B_-\theta_{-\alpha}(z(W,\rho)), \qquad \alpha = \alpha(W,\rho), \tag{1.8}$$

where X_α, $X_{-\alpha}$ are root vectors satisfying

$$[X_\alpha,X_{-\alpha}] = H_\alpha, \qquad \beta(H_\alpha) = 2\frac{(\beta,\alpha)}{(\alpha,\alpha)}, \qquad \theta_{-\alpha}(z) = \exp(zX_{-\alpha}) \in U_{-\alpha}.$$

The local coordinate $z(W_0,\alpha)$ on $S(B_+,B_-)$ is a local coordinate on a line of type α in the sense of [St74]. This will be useful for the subregular germ. We have:

(a) $z(W,\rho) = 0$ if and only if $B(W) = B(W')$;
(b) the collection of the $z(W,\rho)$, for all pairs (W,ρ), uniquely determines the star $(B(W))_W \in S(B_+,B_-)$;
(c) the coordinate functions $z(W,\rho)$ are regular and generate the coordinate ring of $S(B_+,B_-)$.

Remark 2026.5. Three conventions in this subsection were inconsistent in 1994 and are corrected here: the root condition was written with $t^{-1}\theta_\rho(\xi)t$; the pair condition read $\rho = \omega\alpha$, which contradicts $\alpha_{i+1} = \omega_i\rho_{i+1}$ below; and the sign of z in (1.8) was opposite to that in (1.9) and (1.10). The versions above agree with (1.9) and (1.10). The open immersion $U_{-\alpha} \hookrightarrow P_{-,\alpha}/B_-$ was written as an isomorphism.

The coordinates $z(W,\rho)$ are not algebraically independent. Their relations are most naturally described using galleries in the spherical building of Ω, which we identify with that of T_+. Recall that a **gallery** is a sequence $W_0, W_1, \ldots, W_r$ of Weyl chambers such that W_i, W_{i+1} are distinct but share a wall $\rho_{i+1}^{\perp}$. We always take W_0 to be the positive Weyl chamber of T_+ with respect to B_+ and define $\omega_i \in \Omega$ by $W_i = \omega_i^{-1} W_0$. Let $\alpha_{i+1} := \omega_i \rho_{i+1}$, so $\alpha_{i+1} = \alpha(W_i, \rho_{i+1})$. If $z_i := z(W_{i-1}, \rho_i)$ for $1 \le i \le r$, then

$$B(W_{i-1}) = h^{-1} B_- h, \qquad B(W_i) = h^{-1} \theta_{-\alpha_i}(-z_i) B_- \theta_{-\alpha_i}(z_i) h, \tag{1.9}$$

for some $h \in N_+(\overline{F})$ depending on i. Define $n, n_{W_j} \in N_+(\overline{F})$, for $1 \le j \le r$, by $B(W_j) = B_-^{n_{W_j} n}$ and

$$n_{W_j} = \exp(z_j X_{-\alpha_j}) \exp(z_{j-1} X_{-\alpha_{j-1}}) \cdots \exp(z_1 X_{-\alpha_1}). \tag{1.10}$$

In particular, if $W_r = W_0$ (so that the gallery is closed), then

$$\exp(z_r X_{-\alpha_r}) \exp(z_{r-1} X_{-\alpha_{r-1}}) \cdots \exp(z_1 X_{-\alpha_1}) = 1. \tag{1.11}$$

In particular, if W, W' are adjacent across $\rho^{\perp}$ then $z(W,\rho) = -z(W',\rho)$. We now state an observation of Langlands [La83, pp. 483–484].

Lemma 1.12. *The following are equivalent.*

(a) A star $(B(W))_{W \in \mathscr{W}} \in S(B_+, B_-)$ is regular.
(b) $B(W) \neq B(W')$ for all adjacent W, W'.
(c) $z(W,\rho) \neq 0$ for all pairs (W,ρ).

We now define coordinate patches on a partial resolution S_1 of S [Ha92a]. Recall that the short roots, and likewise the long roots, form a single orbit under the Weyl group. Define

$$S'(B_+, B_-) := \{(B(W))_W \in S(B_+, B_-) \mid \forall \alpha \in \Delta,\ \exists \omega \in \Omega \text{ such that } z(\omega^{-1} W_0, \omega^{-1}\alpha) \neq 0\}.$$

If a star in $S(B_+, B_-)$ is not in $S'(B_+, B_-)$, then for *some* $\alpha \in \Delta$ we have $z(\omega^{-1} W_0, \omega^{-1}\alpha) = 0$ for all $\omega \in \Omega$. By Lemma 1.12, such a star is not regular. Hence $S_0(B_+, B_-) \subset S'(B_+, B_-)$.

Let $\mathbb{A}^{\Delta}$ denote the $|\Delta|$-fold product of affine lines and let T_+ act on $\mathbb{A}^{\Delta}$ by

$$t : (z(\alpha))_{\alpha \in \Delta} \longmapsto (\alpha(t)^{-1} z(\alpha))_{\alpha \in \Delta},$$

and on $S'(B_+, B_-)$ by

$$t : (z(W,\rho))_{(W,\rho)} \longmapsto (\alpha(t) z(W,\rho))_{(W,\rho)}, \qquad \alpha = \alpha(W,\rho).$$

The action on $S'(B_+, B_-)$ is free, since for each α some coordinate of type α is non-zero. Let

$$S_1(B_+, B_-) := S'(B_+, B_-) \times_{T_+} \mathbb{A}^{\Delta} := (S'(B_+, B_-) \times \mathbb{A}^{\Delta})/T_+. \tag{1.13}$$

There is a "blow-down" map

$$S_1(B_+, B_-) \to S(B_+, B_-), \tag{1.14}$$

given by $(s', z) T_+ \longmapsto s$, where $s' = (z_1(W,\rho))_{(W,\rho)}$, $z = (z(\alpha))_{\alpha \in \Delta}$, and

$$s = (z(W,\rho))_{(W,\rho)} = (z(\alpha(W,\rho)) z_1(W,\rho))_{(W,\rho)}.$$

This is well-defined, i.e., independent of the chosen representative z of the T_+-orbit in $\mathbb{A}^{\Delta}$. The subscript 1 on $z_1(W,\rho)$ distinguishes it from its image $z(W,\rho) = z(\alpha(W,\rho)) z_1(W,\rho)$ in $S(B_+, B_-)$. Later we fix a canonical choice of a generic representative $(z(\alpha))_{\alpha \in \Delta}$: $z(\alpha) = \lambda / x(\alpha)$, where $x(\alpha)$ is as in (1.17) below (equivalently, the representative which makes $w(\alpha) = 1$ for α simple [Ha92a, Ch. II, §2]).

Let $S'(B_+) \subset S(B_+)$ be the image of $S'(B_+,B_-) \times N_+$ in $S(B_+)$ under (1.7), and let $S_1(B_+)$ be the image of $S_1(B_+,B_-) \times N_+$ under

$$((B(W))_W, (z(\alpha))_{\alpha\in\Delta}, n) \longmapsto ((B(W)^n)_W, (z(\alpha))_{\alpha\in\Delta}).$$

Therefore

$$S'(B_+) \cong S'(B_+,B_-) \times N_+, \tag{1.15a}$$

and

$$S_1(B_+) \cong S_1(B_+,B_-) \times N_+. \tag{1.15b}$$

The apparent dependence of $S_1(B_+)$ on B_- is not serious: by [La83, Lemma 3.6], if we replace B_- by another Borel subgroup B'_- opposite to B_+ then the corresponding variety is canonically isomorphic to $S_1(B_+)$. Let B'_+ be another Borel subgroup and let B_- be opposite to both B_+ and B'_+. There is a birational map from $S_1(B_+)$ to $S_1(B'_+)$, defined in [La83]. Using these maps to glue the $S_1(B_+)$, we obtain a variety S_1. The map $S_0(B_+) \to S_1(B_+)$, $s \mapsto (s,(1)_\alpha)T_+$, extends to a morphism $S_0 \to S_1$. This yields a map $X_0 \to G \times S_1$.

Definition 1.16. Let $S''(B_+) \subset S_1(B_+)$ be the open subvariety on which every coordinate $z_1(W,\rho)$ is non-zero, and let $S'' \subset S_1$ be the union of the $S''(B_+)$. It contains the image of S_0; on it, the coordinates $z(\alpha)$ are unrestricted. Let X be the Zariski closure of X_0 in $G \times S$, X_1 the closure of X_0 in $G \times S_1$, and Y_1 the closure of Y_0 in X_1. Let $X_1(B_+)$ denote the intersection of X_1 with $G \times S_1(B_+)$, and similarly for $Y_1(B_+)$. Define X'', Y'', $X''(B_+)$, $Y''(B_+)$ in the same way, with S'' and $S''(B_+)$ in place of S_1 and $S_1(B_+)$.

Remark 2026.6 (the definition of S''). The 1994 text defined S'' as "the image of S_0" (in S_1 in §0, in Y_1 in Definition 1.16). That set consists of regular stars. Over $\lambda = 0$ the only unipotent element of $\cap_W B(W)$ for a regular star is 1, so the base divisors of §§3–8, on which some $z(\alpha) = 0$, would not lie in Y''. Definition 1.16 is the definition under which the coordinates (1.17), (1.20) and the monomial variety Z of §3 describe Y''. It is presumably the definition of [Ha92a], which was not consulted (level 3).

In particular, the first coordinate of a point $P \in Y''(B_+,B_-) \subset G \times S''$ is an element b of $\cap_W B_-^{n_W} \subset B_-$, which we may write as an ordered product

$$b = t_0 \prod_{\rho\in\Sigma_G^+} \exp(x(\rho)X_\rho), \tag{1.17}$$

where Σ_G^+ denotes the set of roots of T_+ which are positive with respect to B_-, on which we fix an ordering once and for all.

Coordinates on Y''. Consider a point $x \in X_1 \subset G \times S_1$. On the patch of X_1 meeting $G \times S_1(B_+,B_-) \times N_+ \cong G \times S_1(B_+)$, we write x as $(b,(B(W))_W,(z(\alpha))_{\alpha\in\Delta},n)$, where $b \in \cap_W B(W) \subset B_-$ and $B(W_0) = B_-$. The condition $b \in \cap_W B(W)$ is inherited from the definition of X_0.

Other coordinates on $X''(B_+)$ may be defined in terms of a point $(t^g,(B_G(W)^g)_W)$ in $G \times S(B_+,B_-)$, where $t \in T_G(\overline{F})$ is regular. We may write

$$g = g_0 t_+ n_- n_+,$$

where $T_G^{g_0} = T_+$, $t_+ \in T_+(\overline{F})$, $n_- \in N_-(\overline{F})$, and $n_+ \in N_+(\overline{F})$. Let $t'_+ := t^{g_0} \in T_+(\overline{F})$, and write $t^g = (t'_+ n'_-)^{n_+}$ with $n'_- \in N_-(\overline{F})$. Then

$$(t^g,(B_G(W)^g)_W) = (t'_+ n'_-, (B_-^{n_W})_W)^{n_+},$$

where $n_W \in N_+(\overline{F})$. We write

$$n'_- = \prod_{\rho \in \Sigma_G^+} \exp(x(\rho)X_\rho), \qquad n_+ = \prod_{\rho \in \Sigma_G^+} \exp(v(\rho)X_{-\rho}),$$

where, by (1.10),

$$n_W = \exp(z(W_{r-1}, \rho_r)X_{-\alpha_r}) \cdots \exp(z(W_0, \rho_1)X_{-\alpha_1}), \tag{1.18}$$

for a gallery $W_0, \ldots, W_r = W$ from the positive Weyl chamber, with $W = \omega^{-1}W_0$ and $\omega = s_{\alpha_r} \cdots s_{\alpha_1}$. Recall that $z_1(W, \rho)$ is defined, for any pair (W, ρ), by

$$z(W, \rho) = z(\alpha(W, \rho))z_1(W, \rho). \tag{1.19}$$

Define the coordinate function $w(\rho)$, $\rho > 0$, by $w(\alpha) = 1$ if α is simple, and otherwise by

$$(t'_+)^{-1}(t'_+)^{n_-} = \prod_{\rho \in \Sigma_G^+} \exp\Big(\frac{w(\rho)\lambda}{\prod_{\alpha \in \Delta} z(\alpha)^{m(\alpha)}} X_\rho\Big), \tag{1.20}$$

where $\rho = \sum_{\alpha \in \Delta} m(\alpha)\alpha$. Comparing with the definition of $x(\rho)$, this says $\lambda w(\rho) = x(\rho)\prod_\alpha z(\alpha)^{m(\alpha)}$, which is the relation (3.1) of §3. This describes a point in the image of

$$C' \times T_G \backslash G \hookrightarrow Y_1(B_+, B_-) \times N_+ \subset X_1(B_+, B_-) \times N_+,$$

where $C' = C \cap T'_G = C - \{1\}$.

Remark 2026.7. In 1994, (1.18) was written with $z(W_i, \rho_i)$ in place of $z(W_{i-1}, \rho_i)$; the version above agrees with (1.10).

Coordinates on Y_1. We begin with the following.

Lemma 1.21 ([Ha92a, Ch. II, Lemma 3.3])**.** *If $t \in T_+$ and $n = \prod_{\rho > 0} \theta_\rho(n_\rho)$ (ordered product) lies in the unipotent radical generated by the U_ρ, $\rho > 0$, then*

$$t^n = t \prod_{\rho > 0} \theta_\rho(y(\rho)), \qquad y(\rho) = \sum_{\rho_1 + \cdots + \rho_m = \rho} c_{\rho_1, \ldots, \rho_m}(t)\, n_{\rho_1} \cdots n_{\rho_m},$$

the sum ranging over all decompositions of ρ into positive roots, where each $c_{\rho_1,\ldots,\rho_m}(t)$ is a polynomial in the values $\sigma(t)^{\pm 1}$ of roots, divisible by some factor $1 - \sigma_1(t)^{-1}$. Moreover, $c_\rho(t) = 1 - \rho(t)^{-1}$.

We apply this lemma to obtain coordinates of a point

$$(t^g, (B(W))_W, (z(\alpha))_{\alpha \in \Delta})^n \in \chi^{-1}(C) \times S_1(B_+, B_-) \times N_+.$$

For $G = G_2$ we give the coordinate relations between the coordinates in (1.20) and the n_ρ and $z(W,\alpha)$ (see [Ha92a, Ch. II]):

$$
\begin{aligned}
x(\alpha) &= (1-\alpha^{-1})n_\alpha = (1-\alpha^{-1})/z(W_0,\alpha),\\
x(\beta) &= (1-\beta^{-1})n_\beta = (1-\beta^{-1})/z(W_0,\beta),\\
x(\gamma) &= (1-\gamma^{-1})n_\gamma + c_1\cdot\frac{(1-\alpha^{-1})}{z(W_0,\alpha)z(W_0,\beta)z(\sigma_\beta^{-1}W_0,\alpha)}+\cdots,\\
x(\delta) &= (1-\delta^{-1})n_\delta + c_2\cdot\frac{(1-\alpha^{-1})}{z(W_0,\alpha)z(W_0,\beta)z(\sigma_\beta^{-1}W_0,\alpha)}+\cdots,\\
x(\varepsilon) &= (1-\varepsilon^{-1})n_\varepsilon + c_3\cdot\frac{(1-\alpha^{-1})n_\delta}{z(W_0,\alpha)}+\cdots,\\
x(\zeta) &= (1-\zeta^{-1})n_\zeta + c_4\cdot\frac{(1-\alpha^{-1})n_\gamma^2}{z(W_0,\alpha)} + c_5\cdot\frac{(1-\alpha^{-1})n_\delta}{z(W_0,\alpha)z(W_0,\beta)}\\
&\quad + c_6\cdot\frac{(1-\beta^{-1})n_\varepsilon}{z(W_0,\beta)} + c_7\cdot\frac{(1-\gamma^{-1})n_\delta}{z(W_0,\beta)z(\sigma_\beta^{-1}W_0,\alpha)}+\cdots,
\end{aligned}
\tag{1.22}
$$

where α^{-1} abbreviates $\alpha(t_0)^{-1}$ and $t_0 = \exp(\lambda X)$. Consequently, with $\rho(X)$ the derivative of $\rho(\exp(\lambda X))$ at $\lambda = 0$ and $z(\alpha) = \lambda/x(\alpha)$,

$$
\begin{aligned}
w(\gamma) &= \frac{x(\gamma)z(\beta)z(\alpha)}{\lambda} = \Big(\gamma(X)n_\gamma + c_1\frac{\alpha(X)}{z(W_0,\alpha)z(W_0,\beta)z(\sigma_\beta^{-1}W_0,\alpha)}\Big)\frac{z(W_0,\alpha)z(W_0,\beta)}{\alpha(X)\beta(X)}+\cdots,\\
w(\delta) &= \frac{x(\delta)z(\beta)z(\alpha)^2}{\lambda} = \Big(\delta(X)n_\delta + c_2\frac{\alpha(X)}{z(W_0,\alpha)z(W_0,\beta)z(\sigma_\beta^{-1}W_0,\alpha)}\Big)\frac{z(W_0,\alpha)^2z(W_0,\beta)}{\alpha(X)^2\beta(X)}+\cdots,\\
w(\varepsilon) &= \frac{x(\varepsilon)z(\beta)z(\alpha)^3}{\lambda} = \Big(\varepsilon(X)n_\varepsilon + c_3\frac{\alpha(X)n_\delta}{z(W_0,\alpha)}\Big)\frac{z(W_0,\alpha)^3z(W_0,\beta)}{\alpha(X)^3\beta(X)}+\cdots,\\
w(\zeta) &= \frac{x(\zeta)z(\beta)^2z(\alpha)^3}{\lambda} = \Big(\zeta(X)n_\zeta + c_4\frac{\alpha(X)n_\gamma^2}{z(W_0,\alpha)} + c_5\frac{\alpha(X)n_\delta}{z(W_0,\alpha)z(W_0,\beta)} + c_6\frac{\beta(X)n_\varepsilon}{z(W_0,\beta)}\\
&\quad + c_7\frac{\gamma(X)n_\delta}{z(\sigma_\beta^{-1}W_0,\alpha)z(W_0,\beta)}\Big)\frac{z(W_0,\alpha)^3z(W_0,\beta)^2}{\alpha(X)^3\beta(X)^2}+\cdots,
\end{aligned}
$$

where "$\cdots$" denotes terms that vanish at $\lambda = 0$.

Remark 2026.8 (the relations (1.22)). The relations (1.22) are quoted from [Ha92a, Ch. II], which was not consulted; the constants $c_1,\dots,c_7$ are not determined here. Two misprints of 1994 are corrected above: the c_1-term of $w(\gamma)$ lacked the factor $z(\sigma_\beta^{-1}W_0,\alpha)$ that appears in $x(\gamma)$, and the last term of $w(\zeta)$ lacked c_7 and had n_ζ for n_δ. The 1994 text concluded that, on $Y_1(B_+,B_-)$, the seven functions $z(W_0,\alpha)$, $z(W_0,\beta)$, $z(\sigma_\beta^{-1}W_0,\alpha)$, n_γ, n_δ, n_ε, n_ζ may replace the coordinates of (1.17) and (1.20). Together with λ these are eight functions on a variety of dimension 7, since $Y_1(B_+,B_-)$ has dimension $1+\dim S(B_+,B_-) = 1+6$. So they cannot be algebraically independent, and the correct list of coordinates near $\lambda = 0$ is left open. This affects only the computations of the dependence on X in §§6–8, which are themselves incomplete (Remark 2026.29).

§ 2 GALOIS ACTION ON THE COORDINATES: EXPLICIT COMPUTATIONS FOR G_2

This section describes how $\Gamma = \mathrm{Gal}(\overline{F}/F)$ acts on stars and on the coordinates of §1, when the torus T_G is defined over F but not split. The F-rational points of Y_C, over which the orbital integrals are taken, are the fixed points of this action. For G_2 we compute the action of a Galois element that acts on T_G as a simple reflection.

Let G denote a connected semisimple F-quasi-split group with no outer automorphisms. If T is a maximal torus of G defined over F then Γ preserves T and the set Σ of roots of G with respect to T, $\sigma \in \Gamma$ sending U_ρ to $U_{\sigma\rho}$. In general Γ does not preserve the set Δ of simple roots with respect to a Borel subgroup B containing T, unless B is defined over F. The character lattice $X := X^*(T) = X^*(T/\overline{F})$ is a $\mathbb{Z}[\Gamma]$-module, and the action of Γ on $X_{\mathbb{R}} := X^*(T) \otimes \mathbb{R}$ preserves the set of root hyperplanes. Consequently Γ acts on the set of Weyl chambers in $X_{\mathbb{R}}$; it preserves the positive Weyl chamber if B is defined over F.

As in §1, we fix a Borel subgroup $\underline{B}$ of G defined over F, a maximal torus $\underline{T} \subset \underline{B}$ defined over F, and the positive Weyl chamber W_0 for the order which makes the roots occurring in $\underline{B}$ positive. Let T_G denote our fixed maximal torus of G defined over F and let B_G denote a Borel subgroup containing T_G. For any $\sigma \in \Gamma$ both B_G and $\sigma(B_G)$ contain T_G, since T_G is defined over F. The Galois action on $\underline{T}$ induces a homomorphism

$$\rho : \Gamma \to \mathrm{Aut}(X, \Delta, X^\vee, \Delta^\vee), \tag{2.1a}$$

where $X = X^*(\underline{T})$ is the character lattice, $X^\vee = X_*(\underline{T})$ the cocharacter lattice, $\Delta = \Delta(\underline{T}, \underline{B})$ the simple roots and $\Delta^\vee$ the simple coroots. This induces ([Sp79]) a homomorphism

$$\tilde{\rho} : \Gamma \to \mathrm{Aut}(G, \underline{B}, \underline{T}, \{X_\alpha\}_{\alpha \in \Delta}), \tag{2.1b}$$

where X_α denotes a root vector for α in the Lie algebra of G.

The Borel subgroups $B_G(W)$, $W \in \mathscr{W}$, are the Borel subgroups of G which contain T_G, so they are permuted by Γ. Define $\sigma_T : \mathscr{W} \to \mathscr{W}$ by

$$\sigma(B_G(W)) = B_G(\sigma_T W),$$

for $\sigma \in \Gamma$. This defines a continuous homomorphism $\Gamma \to \Omega$, $\sigma \longmapsto \sigma_T$, which factors through $\mathrm{Gal}(E/F)$, where E/F denotes the splitting field of T_G. We may write $B_G = \underline{B}^g$ for some $g \in G(\overline{F})$ with $T_G = \underline{T}^g$; then $g^{-1}\sigma(g) \in N_G(T_G)(\overline{F})$ for all $\sigma \in \Gamma$, and its image in Ω is the Weyl element that gives σ_T. The action σ_T on chambers induces an action, also denoted σ_T, on roots, taking the walls $\rho^\perp$ of W to the walls $\sigma_T(\rho)^\perp$ of $\sigma_T W$.

Remark 2026.9. In 1994 the condition read $g^{-1}\sigma(g) \in T_G(\overline{F})$ for all σ. That would make B_G defined over F, which is impossible for an elliptic T_G. It also said the homomorphism factors through $\mathrm{Gal}(\overline{F}/E)$, which is its kernel; it factors through $\mathrm{Gal}(E/F)$.

We define an action of Γ on the set S_0 of regular stars by

$$\sigma(B(W)) := B_G(W)^{\sigma(h)} = \sigma(B(\sigma_T^{-1} W)), \tag{2.2}$$

where $B(W) = B_G(W)^h$. The F-rational points of $S_0(\overline{F})$ for this Γ-action are the regular stars $(B(W))_W$ with $B(W) = B_G(W)^h$ and $\sigma(h)h^{-1} \in T_G(\overline{F})$ for all $\sigma \in \Gamma$.

For $\gamma \in T_G(F)$, let

$$\phi_\gamma^{-1}(F) := \phi^{-1}(\gamma)^\Gamma = \{x \in X_0(F) \mid \phi(x) = \gamma\},$$

where ϕ is as in Definition 1.3. The projection π gives a bijection

$$\phi_\gamma^{-1}(F) \to \{g^{-1}\gamma g \mid g \in (T_G \backslash G)(F)\}, \tag{2.3}$$

onto the set of F-points of the stable conjugacy class of γ, which is in bijection with $(T_G \setminus G)(F)$.

The action of $\sigma \in \Gamma$ on a point $(B_-^{n_W n})_W \in S(B_+)$, with $n, n_W \in N_+$, which we identify with the point $((B(W))_W, n) \in S(B_+, B_-) \times N_+$, is given by

$$\sigma : ((B(W))_W, n) \longmapsto ((B'(W))_W, n'), \tag{2.4a}$$

where $\sigma(B_-) = B_-^{c_\sigma}$, $c_\sigma \in N_G(T_+)(\overline{F})$, and

$$B'(W)^{n'} = \sigma\big(B(\sigma_T^{-1}(W))^n\big) = (\sigma B_-)^{\sigma(n_{\sigma_T^{-1}(W)} n)}, \tag{2.4b}$$

and

$$n' = c_\sigma \sigma(n_{\sigma_T^{-1}(W_0)} n), \qquad B'(W_0) = B_- \ . \tag{2.4c}$$

How does Γ act on the point of $Y_1(B_+)$ determined by $(b, (B_-^{n_W})_W)^n$, with $b \in B_-$ and $n_W, n \in N_+(\overline{F})$? Let $\alpha(W, \rho) \in \Delta$ denote the simple root associated to a pair (W, ρ) as above. To be consistent with (2.4a), define an action σ_T^* of $\sigma \in \Gamma$ on Δ by

$$(\sigma_T^*)^{-1}(\alpha(W, \rho)) = \alpha(\sigma_T^{-1}(W), \sigma_T^{-1}(\rho)). \tag{2.5}$$

The action of Γ on $S_1(B_+) \cong (S'(B_+, B_-) \times_{T_+} \mathbb{A}^\Delta) \times N_+$ is given by

$$\sigma : ((B(W))_W, (z(\alpha))_{\alpha \in \Delta}, n) \longmapsto ((B'(W))_W, (z'(\alpha))_{\alpha \in \Delta}, n'), \qquad z'(\alpha) = \sigma\big(z((\sigma_T^*)^{-1}(\alpha))\big),$$

with $B'(W)$ and n' as above. By the formulas for the Galois action on $S(B_+)$, we have

$$\sigma : (b, (B_-^{n_W})_W)^n \longmapsto \big(\sigma(b), (\sigma(B_-)^{\sigma(n_{\sigma_T^{-1}(W)})})_W\big)^{\sigma(n)}. \tag{2.6}$$

Here b is as in (1.17) and n_W is given by (1.18). A point of $S(B_+, B_-)$ is determined by its coordinates $z(W, \rho)$, so by (2.2) and (2.6), σ maps $(z(W, \rho))$ to $(z'(W, \rho))$, where

$$z'(W, \rho) = \sigma\big(z(\sigma_T^{-1}(W), \sigma_T^{-1}(\rho))\big). \tag{2.7}$$

Suppose now that we are on a patch $S(B_+, B_-)$ with $T_+ = B_+ \cap B_-$ defined over F, and let $c_\sigma \in N_G(T_+)(\overline{F})$ with $\sigma(B_-) = c_\sigma^{-1} B_- c_\sigma$. We may then write

$$\sigma : (b, (B_-^{n_W})_W)^n \longmapsto (b', (B_-^{n'_W})_W)^{n'}, \tag{2.8}$$

where, writing $m_\sigma := c_\sigma \, \sigma(n_{\sigma_T^{-1}(W_0)})$,

$$b' = m_\sigma \, \sigma(b) \, m_\sigma^{-1}, \qquad n'_W = c_\sigma \, \sigma\big(n_{\sigma_T^{-1}(W)} \, n^{-1}_{\sigma_T^{-1}(W_0)}\big) \, c_\sigma^{-1}, \qquad n' = m_\sigma \, \sigma(n).$$

Conjugation by c_σ commutes with the permutation of the Borel subgroups of a star by σ_T. If B_- is defined over F then $c_\sigma = 1$ and the W_0-component of $(B_-^{n'_W})_W$ is B_-. In particular, if the point of $Y_1(B_+)$ determined by $(b, (B_-^{n_W})_W, (z(\alpha))_{\alpha \in \Delta}, n)$ has coordinates $\{x(\rho) \mid \rho > 0\}$, then its image under σ has coordinates $\{x'(\rho) \mid \rho > 0\}$, determined by

$$m_\sigma \, \sigma(b) \, m_\sigma^{-1} = b' = n_0 t_0 \prod_{\rho \in \Sigma^+} \exp(x'(\rho) X_\rho), \tag{2.9}$$

for some $n_0 \in N_+$, where Σ^+ denotes the set of roots of T_+ which are positive with respect to B_-. Chevalley's commutator formula then gives explicit formulas for $x'(\rho)$.

Remark 2026.10. In 1994, (2.6) carried the exponent $\sigma(n_{\sigma_T^{-1}(W_0)}n)$ outside the star as well as inside it, and the formula for b' in (2.8) and (2.9) read $\sigma(n_{\sigma_T^{-1}(W_0)})c_\sigma b c_\sigma \sigma(n^{-1}_{\sigma_T^{-1}(W_0)})^{-1}$: it had b in place of $\sigma(b)$, c_σ in place of c_σ^{-1}, and a double inverse. The formulas above follow by writing $(b,(B_-^{n_W}))^n = (n^{-1}bn,(B_-^{n_W n}))$ and applying σ (level 1). Throughout, the normalization of c_σ is $\sigma(B_-) = B_-^{c_\sigma}$.

Lemma 2.10 ([Ha89, proof of Lemma 5.3])**.** (a) *Let α be a simple root of a quasi-split group G. If $(u,(B(W))_W) \in E_{\Delta-\alpha}$, then every Borel subgroup $B(W)$ lies on the same line of the Dynkin curve of u. Here $E_{\Delta-\alpha}$ is a 1-regular Richardson branch of $\phi^{-1}(1)$, as in §3.*

(b) *The action of Γ on the points of $E_{\Delta-\alpha}$ is determined by its action on the lines of the Dynkin curve.*

Now we restrict to the case $G = G_2$. Recall the Dynkin curve of a subregular unipotent element of G_2 [St74]: four projective lines, one line ℓ_α meeting three lines ℓ'_β, ℓ''_β, ℓ'''_β, as in Figure 1.

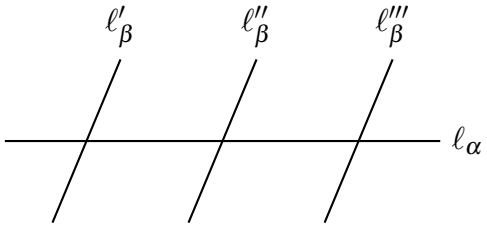


FIGURE 1. The Dynkin curve of a subregular unipotent element of G_2, drawn from the description in the text.

Remark 2026.11. The figure of the Dynkin curve is missing from the 1994 source (the source has empty displays). Figure 1 is reconstructed from the description of the Galois actions below. The assignment of the types α and β to the central and outer lines is taken from that description and has not been checked against [St74].

Possible Galois actions on this configuration include: (i) exchanging ℓ'_β with ℓ'''_β, fixing ℓ''_β, and flipping ℓ_α; (ii) flipping ℓ'_β, ℓ''_β, ℓ'''_β and fixing ℓ_α; (iii) cyclically permuting ℓ'_β, ℓ''_β, ℓ'''_β and "rotating" ℓ_α.

Label and order the positive roots of the maximal torus $T_+ \subset B_-$ of G_2 as

$$\alpha \text{ (short)},\ \beta \text{ (long)},\ \gamma = \alpha+\beta,\ \delta = 2\alpha+\beta,\ \varepsilon = 3\alpha+\beta,\ \zeta = 3\alpha+2\beta.$$

We have

$$\theta_\delta(r)\theta_\gamma(s) = \theta_\gamma(s)\theta_\delta(r)\theta_\zeta(3rs),$$

up to the sign of the structure constant, so

$$\begin{aligned}\theta_\varepsilon(x)^{\theta_{-\alpha}(z)} &= \theta_\varepsilon(x)\theta_\delta(c_1xz)\theta_\gamma(c_2xz^2)\theta_\beta(c_3xz^3)\\ &= \theta_\beta(c_3xz^3)\theta_\gamma(c_2xz^2)\theta_\delta(c_1xz)\theta_\varepsilon(x)\theta_\zeta(c_4x^2z^3),\end{aligned}$$

where the c_i are constants determined by the Chevalley basis. Likewise,

$$\theta_\delta(x)^{\theta_{-\alpha}(z)} = \theta_\beta(d_2xz^2)\theta_\gamma(d_1xz)\theta_\delta(x)\theta_\zeta(d_3x^2z),$$

$$\theta_\gamma(x)^{\theta_{-\alpha}(z)} = \theta_\gamma(x)\theta_\beta(exz) = \theta_\beta(exz)\theta_\gamma(x), \qquad \theta_\beta(x)^{\theta_{-\alpha}(z)} = \theta_\beta(x),$$

and, by a computation in $SL(2)$,

$$\theta_\alpha(x)^{\theta_{-\alpha}(z)} = \theta_{-\alpha}\Big(\frac{-xz^2}{1+xz}\Big)\,\alpha^\vee(1+xz)\,\theta_\alpha\Big(\frac{x}{1+xz}\Big).$$

In general, if $\sigma \in \Gamma$ acts on $T_G(\overline{F})$ as a simple reflection s_ρ, $\rho \in \Delta$, we write $\sigma = \sigma_\rho$. This amounts to taking $\sigma(n_{\sigma_T^{-1}(W_0)}) = \theta_{-\rho}(z)$, where $z = \sigma(z(\rho)z_1(W_0,\rho))$ [Ha89, §3]. We compute the Galois action on the coordinates of a point P in $Y''(B_+,B_-)$, assuming that $B_+ \cap B_- = T_+$ is defined over F and $c_\sigma = 1$ in (2.9). (Reflections c_σ are treated in (2.16)–(2.17).)

Take $\sigma = \sigma_\alpha$, and order the product with α first. Then

$$\prod_{\rho>0} \theta_\rho(x(\rho))^{\theta_{-\alpha}(z)} = \theta_{-\alpha}(y_\alpha)t \prod_{\rho>0} \theta_\rho(\phi_\rho^\alpha(\underline{x},z)), \tag{2.11}$$

with $t = \alpha^\vee(1+x(\alpha)z)$, $y_\alpha = -x(\alpha)z^2/(1+x(\alpha)z)$, $\underline{x} = (x(\alpha),\dots,x(\zeta))$, and, up to conjugation of the factors θ_ρ, $\rho \neq \alpha$, by t,

$$\begin{aligned}
\phi_\alpha^\alpha(\underline{x},z) &= x(\alpha)/(1+x(\alpha)z),\\
\phi_\beta^\alpha(\underline{x},z) &= x(\beta) + ezx(\gamma) + d_2z^2x(\delta) + c_3z^3x(\varepsilon),\\
\phi_\gamma^\alpha(\underline{x},z) &= x(\gamma) + d_1zx(\delta) + c_2z^2x(\varepsilon),\\
\phi_\delta^\alpha(\underline{x},z) &= x(\delta) + c_1zx(\varepsilon),\\
\phi_\varepsilon^\alpha(\underline{x},z) &= x(\varepsilon),\\
\phi_\zeta^\alpha(\underline{x},z) &= x(\zeta) + d_3zx(\delta)^2 + c_4z^3x(\varepsilon)^2 + (\text{cross terms in } x(\delta)x(\varepsilon)).
\end{aligned}$$

Therefore, if $c_{\sigma_\alpha} = 1$, we have

$$\sigma_\alpha : x(\rho) \longmapsto \sigma\big(\phi_\rho^\alpha(\underline{x},z)\big), \tag{2.12}$$

where $z = \sigma_\alpha(z(\alpha)z_1(W_0,\alpha))$.

Remark 2026.12 (formulas (2.11)–(2.14)). These formulas are schematic. The constants are not computed, and the reordering of the factors produces further cross terms that are not written out. Several misprints of 1994 are corrected:

- $\phi_\alpha^\alpha = x(\alpha)$ is replaced by $x(\alpha)/(1+x(\alpha)z)$, from the $SL(2)$ computation above, and the torus factor t, which rescales the other coordinates, is recorded;
- the analogous y_β is $-xz^2/(1+xz)$, not $-x^2z/(1+xz)$;
- in ϕ_δ^α, c_2 is replaced by c_1;
- in the commutator of $\theta_\beta(r)$ and $\theta_\alpha(s)$ the exponents were interchanged, because the root $i\alpha + j\beta$ carries s^ir^j;
- after Lemma 2.10, $\theta_\rho(z)$ is replaced by $\theta_{-\rho}(z)$.

With the normalization of Remark 2026.10, $b' = m_\sigma\sigma(b)m_\sigma^{-1} = \sigma(b)^{\theta_{-\alpha}(-z)}$, so z should be read as $-z$ in (2.11)–(2.14). The formulas were not used in a computation in 1994, and they are not used in one here.

Take $\sigma = \sigma_\beta$ and note

$$\theta_\gamma(r)\theta_\alpha(s) = \theta_\alpha(s)\theta_\gamma(r)\theta_\delta(2rs)\theta_\varepsilon(3rs^2)\theta_\zeta(3r^2s),$$

$$\theta_\beta(r)\theta_\alpha(s) = \theta_\alpha(s)\theta_\beta(r)\theta_\gamma(rs)\theta_\delta(rs^2)\theta_\varepsilon(rs^3)\theta_\zeta(r^2s^3),$$

up to the signs of the structure constants. By Chevalley's commutator formula,

$$\theta_\alpha(x)^{\theta_{-\beta}(z)} = \theta_\alpha(x), \qquad \theta_\beta(x)^{\theta_{-\beta}(z)} = \theta_{-\beta}(y_\beta)\,\beta^\vee(1+xz)\,\theta_\beta\Big(\frac{x}{1+xz}\Big), \quad y_\beta := \frac{-xz^2}{1+xz},$$

$$\theta_\gamma(x)^{\theta_{-\beta}(z)} = \theta_\gamma(x)\theta_\alpha(czx) = \theta_\alpha(czx)\theta_\gamma(x)\theta_\delta(2czx^2)\theta_\varepsilon(3c^2z^2x^3)\theta_\zeta(3czx^3),$$

$$\theta_\delta(x)^{\theta_{-\beta}(z)} = \theta_\delta(x), \qquad \theta_\varepsilon(x)^{\theta_{-\beta}(z)} = \theta_\varepsilon(x), \qquad \theta_\zeta(x)^{\theta_{-\beta}(z)} = \theta_\zeta(x)\theta_\varepsilon(dzx) = \theta_\varepsilon(dzx)\theta_\zeta(x).$$

Therefore

$$\prod_{\rho>0}\theta_\rho(x(\rho))^{\theta_{-\beta}(z)} = \theta_{-\beta}(y_\beta)\,t\prod_{\rho>0}\theta_\rho(\phi_\rho^\beta(\underline{x},z)), \tag{2.13}$$

where, schematically,

$$\begin{aligned}
\phi_\alpha^\beta(\underline{x},z) &= x(\alpha) + cx(\gamma)z + \cdots,\\
\phi_\beta^\beta(\underline{x},z) &= x(\beta)/(1+x(\beta)z),\\
\phi_\gamma^\beta(\underline{x},z) &= x(\gamma) + \cdots,\\
\phi_\delta^\beta(\underline{x},z) &= x(\delta) + 2czx(\gamma)^2 + \cdots,\\
\phi_\varepsilon^\beta(\underline{x},z) &= x(\varepsilon) + dzx(\zeta) + 3c^2x(\gamma)^3z^2 + \cdots,\\
\phi_\zeta^\beta(\underline{x},z) &= x(\zeta) + 3czx(\gamma)^3 + \cdots.
\end{aligned}$$

Therefore, if $c_{\sigma_\beta} = 1$, we have

$$\sigma_\beta : x(\rho) \longmapsto \sigma\big(\phi_\rho^\beta(\underline{x},z)\big), \tag{2.14}$$

where $z = \sigma_\beta(z(\beta)z_1(W_0,\beta))$.

Formulas (2.12) and (2.14) help to determine the set of F-rational points of Y_C. We also need the effect of conjugating b by c_σ. For this we recall a basic lemma [Ca89].

Lemma 2.15. *Let $\phi_\rho : SL(2) \to G$ be the homomorphism associated to the root ρ and let $\theta_\rho : \mathbb{G}_a \to U_\rho$ denote the root group embedding, so $\theta_\rho(x) = \phi_\rho\left(\begin{smallmatrix}1 & x\\ 0 & 1\end{smallmatrix}\right)$. Let $s_\rho := s_\rho(1)$, where*

$$s_\rho(x) = \theta_\rho(x)\theta_{-\rho}(-x^{-1})\theta_\rho(x) = \phi_\rho\begin{pmatrix}0 & x\\ -1/x & 0\end{pmatrix}.$$

Then $s_\rho\theta_\sigma(x)s_\rho^{-1} = \theta_{w_\rho(\sigma)}(\eta_{\rho,\sigma}x)$, where $\eta_{\rho,\sigma} = \pm 1$ is as in [Ca89, pp. 94–96] *and w_ρ is the reflection of the Weyl group associated to ρ.*

As a consequence,

$$s_\alpha \cdot \prod_{\rho>0}\theta_\rho(x(\rho))\cdot s_\alpha^{-1} = \theta_{-\alpha}(x'(\alpha))\prod_{\rho>0,\rho\neq\alpha}\theta_\rho(x'(\rho)), \tag{2.16}$$

where $x'(\alpha) = \eta_{\alpha,\alpha}x(\alpha)$, $x'(\beta) = \eta_{\alpha,\varepsilon}x(\varepsilon)$, $x'(\gamma) = \eta_{\alpha,\delta}x(\delta)$, $x'(\delta) = \eta_{\alpha,\gamma}x(\gamma)$, $x'(\varepsilon) = \eta_{\alpha,\beta}x(\beta)$, and $x'(\zeta) = \eta_{\alpha,\zeta}x(\zeta) + f_1x(\delta)x(\gamma) + f_2x(\beta)x(\varepsilon)$, where f_1, f_2 are constants. Likewise,

$$s_\beta \cdot \prod_{\rho>0}\theta_\rho(x(\rho))\cdot s_\beta^{-1} = \theta_{-\beta}(x'(\beta))\prod_{\rho>0,\rho\neq\beta}\theta_\rho(x'(\rho)), \tag{2.17}$$

where $x'(\alpha) = \eta_{\beta,\gamma}x(\gamma) + f_3x(\alpha)x(\beta)$, $x'(\beta) = \eta_{\beta,\beta}x(\beta)$, $x'(\gamma) = \eta_{\beta,\alpha}x(\alpha)$, $x'(\delta) = \eta_{\beta,\delta}x(\delta) + f_4x'(\alpha)x'(\gamma)$, $x'(\varepsilon) = \eta_{\beta,\zeta}x(\zeta)$, and $x'(\zeta) = \eta_{\beta,\varepsilon}x(\varepsilon)$. These follow from s_α: $\beta \mapsto \varepsilon$, $\gamma \mapsto \delta$, $\delta \mapsto \gamma$, $\varepsilon \mapsto \beta$, $\zeta \mapsto \zeta$, and s_β: $\alpha \mapsto \gamma$, $\gamma \mapsto \alpha$, $\delta \mapsto \delta$, $\varepsilon \mapsto \zeta$, $\zeta \mapsto \varepsilon$ (checked, level 1). The reordering

needed to move $\theta_{-\alpha}$ (respectively $\theta_{-\beta}$) to the front produces further terms, which are not written out. In 1994 the signs in (2.17) were written $\eta_{\alpha,\cdot}$.

§ 3 CONSTRUCTION OF REGULAR PATCHES

This section describes the local structure of Y''. Locally it is the product of an affine space with a monomial variety Z, an affine toric variety. We introduce the fundamental and essential patches of Z, prove that for G_2 they are regular and cover Z up to codimension two, and prove that Z is normal and Cohen–Macaulay for G_2.

Let G denote a connected semisimple group over $\overline{F}$. Let $\chi : G \to T_G/\Omega$ denote the adjoint quotient [Sl80]. There is an open patch

$$Y''(B_+,B_-)\times N_+ \subset \chi^{-1}(C)\times S''(B_+,B_-)\times N_+ \subset G\times S''(B_+,B_-)\times N_+$$

of Y'' which is isomorphic to an open subset of the product $\mathbb{A}^p \times Z$, where $Z \subset \mathbb{A}^{2p+1}$ is the "monomial" affine variety of dimension $p+1$ with coordinates

$$w(\rho)\ (\rho\in\Sigma_G^+-\Delta),\qquad z(\alpha)\ (\alpha\in\Delta),\qquad x(\rho)\ (\rho\in\Sigma_G^+),\qquad \lambda,$$

defined as the Zariski closure of the set of points with $\lambda \neq 0$ satisfying, for all $\rho \in \Sigma_G^+$,

$$\lambda w(\rho) = x(\rho)\prod_{\alpha\in\Delta} z(\alpha)^{m(\alpha)}, \tag{3.1}$$

where we set $w(\alpha)=1$ for $\alpha\in\Delta$, and $m(\alpha)=m_\rho(\alpha)\geq 0$ is the coefficient of α in $\rho = \sum_{\alpha\in\Delta} m(\alpha)\alpha$. Here Δ denotes the simple roots of T_+ in B_-. In particular $\lambda = x(\alpha)z(\alpha)$ for α simple. On $\lambda\neq 0$ the coordinates λ, $z(\alpha)$ ($\alpha\in\Delta$), and $w(\rho)$ ($\rho\notin\Delta$) are free, the $z(\alpha)$ are non-zero, and the $x(\rho)$ are determined by (3.1). So Z is the closure of the orbit of a torus of dimension $p+1$: it is the affine toric variety $\operatorname{Spec} k[S]$, where S is the semigroup of exponent vectors of the monomials λ, $z(\alpha)$, $w(\rho)$, $x(\rho)$ in the torus coordinates $(\lambda, z(\alpha), w(\rho))$. The projection of the patch onto Z lets us pull back open subvarieties $Z_U \subset Z$ to open subvarieties $Y''_U \subset Y''$. The variety Z is regular on $\lambda\neq 0$.

Example. For G_2, $p=6$, $Z\subset\mathbb{A}^{13}$ has dimension 7, and the torus coordinates are $\lambda, z(\alpha), z(\beta), w(\gamma), w(\delta), w(\varepsilon), w(\zeta)$. For instance $x(\zeta) = \lambda w(\zeta) z(\alpha)^{-3} z(\beta)^{-2}$, so the exponent vector of $x(\zeta)$ is $(1,-3,-2,0,0,0,1)$.

We first claim that, for $G=G_2$, the singularities of Z lie in

$$K:\quad x(\alpha)=x(\beta)=x(\gamma)=z(\alpha)=w(\zeta)=0$$

or in

$$K_0:\quad x(\alpha)=x(\beta)=x(\gamma)=x(\delta)=x(\varepsilon)=x(\zeta)=w(\zeta)=0.$$

The points of $Y''(B_+,B_-)$ with coordinates in K_0 have $\lambda=0$ and map to the identity under π, hence correspond to the trivial class. To prove the claim, we exhibit four regular patches, give generators of the local coordinate ring, and express the remaining coordinates as regular functions of the generators on the patch.

$x(\alpha)\neq 0$. Generators: $x(\alpha)$, $x(\beta)$, $z(\beta)$, $x(\gamma)$, $x(\delta)$, $x(\varepsilon)$, $x(\zeta)$. Remaining coordinates: $\lambda = x(\beta)z(\beta)$, $z(\alpha) = x(\beta)z(\beta)/x(\alpha)$, $w(\gamma) = x(\gamma)z(\beta)/x(\alpha)$, $w(\delta) = x(\delta)z(\alpha)z(\beta)/x(\alpha)$, $w(\varepsilon) = x(\varepsilon)z(\alpha)^2 z(\beta)/x(\alpha)$, $w(\zeta) = x(\zeta)z(\alpha)^2 z(\beta)^2/x(\alpha)$.

$x(\beta)\neq 0$. Generators: $x(\alpha)$, $x(\beta)$, $z(\alpha)$, $x(\gamma)$, $x(\delta)$, $x(\varepsilon)$, $x(\zeta)$. Remaining coordinates: $\lambda = x(\alpha)z(\alpha)$, $z(\beta) = x(\alpha)z(\alpha)/x(\beta)$, $w(\gamma) = x(\gamma)z(\alpha)/x(\beta)$, $w(\delta) = x(\delta)z(\alpha)^2/x(\beta)$, $w(\varepsilon) = x(\varepsilon)z(\alpha)^3/x(\beta)$, $w(\zeta) = x(\zeta)x(\alpha)z(\alpha)^4/x(\beta)^2$.

$x(\gamma)\neq 0$. Generators: $x(\alpha)$, $x(\beta)$, $x(\gamma)$, $w(\gamma)$, $x(\delta)$, $x(\varepsilon)$, $x(\zeta)$. Remaining coordinates: $\lambda = x(\alpha)x(\beta)w(\gamma)/x(\gamma)$, $z(\alpha) = x(\beta)w(\gamma)/x(\gamma)$, $z(\beta) = x(\alpha)w(\gamma)/x(\gamma)$, $w(\delta) = x(\delta)w(\gamma)^2x(\beta)/x(\gamma)^2$, $w(\varepsilon) = x(\varepsilon)w(\gamma)^3x(\beta)^2/x(\gamma)^3$, $w(\zeta) = x(\zeta)x(\alpha)x(\beta)^2w(\gamma)^4/x(\gamma)^4$.

$w(\zeta)\neq 0$. Generators: $z(\alpha)$, $z(\beta)$, $w(\gamma)$, $w(\delta)$, $w(\varepsilon)$, $w(\zeta)$, $x(\zeta)$. Remaining coordinates: $\lambda = x(\zeta)z(\alpha)^3z(\beta)^2/w(\zeta)$, $x(\alpha) = x(\zeta)z(\alpha)^2z(\beta)^2/w(\zeta)$, $x(\beta) = x(\zeta)z(\alpha)^3z(\beta)/w(\zeta)$, $x(\gamma) = w(\gamma)x(\zeta)z(\alpha)^2z(\beta)/w(\zeta)$, $x(\delta) = w(\delta)x(\zeta)z(\alpha)z(\beta)/w(\zeta)$, $x(\varepsilon) = w(\varepsilon)x(\zeta)z(\beta)/w(\zeta)$.

On Z the monomial identities $x(\delta)z(\alpha)^2 = x(\beta)w(\delta)$, $x(\varepsilon)z(\alpha)^3 = x(\beta)w(\varepsilon)$ and $x(\zeta)z(\alpha)^3z(\beta) = x(\beta)w(\zeta)$ hold, because both sides are the same character of the torus. Hence, if $x(\alpha)=x(\beta)=x(\gamma)=w(\zeta)=0$ and $z(\alpha)\neq 0$, then also $x(\delta)=x(\varepsilon)=x(\zeta)=0$. So the complement of the four patches is contained in $K\cup K_0$, and the claim follows.

Remark 2026.13. Three of the patch formulas were misprinted in 1994. The correct values are $w(\zeta) = x(\zeta)x(\alpha)z(\alpha)^4/x(\beta)^2$ on $x(\beta)\neq 0$ and $w(\zeta) = x(\zeta)x(\alpha)x(\beta)^2w(\gamma)^4/x(\gamma)^4$ on $x(\gamma)\neq 0$; the 1994 versions were $x(\zeta)z(\alpha)^5x(\alpha)^2/x(\beta)^3$ and $x(\zeta)x(\alpha)x(\beta)^2w(\gamma)^4/x(\gamma)$. The 1994 text also gave a fifth patch, $z(\alpha)\neq 0$ with $x(\varepsilon)\neq 0$ (or $x(\delta)\neq 0$, or $x(\zeta)\neq 0$). By the identities above it lies inside $\{x(\beta)\neq 0\}$, so it is omitted; its formula for $w(\zeta)$ had a spurious factor $z(\alpha)^{-3}$. All formulas were checked symbolically (script `g2_patch_jacobians.py`).

The claim is weaker than it looks: K contains the divisor $E_{\emptyset,\alpha,\delta}$ of §4, along which Z is generically regular. The exact singular locus is described in Remark 2026.23. Since Z is a normal toric variety (Proposition 3.7), it has a resolution of singularities $\tilde{Z}\to Z$ given by a regular subdivision of its fan, which is an isomorphism over the regular locus. The divisor $\lambda=0$ in $\tilde{Z}$ has normal crossings, but its branches occur with multiplicities $a(E)$ that may exceed 1 (Table 4.3A). Let $\tilde{Y''}\to Y''$ be the corresponding resolution of Y''. The exceptional divisors of $\tilde{Y''}\to Y''$ have not been analyzed (Remark 2026.23).

Fundamental and essential patches. Let G again be an arbitrary connected semisimple group. Let $\Delta=\{\alpha_1,\dots,\alpha_\ell\}$, which we sometimes identify with $\{1,\dots,\ell\}$, $\ell := \operatorname{rk} G$. Let $S_1,S_2\subset\{1,\dots,\ell\}$ be disjoint sets. If they are non-empty, let $i_{min} := \min_{i\in S_1} i$, $j_{min} := \min_{j\in S_2} j$, and similarly i_{max}, j_{max}. If S_1,S_2 are non-empty, we assume (i) $i_{max}<j_{min}$ or $j_{max}<i_{min}$, and (ii) if $S_2\neq\emptyset$ then $S_1\cup S_2\neq\Delta$.

For each map $\sigma: S_2\to\Sigma_G^+-\Delta$, define the subset $Z_{S_1,S_2,\sigma}\subset Z$ by the conditions

(3.2a) $z(\alpha)\neq 0$ for all $\alpha\notin S_1\cup S_2$,

(3.2b) $x(\alpha)\neq 0$ for all $\alpha\in S_1$,

(3.2c) $x(\sigma(j))\neq 0$ for all $j\in S_2$,

(3.2d) $w(\sigma(j))\neq 0$ for all $j\in S_2$.

These are the **fundamental patches** of Z in the sense of [Ha92a]. The collection depends, a priori, on the identification of Δ with $\{1,\dots,\ell\}$. For $G=G_2$ the divisor $\lambda=0$ meets each $Z_{S_1,S_2,\sigma}$ in a single irreducible branch, denoted $E_{S_1,S_2,\sigma}$, occurring with some multiplicity $a(E_{S_1,S_2,\sigma})$ (Table 4.3A). The $E_{S_1,S_2,\sigma}$ are the **fundamental branches** of $\lambda=0$ on Z. If $S_2=\{\rho\}$ is a singleton and $\tau=\sigma(\rho)$, we write $E_{S_1,\rho,\tau}$ for $E_{S_1,S_2,\sigma}$. If $S_2=\emptyset$ we write Z_{S_1} and call it a **Richardson patch**, and E_{S_1} a **Richardson branch**, since by [Ha92a, Ch. II, Proposition 9.1] $\pi(E_{S_1})$ is the closure of the Richardson class of the parabolic subgroup $P_{\Delta-S_1}$ whose Levi subgroup has simple roots $\Delta-S_1$. In

particular the **regular branch** is E_Δ ($S_1 = \Delta$), with $\pi(E_\Delta) = \overline{\mathcal{O}}_{reg} = V$, the unipotent variety, and $E_\emptyset$ is the branch of the trivial class. Since E_Δ is regular, $\pi : E_\Delta \to V$ is a resolution of singularities similar to the Springer resolution [Sl80], [St74].

Remark 2026.14. The 1994 text called $E_\emptyset$ the regular branch. With the convention of this paper ("Hales' E_S is my $E_{\Delta-S}$") the regular branch is E_Δ; $E_\emptyset$ maps to the identity (Table 4.3A).

Fix a map $\sigma : S_2 \to \Sigma_G^+ - \Delta$ satisfying the following.

(3.3a) For $j, j' \in S_2$, the simple root expansions of $\sigma(j)$ and $\sigma(j')$ contain no common simple roots unless $j = j'$ or j, j' are consecutive.

(3.3b) If j, j' are consecutive in S_2 then $\sigma(j) = \sum_{k=j}^{j'} m_{jk}\alpha_k$ for some integers $m_{jk} \geq 0$.

(3.3c) There is an $a \geq 1$ for which $\sigma(j_{max}) = a\alpha_{j_{max}} + \sum_{\alpha \notin S_1 \cup S_2} m(\alpha)\alpha$, for some integers $m(\alpha) \geq 0$.

Such a map, and the corresponding fundamental patch, are called **essential**. Patches which are not essential are expected to be redundant in a covering of Z up to codimension 2, but in general there are also redundant essential patches, so the definition is provisional. These patches generalize the sequences $\gamma(i, j), \ldots$ of roots used in [Ha89].

Notation 3.3d. The notation for these divisors differs from that of [Ha92a], [Ha89], though the notation for the patches seems to agree: Hales' E_S is our $E_{\Delta-S}$.

Conjecture 3.4. *If G is not of type D_n, E_n, F_4, then Z is covered by the essential patches $Z_{S_1,S_2,\sigma}$ except for a subset of codimension ≥ 2, and the $Z_{S_1,S_2,\sigma}$ are all regular.*

Remark 2026.15. The 1994 manuscript stated that Conjecture 3.4 "is known" for A_n and C_n [Ha92a], [Ha89], and false for D_n. These claims have not been checked; [Ha89] treats only $GSp(4)$. The 1994 statement also included "Z is regular in codimension one"; that holds for every G for which Z is normal (Proposition 3.7 and Conjecture 3.7A).

Proposition 3.4A. *Conjecture 3.4 holds for G_2. More precisely, the twelve patches listed in §4 are regular, with the generators given there, and every torus-invariant prime divisor of Z meets one of them.*

Proof. Regularity: on each patch, every coordinate of Z is a monomial in the listed generators, with negative exponents only on generators that are inverted on the patch; this was checked symbolically for all twelve patches (script `g2_patch_regularity.py`). Covering: the complement of a union of torus-invariant open sets is torus-invariant and closed, so it has codimension at least 2 unless it contains a torus-invariant prime divisor. For a facet of the cone of S with primitive inner normal u, the patch $\{m \neq 0\}$ meets the divisor D_u exactly when $\langle m, u\rangle = 0$. The eleven facets of the cone were computed, and each is orthogonal to the monomial of at least one patch (script `g2_toric_divisors.py`). □

Lemma 3.5 (Hales [Ha92a], corrected)**.** *For $S_1 \subset \Delta$ non-empty, Z_{S_1} is regular, with generators*

$$\lambda, \quad z(\alpha)\ (\alpha \in \Delta - S_1), \quad x(\rho)\ (\rho \in \Sigma_G^+ - \Sigma_{\Delta-S_1}^+), \quad w(\rho)\ (\rho \in \Sigma_{\Delta-S_1}^+ - \Delta),$$

where $\Sigma_{\Delta-S_1}^+$ denotes the positive roots of the root subsystem spanned by $\Delta - S_1$. (The $x(\rho)$ include $x(\alpha)$ for $\alpha \in S_1$.) It is $(p+1)$-dimensional, and the divisor $\lambda = 0$ occurs on it with multiplicity one.

Proof. On Z_{S_1} we have $z(\alpha) = \lambda/x(\alpha)$ for $\alpha \in S_1$. For a positive root ρ write $m_{S_1}(\rho) = \sum_{\alpha\in S_1} m_\rho(\alpha)$. By (3.1),

$$w(\rho) = x(\rho)\,\lambda^{m_{S_1}(\rho)-1} \prod_{\alpha\notin S_1} z(\alpha)^{m_\rho(\alpha)} \prod_{\alpha\in S_1} x(\alpha)^{-m_\rho(\alpha)}.$$

If $\rho \notin \Sigma^+_{\Delta-S_1}$ then $m_{S_1}(\rho) \geq 1$, and $w(\rho)$ is regular in $x(\rho)$ and the generators. If $\rho \in \Sigma^+_{\Delta-S_1}$ then $m_{S_1}(\rho) = 0$ and $x(\rho) = \lambda w(\rho)\prod_{\alpha\notin S_1} z(\alpha)^{-m_\rho(\alpha)}$ is regular in $w(\rho)$ and the generators. So all coordinates are regular functions of the listed ones, whose number is $1 + (\ell - |S_1|) + (p - |\Sigma^+_{\Delta-S_1}|) + (|\Sigma^+_{\Delta-S_1}| - (\ell - |S_1|)) = p + 1 = \dim Z$. Hence Z_{S_1} is an open subset of affine space, with λ one of the coordinates. □

Remark 2026.16. As printed in 1994, the lemma had $\Sigma^+_{S_1}$ in place of $\Sigma^+_{\Delta-S_1}$, so the roles of $x(\rho)$ and $w(\rho)$ were interchanged. For $S_1 = \Delta$ that version gives the generators λ, $x(\alpha)$, $x(\beta)$, $w(\gamma), \ldots$, for which $x(\gamma) = w(\gamma)x(\alpha)x(\beta)/\lambda$ is not regular. The generator count also failed unless $S_1 = \Delta$. The proof above is new.

For brevity, let m_α denote the multiplicity with which $z(\alpha) = 0$ occurs in $\lambda = 0$.

Lemma 3.6. *Let $\alpha \in \Delta$ and $\sigma \in \Sigma^+_G - \Delta$, and suppose that $Z_{\emptyset,\alpha,\sigma}$ is an essential patch and that every positive root is comparable with σ in the partial order $\rho \leq \sigma$ iff $\sigma - \rho$ is a non-negative combination of simple roots. Then $Z_{\emptyset,\alpha,\sigma}$ is regular and its coordinate ring has generators*

$$z(\beta)\ (\beta \in \Delta), \qquad x(\rho)\ (\rho \geq \sigma), \qquad w(\rho)\ (\rho \leq \sigma,\ \rho \notin \Delta).$$

The divisor $\lambda = 0$ occurs as $z(\alpha) = 0$ with multiplicity m, where $m \geq 1$ is the coefficient of α in σ. The hypothesis holds for G_2.

Proof. Write $\sigma = m\alpha + \sum_{\beta\neq\alpha} m_\beta \beta$. On $Z_{\emptyset,\alpha,\sigma}$ we have $z(\beta) \neq 0$ for $\beta \neq \alpha$ simple, $x(\sigma) \neq 0$, and $w(\sigma) \neq 0$. By (3.3c) and (3.1), $\lambda = x(\sigma)w(\sigma)^{-1}z(\alpha)^m \prod_{\beta\neq\alpha} z(\beta)^{m_\beta}$, so $\lambda = 0$ is the divisor $z(\alpha) = 0$ with multiplicity m. If $\rho \leq \sigma$, say $\sigma = \rho + \sum_\beta n(\beta)\beta$ with $n(\beta) \geq 0$, then

$$x(\rho) = \frac{x(\sigma)w(\rho)}{w(\sigma)} \prod_\beta z(\beta)^{n(\beta)},$$

which is regular in the $w(\rho)$ with $\rho \leq \sigma$. If $\rho \geq \sigma$ then

$$w(\rho) = \frac{w(\sigma)x(\rho)}{x(\sigma)} \prod_\beta z(\beta)^{m_\rho(\beta) - m_\sigma(\beta)},$$

with non-negative exponents, which is regular in the $x(\rho)$ with $\rho \geq \sigma$. For simple β, $x(\beta) = \lambda/z(\beta)$ is regular because $m \geq 1$. The number of generators is $\ell + |\{\rho \geq \sigma\}| + |\{\rho < \sigma\}| - \ell = p + 1$ when every root is comparable with σ. For G_2 the hypothesis is checked directly: $\delta - \gamma$, $\varepsilon - \gamma$, $\zeta - \gamma$, $\varepsilon - \delta$, $\zeta - \delta$, $\zeta - \varepsilon$ are all non-negative combinations of α and β, and the simple roots are below every non-simple root. □

Remark 2026.17. In 1994 the statement read "$x(\rho)$ if $\rho > \sigma$", while the proof used $\rho \geq \sigma$; with $\rho > \sigma$ the generator count is one short. The comparability hypothesis is new; without it the count fails in general, since roots incomparable with σ are not covered by either list.

Proposition 3.7. *For $G = G_2$, the variety Z is normal and Cohen–Macaulay.*

Proof. $Z = \operatorname{Spec} k[S]$ is an affine toric variety. It is normal if and only if S is saturated in the lattice it generates [Ke72, Ch. I, Theorem 1]. For G_2 that lattice is $\mathbb{Z}^7$, since the torus coordinates themselves lie in S. Every generator of S has degree 0 or 1 in λ. By Carathéodory's theorem every lattice point of the cone of S lies in a simplicial subcone spanned by seven linearly independent generators, and is the sum of a lattice point of the half-open fundamental parallelepiped of that subcone and a non-negative integral combination of its generators. The parallelepiped points have λ-degree at most 6, and, since $z(\alpha)$ and $z(\beta)$ are the only generators with a positive z-coordinate, their z-coordinates are at most 0; this confines them to a finite set. So S is saturated if every lattice point of the cone in that finite set lies in S. This was checked exhaustively in exact integer arithmetic (script `g2_saturation_full.py`): 658,651 lattice points of λ-degree 1 to 6, all in S. Normal semigroup rings are Cohen–Macaulay by Hochster's theorem [Ho72] (see also [Ke72, Ch. I, Theorem 14]). □

Conjecture 3.7A. *For every connected semisimple group G, the variety Z is normal (hence Cohen–Macaulay, and regular in codimension one).*

Remark 2026.18 (Lemma 3.7 of 1994). The 1994 manuscript stated Proposition 3.7 for every connected semisimple group, and added that Z is a set-theoretic complete intersection. Its proof asserted that the semigroup is saturated without checking it; that part is now Conjecture 3.7A. The complete-intersection claim was justified by counting the p relations (3.1). Those relations do not cut out Z: for G_2 they also vanish on the 10-dimensional set $\lambda = z(\alpha) = z(\beta) = 0$. The claim is withdrawn.

§ 4 Regularity of patches for $G = G_2$

We now write all the patches of §3 down for G_2 and compute the Igusa constants of the branches of $\lambda = 0$. Recall $\Sigma_G^+ = \{\alpha, \beta, \gamma := \alpha+\beta, \delta := 2\alpha+\beta, \varepsilon := 3\alpha+\beta, \zeta := 3\alpha+2\beta\}$, with α short and β long, so $p = 6$ and $\dim Z = 7$. We have variables

$$\begin{aligned} &x(\alpha), x(\beta), x(\gamma), x(\delta), x(\varepsilon), x(\zeta), \\ &z(\alpha), z(\beta), w(\gamma), w(\delta), w(\varepsilon), w(\zeta), \\ &\lambda, \end{aligned} \tag{4.1}$$

subject to the relations (3.1).

With B_+ chosen as in §3, we define $\omega_{Y''(B_+)}$ to be the form

$$\omega_{Y''(B_+)} = d\lambda \wedge dx(\alpha) \wedge dx(\beta) \wedge dx(\gamma) \wedge dx(\delta) \wedge dx(\varepsilon) \wedge dx(\zeta) \wedge dv,$$

where dv is an invariant form on N_+. We use it to compute the Igusa constants β as in [Ha89].

By Lemma 3.5, the Richardson patches Z_{S_1} are regular. Since $S_1 \cup S_2 \neq \Delta$ if $S_2 \neq \emptyset$, the remaining patches are those with $S_1 = \emptyset$ and either $S_2 = \{\alpha\}$, $\sigma(\alpha) \in \{\gamma, \delta, \varepsilon, \zeta\}$, or $S_2 = \{\beta\}$, $\sigma(\beta) \in \{\gamma, \delta, \varepsilon, \zeta\}$. (The 1994 text also allowed $\sigma(\alpha) = \alpha$ and $\sigma(\beta) = \beta$; with $w(\alpha) = w(\beta) = 1$ these give the Richardson patches $Z_{\{\alpha\}}$ and $Z_{\{\beta\}}$ again, intersected with $z(\beta) \neq 0$, respectively $z(\alpha) \neq 0$.) By Lemma 3.6 the generators of the local coordinate ring are as follows; each patch is regular, and the branch of $\lambda = 0$ on it is the divisor $z(\alpha) = 0$, respectively $z(\beta) = 0$, with the multiplicity a shown.

patch	generators	a
$Z_{\emptyset,\alpha,\gamma}$, $Z_{\emptyset,\beta,\gamma}$	$z(\alpha), z(\beta), x(\gamma), w(\gamma), x(\delta), x(\varepsilon), x(\zeta)$	1, 1
$Z_{\emptyset,\alpha,\delta}$, $Z_{\emptyset,\beta,\delta}$	$z(\alpha), z(\beta), w(\gamma), x(\delta), w(\delta), x(\varepsilon), x(\zeta)$	2, 1
$Z_{\emptyset,\alpha,\varepsilon}$, $Z_{\emptyset,\beta,\varepsilon}$	$z(\alpha), z(\beta), w(\gamma), w(\delta), w(\varepsilon), x(\varepsilon), x(\zeta)$	3, 1
$Z_{\emptyset,\alpha,\zeta}$, $Z_{\emptyset,\beta,\zeta}$	$z(\alpha), z(\beta), w(\gamma), w(\delta), w(\varepsilon), w(\zeta), x(\zeta)$	3, 2

The two patches on each line have the same generators but are different open sets: $Z_{\emptyset,\alpha,\sigma}$ has $z(\beta) \neq 0$ and $Z_{\emptyset,\beta,\sigma}$ has $z(\alpha) \neq 0$, by (3.2a). Together with $Z_{\{\alpha,\beta\}}$, $Z_{\{\alpha\}}$, $Z_{\{\beta\}}$ and $Z_{\emptyset}$ these are the twelve patches of Proposition 3.4A.

There is redundancy among the branches. Comparing the valuations they define (Table 4.3A) one finds

$$E_{\emptyset,\beta,\gamma} = E_{\emptyset,\beta,\delta} = E_{\emptyset,\beta,\varepsilon} = E_{\{\beta\}}, \qquad E_{\emptyset,\alpha,\gamma} = E_{\{\alpha\}}, \qquad E_{\emptyset,\alpha,\varepsilon} = E_{\emptyset,\alpha,\zeta}.$$

So $\lambda = 0$ has exactly seven irreducible branches on Z.

Remark 2026.19. The 1994 text noted only the first set of identities. The multiplicity of $E_{\emptyset,\alpha,\zeta}$ was correctly given as 3, the coefficient of α in ζ.

We denote the branch of $\{\lambda = 0\} \subset Z$ occurring in $Z_{S_1,S_2,\sigma}$ by $E_{S_1,S_2,\sigma}$. The same notation will be used for the corresponding branch in Y_U'' and for its Zariski closure in Y_1. To determine which of the κ-stable germs, if any, the divisor $E_{S_1,S_2,\sigma}$ contributes to by Formula 0.6, we must compute the Igusa constant

$$\beta(\tilde{E}_{S_1,S_2,\sigma}) = \frac{b(\tilde{E}_{S_1,S_2,\sigma})}{a(\tilde{E}_{S_1,S_2,\sigma})}, \tag{4.2}$$

where $\tilde{E}$ denotes the strict transform of E in the desingularized variety Y_C.

We record two observations regarding the Igusa constants $\beta(E)$.

(I) If $\phi : V \to W$ is an isomorphism of smooth varieties which sends a divisor $E \subset V$ to a divisor $F = \phi(E) \subset W$, then $\beta(E) = \beta(F)$.

(II) The numbers $a(E) = \mathrm{ord}_E(\lambda)$ and $b(E) - 1 = \mathrm{ord}_E(\omega)$ depend only on the discrete valuation ord_E of the function field, and ω is a rational top-degree form. A birational morphism $\xi : V' \to V$ that is an isomorphism over the generic point of E does not change this valuation. Hence $\beta(\tilde{E}) = \beta(E)$ for the strict transform $\tilde{E}$ of any branch E that meets the regular locus of V.

Remark 2026.20. The 1994 version of (II) used the blow-up formula $\xi'^*(E) = \tilde{E} + \delta E_{V'}$ and concluded that $\beta(\tilde{E}) = \beta(E)$ "provided E does not contain any of the images of the centers of the blow ups". That proviso is unnecessary, by the valuation argument above. Since Z is normal (Proposition 3.7), every branch of $\lambda = 0$ meets the regular locus of Z.

The only divisors in $\{\lambda = 0\} \subset \tilde{Z}$ other than the strict transforms $\tilde{E}$ above are exceptional divisors. To compute all the β-values of the branches of $\phi^{-1}(1)$ in Y_C one must therefore compute the β-values of the divisors $E_{S_1,S_2,\sigma}$ on Z and of the exceptional divisors of the resolution. The 1994 manuscript did not compute the latter, on the grounds of Lemma 0.8 and the claim that all the needed F-rational $\mathscr{O}$-divisors are among the $E_{S_1,S_2,\sigma}$. That claim is the proviso of Lemma 0.8; see Remark 2026.23.

If $S_2 = \emptyset$ then $\beta(E_{S_1})$ has been computed in [Ha92a, Ch. II, Proposition 9.1], so some of our calculations are covered by Hales' general calculations. For completeness, we list everything.

Igusa forms on the fundamental patches of Y''. We pull back the patch $Z_{S_1,S_2,\sigma}$ to $Y''(B_+,B_-)$ via the open immersion $Y''(B_+,B_-)\to\mathbb{A}^p\times Z$, and call this inverse image $Y''_{S_1,S_2,\sigma}$. The Igusa form ω_U on $U=Y''_{S_1,S_2,\sigma}$ is related to the Igusa form $\omega_{Z,U}$ on $Z_{S_1,S_2,\sigma}$ by $\omega_U=\omega_{Z,U}\wedge dv$, where dv is regular and non-vanishing. We compute

$$\omega_{Z,U}=d\lambda\wedge dx(\alpha)\wedge dx(\beta)\wedge dx(\gamma)\wedge dx(\delta)\wedge dx(\varepsilon)\wedge dx(\zeta)$$

in terms of the generators of each patch. Each Jacobian below was computed symbolically (script `g2_patch_jacobians.py`).

$Z_{\{\alpha,\beta\}}$: generators $\lambda, x(\alpha),\dots,x(\zeta)$, so $\omega_{Z,U}$ is the coordinate form, $a=b=1$, and $\beta(E_{\{\alpha,\beta\}})=1$.

$Z_{\{\alpha\}}$: generators λ, $x(\alpha)$, $z(\beta)$, $x(\gamma),\dots,x(\zeta)$; here $x(\beta)=\lambda/z(\beta)$ and

$$\omega_{Z,U}=-\frac{\lambda}{z(\beta)^2}\,d\lambda\wedge dx(\alpha)\wedge dz(\beta)\wedge dx(\gamma)\wedge dx(\delta)\wedge dx(\varepsilon)\wedge dx(\zeta).$$

So $a=1$, $b=2$, $\beta(E_{\{\alpha\}})=2$.

$Z_{\{\beta\}}$: symmetrically, $\omega_{Z,U}=-\lambda z(\alpha)^{-2}d\lambda\wedge dz(\alpha)\wedge dx(\beta)\wedge dx(\gamma)\wedge\cdots\wedge dx(\zeta)$, and $\beta(E_{\{\beta\}})=2$.

$Z_\emptyset$: generators λ, $z(\alpha)$, $z(\beta)$, $w(\gamma),\dots,w(\zeta)$ (Lemma 3.5), and

$$\omega_{Z,U}=\frac{\lambda^6}{z(\alpha)^{11}z(\beta)^7}\,d\lambda\wedge dz(\alpha)\wedge dz(\beta)\wedge dw(\gamma)\wedge dw(\delta)\wedge dw(\varepsilon)\wedge dw(\zeta).$$

So $a=1$, $b=7$, $\beta(E_\emptyset)=7$.

$Z_{\emptyset,\alpha,\gamma}$:

$$\omega_{Z,U}=\frac{x(\gamma)^3z(\alpha)z(\beta)}{w(\gamma)^4}\,dz(\alpha)\wedge dz(\beta)\wedge dw(\gamma)\wedge dx(\gamma)\wedge dx(\delta)\wedge dx(\varepsilon)\wedge dx(\zeta).$$

So $a=1$, $b=2$, $\beta=2$.

$Z_{\emptyset,\alpha,\delta}$:

$$\omega_{Z,U}=\frac{x(\delta)^4z(\alpha)^5z(\beta)}{w(\delta)^5}\,dz(\alpha)\wedge dz(\beta)\wedge dw(\gamma)\wedge dw(\delta)\wedge dx(\delta)\wedge dx(\varepsilon)\wedge dx(\zeta).$$

So $a=2$, $b=6$, $\beta=3$.

$Z_{\emptyset,\alpha,\varepsilon}$:

$$\omega_{Z,U}=-\frac{x(\varepsilon)^5z(\alpha)^{10}z(\beta)}{w(\varepsilon)^6}\,dz(\alpha)\wedge dz(\beta)\wedge dw(\gamma)\wedge dw(\delta)\wedge dw(\varepsilon)\wedge dx(\varepsilon)\wedge dx(\zeta).$$

So $a=3$, $b=11$, $\beta=11/3$.

$Z_{\emptyset,\alpha,\zeta}$ and $Z_{\emptyset,\beta,\zeta}$:

$$\omega_{Z,U}=\frac{x(\zeta)^6z(\alpha)^{10}z(\beta)^7}{w(\zeta)^7}\,dz(\alpha)\wedge dz(\beta)\wedge dw(\gamma)\wedge dw(\delta)\wedge dw(\varepsilon)\wedge dw(\zeta)\wedge dx(\zeta). \tag{4.2A}$$

On $Z_{\emptyset,\alpha,\zeta}$ the branch is $z(\alpha)=0$, with $a=3$, $b=11$, $\beta=11/3$. On $Z_{\emptyset,\beta,\zeta}$ it is $z(\beta)=0$, with $a=2$, $b=8$, $\beta=4$.

On $Z_{\emptyset,\beta,\sigma}$ with $\sigma = \gamma, \delta, \varepsilon$ the branch $z(\beta) = 0$ has $a = 1$ and $b = 2$, in agreement with $E_{\emptyset,\beta,\sigma} = E_{\{\beta\}}$.

Remark 2026.21. In the 1994 text the forms on $Z_{\emptyset,\alpha,\varepsilon}$, $Z_{\emptyset,\alpha,\zeta}$ and $Z_{\emptyset,\beta,\zeta}$ had the denominator $w(\delta)$ in place of $w(\varepsilon)$, respectively $w(\zeta)$, and the form on $Z_{\emptyset,\alpha,\varepsilon}$ had $z(\alpha)^9$ and $b = 10$, giving $\beta = 10/3$ and $\beta - 1 = 7/3$ in Table 4.3. The correct values are $z(\alpha)^{10}$, $b = 11$, $\beta = 11/3$; this agrees with $E_{\emptyset,\alpha,\varepsilon} = E_{\emptyset,\alpha,\zeta}$. The unspecified constants c of the 1994 text are ± 1, as shown.

We summarize this in the following table.

(4.3)

S_1	S_2	σ	$\beta(E_{S_1,S_2,\sigma}) - 1$
α,β	$\emptyset$	—	0
α	$\emptyset$	—	1
β	$\emptyset$	—	1
$\emptyset$	$\emptyset$	—	6
$\emptyset$	α	γ	1
$\emptyset$	α	δ	2
$\emptyset$	α	ε	$8/3$
$\emptyset$	α	ζ	$8/3$
$\emptyset$	β	γ	1
$\emptyset$	β	δ	1
$\emptyset$	β	ε	1
$\emptyset$	β	ζ	3

Since Z is an affine toric variety, each branch E of $\lambda = 0$ is a torus-invariant prime divisor, and corresponds to the primitive inner normal $u_E \in \mathbb{Z}^7$ of a facet of the cone of S [Ke72, Ch. I]. In the torus coordinates $(\lambda, z(\alpha), z(\beta), w(\gamma), w(\delta), w(\varepsilon), w(\zeta))$, the vector u_E lists the orders of vanishing of these coordinates along E, and the order of vanishing of any monomial is the pairing of its exponent vector with u_E. Thus $a(E) = \langle \lambda, u_E \rangle$, and $b(E) = \langle m_\omega, u_E \rangle$, where $m_\omega = (7, -10, -6, 1, 1, 1, 1)$ is the exponent vector with $\omega_Z = \lambda^7 z(\alpha)^{-10} z(\beta)^{-6} w(\gamma) w(\delta) w(\varepsilon) w(\zeta)\, \frac{d\lambda}{\lambda} \wedge \cdots \wedge \frac{dw(\zeta)}{w(\zeta)}$, up to sign. This gives a check of Table 4.3 that does not depend on the choice of patch.

TABLE 4.3A. The seven branches of $\lambda = 0$ on Z for $G = G_2$. The vector u_E is given in the torus coordinates $(\lambda, z(\alpha), z(\beta), w(\gamma), w(\delta), w(\varepsilon), w(\zeta))$; the last column gives the coordinates $x(\rho)$ that vanish on E (script `g2_toric_divisors.py`).

branch E	u_E	a	b	β	$\beta - 1$	$x(\rho) = 0$ on E
$E_{\{\alpha,\beta\}}$	$(1,1,1,1,2,3,4)$	1	1	1	0	none
$E_{\{\alpha\}} = E_{\emptyset,\alpha,\gamma}$	$(1,1,0,0,1,2,2)$	1	2	2	1	β
$E_{\{\beta\}} = E_{\emptyset,\beta,\gamma}$	$(1,0,1,0,0,0,1)$	1	2	2	1	α
$E_{\emptyset,\alpha,\delta}$	$(2,1,0,0,0,1,1)$	2	6	3	2	α,β,γ
$E_{\emptyset,\alpha,\varepsilon} = E_{\emptyset,\alpha,\zeta}$	$(3,1,0,0,0,0,0)$	3	11	$11/3$	$8/3$	$\alpha,\beta,\gamma,\delta$
$E_{\emptyset,\beta,\zeta}$	$(2,0,1,0,0,0,0)$	2	8	4	3	all but ζ
$E_{\emptyset}$	$(1,0,0,0,0,0,0)$	1	7	7	6	all

The integral values $\beta - 1 = 0, 1, 2, 3, 6$ are the values $r(\mathcal{O})$ of the regular, subregular, $\tilde{A}_1$, A_1 and trivial classes. The branch with $\beta = 11/3$ has non-integral β; by Igusa's theorem (property (e) in §0) its contribution to the asymptotic expansion has the form $|\lambda|^{8/3}$ times a function of the angular component, and since the left side of (0.1) contains no such terms, this contribution must cancel. That the coefficient vanishes was not checked directly.

Let $H_1 = PGL(3)$ and $H_2 = (SL(2) \times SL(2))/\{\pm 1\}$.

Lemma 4.4. *For $S_1 \subset \Delta$,*

$$\beta(E_{S_1}) = 1 + |\Sigma^+_{\Delta - S_1}| = 1 + \frac{\dim \mathrm{Cent}_G(u) - \ell}{2},$$

where $\ell = \mathrm{rk}\, G$, u is an element of the Richardson class of the parabolic subgroup $P_{\Delta - S_1}$, and E_{S_1} is the Richardson branch on the patch Z_{S_1}. For $G = G_2$,

$$\beta(E_{\emptyset,\alpha,\delta}) = 3 = 1 + |\Sigma^+_{H_2}|, \qquad \beta(E_{\emptyset,\beta,\zeta}) = 4 = 1 + |\Sigma^+_{H_1}|,$$

and these are $1 + r(\mathcal{O})$ for the 2-regular class $\mathcal{O} = \tilde{A}_1$ and the 3-regular class $\mathcal{O} = A_1$, respectively.

Proof. The first part is a special case of [Ha92a, Ch. II, Proposition 9.1]; for G_2 it also follows from the calculations above, using $\dim \mathrm{Cent}_G(u) = \ell + 2r(\mathcal{O})$ and the dimensions 12, 10, 0 of the regular, subregular and trivial classes. The second part is read off from Table 4.3A, with $|\Sigma^+_{H_2}| = 2$, $|\Sigma^+_{H_1}| = 3$, and $\dim \tilde{A}_1 = 8$, $\dim A_1 = 6$ [Spa82]. □

Remark 2026.22. The 1994 statement had "$\dim \mathrm{Cent}_G(u) - 2$", which is correct only for rank 2, and labeled both orbits "2-regular". It described $\mathcal{O}$ as the orbit "induced" from the trivial class of the endoscopic group H_i; since H_i is not a Levi subgroup of G, the intended meaning is the class with $\dim \mathcal{O} = \dim G - \dim H_i$, that is, $r(\mathcal{O}) = |\Sigma^+_{H_i}|$. Lemma 4.4 computes Igusa constants only. That $\pi(E_{\emptyset,\alpha,\delta})$ and $\pi(E_{\emptyset,\beta,\zeta})$ are the closures of $\tilde{A}_1$ and A_1 was asserted in 1994 and is consistent with the homogeneity (0.1A), but the images were not computed independently in this revision.

Remark 2026.23 (exceptional divisors). The fan of Z is the cone $\sigma \subset \mathbb{R}^7$ spanned by the eleven facet normals: the seven vectors u_E of Table 4.3A and the four coordinate vectors of $w(\gamma), \ldots, w(\zeta)$ (the divisors $w(\rho) = x(\rho) = 0$, which do not lie over $\lambda = 0$). The singular locus of Z is the union of the torus orbits attached to the singular faces of σ, those not spanned by part of a $\mathbb{Z}$-basis. Of the 262 faces of dimension at least 2, 32 are singular, and there are five minimal singular faces (script `g2_exceptional_divisors.py`):

$$\{E_{\{\beta\}}, E_{\emptyset,\beta,\zeta}, E_{\emptyset}, D_{w(\zeta)}\}, \quad \{E_{\{\alpha\}}, E_{\emptyset,\alpha,\delta}, E_{\emptyset,\alpha,\zeta}, D_{w(\delta)}\},$$
$$\{E_{\{\alpha,\beta\}}, E_{\{\beta\}}, E_{\{\alpha\}}, E_{\emptyset,\alpha,\delta}, D_{w(\gamma)}\}, \quad \{E_{\{\beta\}}, E_{\emptyset,\alpha,\delta}, E_{\emptyset,\alpha,\zeta}, E_{\emptyset,\beta,\zeta}, D_{w(\varepsilon)}\},$$
$$\{E_{\emptyset,\alpha,\delta}, E_{\emptyset,\alpha,\zeta}, E_{\emptyset}, D_{w(\zeta)}, D_{w(\varepsilon)}\}.$$

A toric resolution subdivides σ, and each new ray u gives an exceptional divisor with $a = \langle \lambda, u \rangle$, $b = \langle m_\omega, u \rangle$. Integral values of β cannot be excluded in advance. For example, $u = (3,2,2,2,3,5,7) = u_{E_{\{\alpha,\beta\}}} + u_{E_{\{\alpha\}}} + u_{E_{\{\beta\}}} + u_{D_{w(\gamma)}}$ lies in the relative interior of the third face (which is 4-dimensional with five rays), and has $a = 3$, $b = 6$ and $\beta = 2$; other lattice points of the singular faces give $\beta = 5/2, 7/3, 7/2, 13/4, 9/2, 13/3$. Which rays occur depends on the resolution. So for $\beta = 2, 3, 4$ the coefficients $F_1(1, \beta, f_\kappa)$ may receive contributions from exceptional divisors, and the proviso in Lemma 0.8 must be checked: one must determine the images under π and the F-points of the exceptional divisors with integral β. This has not been done. The 1994 claim that the singular locus lies in $K \cup K_0$ (§3) is correct, but K is larger than the singular locus.

§ 5 THE IGUSA EXPANSION FOR G_2

Notes on the derivation of the Igusa expansion. Consider an F-morphism

$$\phi : Y \to \mathbb{A},$$

smooth outside the special fiber $\phi^{-1}(0)$, from a smooth F-variety Y to the affine line over F. When we regard $Y(F)$ and $\mathbb{A}(F) = F$ as F-manifolds in the p-adic topology it will be explicitly stated; otherwise the topology on Y is the Zariski topology. Let $f \in C_c^\infty(Y(F))$, considering $Y(F)$ as a p-adic manifold, and consider

$$F(\lambda) := \int_{\phi^{-1}(\lambda)(F)} f(y)|\omega_\lambda|,$$

for $\lambda \in F - \{0\}$. If f is locally a product of quasi-characters in suitable local coordinates on a neighborhood of $\phi^{-1}(0)$ in $Y(F)$, then Igusa has given an asymptotic expansion for $F(\lambda)$ as $\lambda \to 0$. Since $Y(F)$ is totally disconnected, this expansion is an equality valid for all λ in a neighborhood of 0. We explain this next.

Assume that $Y(F)$ may be covered by patches U on which there are coordinates $\mu_1, \dots, \mu_n$ such that the following hold.

(5.1) $\phi^{-1}(0)$ has normal crossings; more precisely, the irreducible components E_i of $\phi^{-1}(0) \cap U$ are of the form $\{\mu_i = 0\}$ on U and $\lambda = \upsilon \prod_i \mu_i^{a_i}$ for some integers $a_i \geq 0$, where υ is a non-vanishing function on U with $|\upsilon| = q^{-m(\upsilon)}$ constant.

(5.2) On U, $f(y)$ is of the form $f(\mu_1, \dots, \mu_n) = c_f \prod_i \kappa_i(\mu_i)$, where c_f is a constant and the κ_i are quasi-characters of $F^\times$ (extended by $\kappa_i(0) = 0$ when $\kappa_i \neq 1$).

(5.3) The form ω is given on U by $\omega = \prod_i \mu_i^{b_i - 1}\, d\mu_1 \wedge \cdots \wedge d\mu_n$.

(5.4) We **assume** that f vanishes off U, and that if $b_i = 0$ then $\kappa_i \neq 1$, $1 \leq i \leq n$.

Remark 2026.24. In the 1994 text the unit in (5.1) was called α (also a root), the constant in (5.2) was called γ (also a root and a group element), the truncation radius in Lemma 5.6 was ε (also a root), and the number of indices in Lemma 5.7 was p (also the residue characteristic and $|\Sigma_G^+|$). These have been renamed υ, c_f, q^{-N} and k.

Let $F(\lambda)$ be given by (0.2). We sketch the derivation of an equality of the form

$$F(\lambda) = \sum_{\theta,\beta} \theta(\lambda)|\lambda|^{\beta-1} \sum_{r=1}^{e(\theta,\beta)} (-\log_q|\lambda|)^{r-1} F_r(\theta,\beta,f). \tag{5.5}$$

The following lemma helps us to compute F_1.

Lemma 5.6. *Let*

$$G(\lambda) = \sum_{\theta,\beta} \sum_{r=1}^{e(\theta,\beta)} a(\theta,\beta,r)(-\log_q|\lambda|)^{r-1}\theta(\lambda)|\lambda|^{\beta-1},$$

where the $a(\theta,\beta,r)$ are constants. If $a(\theta,\beta,r) = 0$ for $r > 1$ then the coefficients b_n in the power series of the truncated Mellin transform

$$\mathscr{M}_N G(s) := \int_{|\lambda| \leq q^{-N}} G(\lambda)|\lambda|^s d\lambda = \sum_{n \geq 0} b_n q^{-ns}$$

are given by

$$b_n = \begin{cases} \mathrm{vol}(\mathscr{O}_F^\times, d\lambda) \sum_{\substack{\theta,\beta \\ \theta \text{ unramified}}} a(\theta,\beta,1)(\theta(\pi)q^{-\beta})^n, & n \geq N, \\ 0, & n < N. \end{cases}$$

Proof. Write $\lambda = \pi^n \nu$ with $\nu \in \mathscr{O}_F^\times$. Then

$$\int_{|\lambda|=q^{-n}} \theta(\lambda)|\lambda|^{\beta-1+s} d\lambda = \theta(\pi)^n q^{-n(\beta-1+s)} q^{-n} \int_{\mathscr{O}_F^\times} \theta(\nu)\, d\nu,$$

and the last integral is $\mathrm{vol}(\mathscr{O}_F^\times)$ if θ is unramified and 0 otherwise. Sum over $n \geq N$. □

To verify (5.5) for $F(\lambda)$ we may verify it for its truncated Mellin transform and use Lemma 5.6 to compare the coefficients of the two power series in q^{-s}. The formula for F_1 is stated below. First, some notation: let $c_j = c_j(a_1,\ldots,a_k)$ denote the coefficients for which the principal part of $\prod_{i\leq k} \frac{1}{1-t^{a_i}}$ at $t=1$ is

$$\sum_j \frac{c_j}{(1-t)^j}.$$

Let

$$A(x) = \sum_j \frac{c_j}{(j-1)!}(x+1)\cdots(x+j-1),$$

and define $A_r(y)$ by $A(x-y) = \sum_r x^{r-1} A_r(y)$, so $A_1(y)$ is the constant term of this polynomial in x. The function $A(x)$ arises from the identity

$$\frac{(j-1)!}{(1-t)^j} = \sum_{n\geq 0} (n+1)(n+2)\cdots(n+j-1)t^n.$$

Lemma 5.7. *Suppose that $Y(F)$, f, ω, and λ satisfy* (5.1)–(5.4)*, and that*

$$U = \{(\mu_1,\ldots,\mu_n) \in F^n \mid |\mu_i| \leq q^{-M_i}\}.$$

Fix (θ,β), and reindex the μ_i, if necessary, so that

$$\theta^{a_i} = \kappa_i, \quad \beta = b_i/a_i, \quad 1 \leq i \leq k,$$

and for each $i > k$ at least one of these equalities fails. Then $F(\lambda)$ satisfies (5.5)*, and $F_1(\theta,\beta,f)$ is the value at $z_{k+1} = \cdots = z_n = 0$ of the analytic continuation of*

$$\int_{D_k \cap U} A_1(M)\, h_k \prod_{i=k+1}^n |\mu_i|^{z_i} |\nu_k|,$$

where $D_k = \bigcap_{i\leq k} E_i$ and

$$M = m(\upsilon) + \sum_{i\leq k} a_i M_i + \sum_{i>k} a_i(-\log_q|\mu_i|), \qquad h_k = \frac{c_f}{\theta(\upsilon)} \prod_{i=k+1}^n \kappa_i(\mu_i)\theta(\mu_i)^{-a_i},$$

$$\nu_k = \frac{1}{|\upsilon|^\beta} \prod_{i=k+1}^n |\mu_i|^{b_i - \beta a_i - 1} d\mu_{k+1}\cdots d\mu_n.$$

If any one of the characters $\kappa_i\theta^{-a_i}$, $i \leq k$, is ramified, then $F_1(\theta,\beta,f) = 0$.

We sketch the proof following [La83]. By our assumptions,

$$\begin{aligned}\mathscr{M}_N(F\theta^{-1})(s) &= \frac{|\upsilon|^s}{\theta(\upsilon)}\int_U f(y)\prod_i \theta(\mu_i)^{-a_i}|\mu_i|^{a_i s+b_i-1}d\mu_1\cdots d\mu_n \\ &= \frac{|\upsilon|^s}{\theta(\upsilon)}c_f\prod_i\int_{|\mu|\le q^{-M_i}}\kappa_i(\mu)\theta(\mu)^{-a_i}|\mu|^{a_i s+b_i-1}d\mu,\end{aligned} \tag{5.8}$$

since $|\omega_\lambda||d\lambda| = \prod_{i=1}^n |\mu_i|^{b_i-1}d\mu_1\cdots d\mu_n$, and N is chosen so that $|\lambda|\le q^{-N}$ on U. Since $\int_{|\mu|\le q^{-M_i}}\kappa'(\mu)|\mu|^z d\mu = 0$ for a ramified character κ', we may assume that every character $\kappa_i\theta^{-a_i}$ is unramified, and we let $\eta_i := \kappa_i(\pi)\theta(\pi)^{-a_i}$, $1\le i\le n$, where π is a fixed uniformizer of $\mathscr{O}_F$. Then

$$\mathscr{M}_N(F\theta^{-1})(s) = c_f\frac{|\upsilon|^s}{\theta(\upsilon)}(1-q^{-1})^n\prod_i\frac{(\eta_i q^{-a_i s-b_i})^{M_i}}{1-\eta_i q^{-a_i s-b_i}}.$$

With the indexing of the lemma, $\eta_i = 1$ and $b_i = \beta a_i$ for $i\le k$, and

$$\mathscr{M}_N(F\theta^{-1})(s) = (1-q^{-1})^k|\upsilon|^{s+\beta}\Big[\prod_{i\le k}\frac{q^{-a_i M_i(s+\beta)}}{1-q^{-a_i(s+\beta)}}\Big]\int_{\substack{|\mu_i|\le q^{-M_i}\\ i>k}}\prod_{i>k}|\mu_i|^{z_i+a_i(s+\beta)}\,h_k\,|\nu_k| \tag{5.9}$$

at $z_{k+1} = \cdots = z_n = 0$. The last integral is a principal value integral over $D_k = \cap_{i\le k}E_i$, and is a rational function of q^{-s}.

Put $t = q^{-s-\beta}$. By definition of the c_j, the principal part at $t=1$ of

$$\prod_{i\le k}\frac{t^{a_i M_i}}{1-t^{a_i}} \quad \text{is that of} \quad t^{\sum_{i\le k}a_i M_i}\sum_j\frac{c_j}{(1-t)^j} = t^{\sum_{i\le k}a_i M_i}\sum_{n\ge 0}A(n)t^n.$$

Multiplying and dividing by $\prod_{i>k}|\mu_i|^{a_i(s+\beta)}|\upsilon|^{s+\beta}$ to account for all of M, the coefficient of t^n for $n\ge M$ is $A(n-M) = \sum_r n^{r-1}A_r(M)$, and it is 0 for $n<M$. Comparing with Lemma 5.6 gives the lemma.

Remark 2026.25. In the 1994 text the principal part was written $\sum_{1\le i\le p}c_j(j-1)!^{-1}(1-t)^{-j}$, with the wrong summation index and a spurious factor $(j-1)!^{-1}$; the factor belongs in $A(x)$. The comparison with Lemma 5.6 uses that the remaining poles of $\prod_i(1-\eta_i q^{-a_i s-b_i})^{-1}$ belong to other pairs (θ,β); this is the standard argument of [La83] and [Ig78], and is not repeated. The proof above remains a sketch. Lemma 5.6 was also left to the reader in 1994; its proof is new.

Geometry of the Igusa expansion. Let E be an irreducible divisor in $Y''\subset G\times S''$ whose image under π is contained in the Zariski closure of a unipotent class $\mathscr{O}$ of G. Such a divisor is called an **$\mathscr{O}$-divisor** if $\mathscr{O}$ is minimal, with respect to the partial ordering by inclusion of Zariski closures, with this property. It will be called an **$\mathscr{O}(F)$-divisor** if E is defined over F and $\pi(E(F))$ meets $\mathscr{O}(F)$ in a dense subset of $\mathscr{O}(F)$ as p-adic manifolds. Let E be an $\mathscr{O}(F)$-divisor.

We record an observation about the fibration $\pi: E\to\overline{\mathscr{O}}$. As a corollary to [DG80, Ch. II, Prop. 3.1], the 1994 manuscript states the following.

Lemma 5.10. *Let E be an $\mathscr{O}$-divisor meeting the patch Z_S, where $\mathscr{O}$ is r-regular.*

(a) The morphism $\pi: E\cap Z_S\to\overline{\mathscr{O}}$ is smooth of relative dimension $2r$ over $\mathscr{O}$.

(b) $\pi(E\cap Z_S)\subset\mathscr{O}$.

(c) Let

$$\{1\} = \mathscr{O}'_0\subset\overline{\mathscr{O}}'_1\subset\cdots\subset\overline{\mathscr{O}}'_m = \overline{\mathscr{O}},\qquad \overline{\mathscr{O}}'_i = \mathscr{O}'_i\cup\overline{\mathscr{O}}'_{i-1},$$

be a chain of unipotent classes $\mathcal{O}_i'$ of G whose closures exhaust $\overline{\mathcal{O}}$, and let $E_i = \pi^{-1}(\mathcal{O}_i') \cap E$. Then $\pi|_{E_i} : E_i \to \mathcal{O}_i'$ is faithfully flat.

Remark 2026.26. The chain in (c) was garbled in 1994 ("$\overline{\mathcal{O}} = \mathcal{O} \cup \mathcal{O}_m$"); the version above is our reading. For G_2 the unipotent closures are totally ordered, so the chain is unique. The relative dimension in (a) agrees with the dimension count of §3: the patch $Y''(B_+, B_-) \times N_+$ is open in $\mathbb{A}^p \times Z$, of dimension $2p+1$, so $\dim E = 2p$, while $\dim \mathcal{O} = \dim G - \dim \mathrm{Cent}_G(u) = (2p+\ell) - (\ell+2r) = 2p-2r$. Part (b) is stated as in 1994 and was not checked.

We apply Igusa's expansion to

$$F(\lambda) = \int_{\phi^{-1}(\lambda)(F)} f(y)|\omega_\lambda|, \tag{5.11}$$

where f is a function on $Y_C(F)$ which is compactly supported on the p-adic manifold $Y_C(F)$ and such that ϕ, $|\omega_\lambda|$, Y_C, and f form Igusa data in the sense of [LS83]. On each fiber $\tilde{E}(u)$ the pull-back $\pi^*(f)$ takes the constant value $f(u)$. By [Ha92a, Ch. I], the regular function $\kappa(t_*)$ extends from $E(u)$ to a regular function on $\tilde{E}(u)$. By Lemma 5.10,

$$\int_{\overline{\mathcal{O}}(F)} f \cdot PV \int_{\tilde{E}(u)(F)} \kappa(\tilde{t}_*)|\omega_{\tilde{E}(u)}||\omega_{\overline{\mathcal{O}}}| = PV \int_{\tilde{E}(F)} \tilde{\pi}^*(f)\kappa(\tilde{t}_*)|\omega_{\tilde{E}}|, \tag{5.12}$$

provided everything in sight satisfies (5.1)–(5.4). By definition, there are disjoint patches $U_i \subset Y_C(F)$, each analytically isomorphic to a box in affine space, such that

$$PV \int_{\tilde{E}(F)} \tilde{\pi}^*(f)\kappa(\tilde{t}_*)|\omega_{\tilde{E}}| = \sum_i PV \int_{\tilde{E}(F)\cap U_i} \tilde{\pi}^*(f)\kappa(\tilde{t}_*)|\omega_{\tilde{E}}|, \tag{5.13}$$

provided that (5.1)–(5.4) hold. The computations in §4 show that $b(E) > 0$ for every branch of $\lambda = 0$ on Z; in particular $b(E) \neq 0$.

Lemma 5.14 (conditional)**.** *Let $G = G_2$. Let E be a base $\mathcal{O}(F)$-divisor of Y_C. If E is F-rational and $E(F) \neq \emptyset$, then $e(\theta_E, \beta_E) = 1$.*

Sketch of the 1994 argument. Suppose $e = e(\theta_E, \beta_E) \geq 2$. By [La83], a principal value integral contributing to $F_1(\theta_E, \beta_E, f_\kappa)$ is

$$\int_{\mathcal{O}(F)} f \int_{D_u(F)} \kappa(\tilde{t}_*)|\omega_{D_u}||\omega_{\mathcal{O}}|,$$

where $D = \cap_{i=1}^e E_i$ is an F-rational intersection of branches with $(\theta_{E_i}, \beta_{E_i}) = (\theta_E, \beta_E)$ (the E_i themselves need not be F-rational). The claim is that the homogeneity degree of $|\omega_{D_u}|$ in the tangent direction X is $\dim D - \dim \mathcal{O} < \dim E - \dim \mathcal{O} = 2r$, which contradicts the homogeneity (0.1A) of the $\mathcal{O}$-germ.

The regular and 1-regular cases are treated in [Ha92a], [LS90]. Let $\mathcal{O} = \mathcal{O}_2$ be the 2-regular class and $E = E_{\emptyset,\alpha,\delta}$, and suppose $e = 2$. By §7, the form on the F-rational fiber $E_{\emptyset,\alpha,\delta}(u)$, $u \in \mathcal{O}$, is given on $Y''_{\emptyset,\alpha,\delta}$ by

$$\omega_{E_{\emptyset,\alpha,\delta}(u)} = g\, dz(\beta)\, dx(\zeta)\, dv(\alpha)\, dv(\zeta), \tag{5.15}$$

where g is a regular function. (These coordinates need not be F-rational; one may average them over the image of the Galois group in Ω using §2, and change variables to obtain F-rational coordinates satisfying (5.1)–(5.4).) By (1.22), this form is homogeneous of degree 2 in X. If E' is another F-rational $\mathcal{O}_2$-divisor with $(\theta_{E'}, \beta_{E'}) = (\theta_E, \beta_E)$, then E meets E' with normal crossings in Y_C, and $D = E \cap E'$ has codimension 1 in E and maps onto $\mathcal{O}$ [DG80, Ch. II, Prop. 3.1]. The form on a fiber

has the shape $\omega_{D(u)} = g_D\,dx\,dv_1\,dv_2$, which is homogeneous of degree < 2 in X. The case $e = 3$ is similar.

For $\mathscr{O} = \mathscr{O}_3$ and $E = E_{\emptyset,\beta,\zeta}$, by §8,

$$\omega_{E_{\emptyset,\beta,\zeta}(u)} = h\,dw(\varepsilon)\,dw(\zeta)\,dx(\zeta)\,dv(\beta)\,dv(\varepsilon)\,dv(\zeta), \tag{5.16}$$

with h regular, homogeneous of degree 3 in X, and the argument is the same. □

Remark 2026.27 (status of Lemma 5.14). The argument above is incomplete in four ways. (i) The homogeneity degrees "by (1.22)" are asserted, not computed. (ii) Lemma 5.7 expresses F_1 as a sum over strata; only the top stratum $D = E \cap E'$ is considered, and the contradiction requires its contribution to be non-zero. (iii) The possible partners E' include the exceptional divisors of Remark 2026.23, which were not analyzed. (iv) The 1994 statement added "In particular, spurious divisors do not contribute to the Igusa expansion"; $e = 1$ does not imply this, and the sentence has been removed. We therefore treat Lemma 5.14 as a hypothesis in what follows. Note that $e(\theta,\beta) = 1$ for all (θ,β) also follows from Lemma 0.4, since the Shalika germ expansion has no logarithmic terms; what Lemma 5.14 would add is that a *single* divisor accounts for the β_E-term.

Lemma 5.17. *Let $G = G_2$ and assume Lemma 5.14 and the proviso of Lemma 0.8. Let E be a base $\mathscr{O}(F)$-divisor of Y_1, and $\tilde{E}$ its preimage under $Y_C \to Y_1$. Then $F_1(\theta_E,\beta_E,f_\kappa)$ is given, up to a constant, by*

$$PV \int_{\tilde{E}(F)} \tilde{\pi}^*(f_\kappa)|\omega_{\tilde{E}}| = \int_{\overline{\mathscr{O}}(F)_{st}} f \int_{\tilde{E}_u(F)} \kappa(\tilde{t}_*)|\omega_{\tilde{E}_u}||\omega_{\overline{\mathscr{O}}}|,$$

where $\omega_{\overline{\mathscr{O}}_{st}}$ is a sum of forms of maximal degree on the equidimensional pieces $\mathscr{O}'_i$ of Lemma 5.10 of the stable class $\overline{\mathscr{O}}(F)_{st} = \cup_i \mathscr{O}'_i(F)$.

Proof. By Lemma 5.14, (0.1) and Formula 0.6, it suffices to check that the conditions of [LS83] are met. This has essentially been done in [La83] and [Ha92a, Ch. I, proof of Prop. 5.2]; we add some details on the function $\kappa(\tilde{t}_*)$ and the cocycle-valued function t_*.

Some information about t_* is given after Formula 0.6. More explicit information about $t_\sigma : Y_1(B_+,B_-) \to T_G(\overline{F})$ is given in [Ha92a, Ch. I, Corollary 5.4]. Since the only F-form of G_2 over a p-adic field is the split form [Ti65], we have $G = G_2 = G_{in} = G_{qs}$, $T_G = T_{in}$, $\underline{T} = T_{qs}$, and $\underline{B} = B_{qs}$, in the notation of [Ha92a]. Let $\alpha \in \Delta$ be a simple root, and suppose $\sigma \in \Gamma$ acts on $T_G(\overline{F})$ through the reflection $s_\alpha \in \Omega$. Choose g, h as in [Ha92a, Ch. I, §5], and take $A_\sigma = w_\sigma^{-1} = s_\alpha \in N_G(T_G)$ in [Ha92a, Ch. I, Corollary 5.7]. There is a constant $t_1 \in T_G(\overline{F})$, independent of the star, such that at a point

$$(t^g, (B(W))_{W\in\mathscr{W}}, (z(\alpha))_{\alpha\in\Delta}, n) \in \chi^{-1}(C) \times S_1(B_+,B_-) \times N_+,$$

the value t_σ is determined by

$$s_\alpha \theta_{-\alpha}(-\xi) n^{-1} w_\sigma^{-1} s_\alpha^{-1} \in N_+^- N_+ (t_1 t_\sigma)^{w_\sigma^{-1} s_\alpha^{-1}}. \tag{5.18}$$

Taking the transpose of both sides of [Ha92a, Ch. I, (5.5)] and comparing with (5.18), one finds $t_1 t_\sigma = \alpha^\vee(\xi)$, where ξ is a regular function of the coordinates of $Y''(B_+,B_-) \times N_+$. Choose B_-, B_+ such that these coordinates are F-rational.

When $H^1(F,T_G)$ is cyclic, of order $m \in \{2,3\}$, which is the case for the biquadratic, cubic and short-reflection S_3 tori (Table A.6), $\kappa(\tilde{t}_*)$ is determined by its value on a generator. This determines a character of $\mathbb{Z}/m\mathbb{Z}$, and hence by local class field theory a character θ of $F^\times / N_{E_{T_G}/F} E_{T_G}^\times$, where E_{T_G} is the Galois closure of the splitting field of T_G. One may construct a (non-canonical) map $\phi_T : H^1(F,T_G) \to F^\times / N_{E_{T_G}/F} E_{T_G}^\times$ such that $\kappa(\tilde{t}_*) = \theta(\phi_T(\tilde{t}_*))$ is a monomial in the coordinates of

$Y''(B_+,B_-)\times N_+$. So (5.1)–(5.4) hold. For the quadratic tori, $H^1(F,T_G)\cong(\mathbb{Z}/2)^2$; there κ is a character of $(\mathbb{Z}/2)^2$, and the same argument applies with θ a quadratic character, but this case was not treated in 1994. □

Remark 2026.28. The 1994 proof asserted that $H^1(\Gamma,T_G)$ is always cyclic of order 2 or 3; this fails for the quadratic tori (Table A.6). It also cited "(5.10)" (a lemma, not an equation) for the comparison with (5.18), and wrote $\varepsilon_{-\alpha}$ for $\theta_{-\alpha}$; and "ξ satisfies (5.2)" was unexplained. The precise source of ξ in [Ha92a] has not been checked.

We close this section by recalling a reduction lemma due to T. Hales, in the form quoted in 1994, and by explaining why it does not apply here.

Reduction Lemma 5.19 (Hales, as quoted in 1994)**.** *Suppose that for each (T,κ) associated to H the germs $\Gamma^{T,\kappa}_{\mathcal{O}}(\gamma)$ have the form*

$$\Gamma^{T,\kappa}_{\mathcal{O}}(\exp(\lambda^2 X))=|\lambda|^{2(r-|\Sigma_H^+|)}m^{T,\kappa}_{\mathcal{O}}(X),$$

for λ small, where $m^{T,\kappa}_{\mathcal{O}}(X)$ satisfies

$$m^{T,\kappa}_{\mathcal{O}}(X)=m^{T,\kappa}_{\mathcal{O}}(X+Y),$$

for $X,Y\in\mathfrak{t}_G$ sufficiently small, X and $X+Y$ regular, and Y in the Lie algebra of the center of an endoscopic group $H\subset G$ with $\mathrm{Cent}(Y,G)=H$. Then (G,H) admits local Δ-transfer at the identity.

Source. [Ha92b, Lemma 6.1]; see also [Ha89, Proposition 2.11]. This lemma implicitly uses [LS90]. □

Remark 2026.29 (the main gap). The 1994 proofs of Theorems 6.1, 7.2 and 8.3 end by applying Lemma 5.19. Its hypothesis cannot be met for the cuspidal endoscopic groups of G_2. Both $H_1=PGL(3)$ and $H_2=(SL(2)\times SL(2))/\{\pm1\}$ are semisimple of rank 2, the rank of G_2, so $\mathrm{Lie}\,Z(H)=0$. Moreover the centralizer in G of an element Y of $\mathfrak{t}_G$ is a Levi subgroup, and the Levi subgroups of G_2 are T, the two $GL(2)$'s and G_2; none of them is $SL(3)$ or $SO(4)$, the subgroups of G_2 that the 1994 text uses as "$H\subset G$". So no admissible Y exists. A machine summary of [Ha89] (consulted, not the paper itself) indicates that Hales' descent there is to elements of the connected center of H, and that for the cuspidal $SO(4)$ of $GSp(4)$ the dependence on the tangent direction is computed explicitly. We could not consult [Ha92b]. In the 1994 text the exponent was $2r$; with $\Gamma^{T,\kappa}$ normalized by the transfer factor it is $2(r-|\Sigma_H^+|)$, by (0.1A). The decomposition $f=f_1+f_2$ that preceded the lemma in 1994, with f_2 "whose support does not meet the orbits outside the Zariski closure of $\mathcal{O}$", is impossible, because regular unipotent elements are dense in every neighborhood of a unipotent element; it has been removed.

We replace the appeal to Lemma 5.19 by an explicit hypothesis. Let (T,κ) be an elliptic pair with $\kappa\neq1$ and endoscopic group H, let $r\in\{1,2,3\}$, and let E be a base divisor with $\beta(E)=1+r$ (Table 4.3A). For $u\in\mathcal{O}_r^{st}(F)$ and regular $X\in\mathfrak{t}(F)$ near 0, put

$$I_E(X,u):=\kappa(\mathrm{inv}(\gamma_G^0,\exp X))\,\Delta_0(X)\int_{\tilde{E}_u(F)}\kappa_1(\tilde{t}_*)|\omega_{\tilde{E}_u}|,$$

where the dependence on X enters through the coordinates (1.22), and Δ_0 is as in §0.

Hypothesis (X_r)**.** There are r-regular unipotent classes $\mathcal{O}_{H,i}$ of $H(F)$ and functions c_i on $\mathcal{O}_r^{st}(F)$, constant on $G(F)$-classes, such that $I_E(X,u)=\sum_i c_i(u)\Gamma^{T_H,st}_{\mathcal{O}_{H,i}}(\exp X_H)$ for all regular X near 0, where X_H corresponds to X under $T\cong T_H$.

For $r = |\Sigma_H^+|$ the classes $\mathcal{O}_{H,i}$ are trivial, the stable germ $\Gamma_1^{T_H,st}$ is constant, and (X_r) says that $I_E(X,u)$ *does not depend on* X. Homogeneity (0.1A) shows only that $I_E(\lambda X,u) = I_E(X,u)$ for $\lambda \in F^\times$ near 1. For $r = 1$, (X_1) asks that the X-dependence of I_E be that of the stable subregular germ of H, given for $SL(2)$ in [SS83] and for $GL(3)$ in [Re84]. Verifying (X_r) requires computing $\Delta_0(X)$ and the fiber integrals as functions of X, as in [Ha89, §6]; this is the main open step of this revision. For $r = 2$ and the biquadratic tori, Δ_0 is computed in Proposition 7.6, and (X_2) is reduced to the explicit identity (7.7) (Corollary 7.8).

§ 6 THE MATCHING FOR THE 1-REGULAR GERM

Let $G = G_2$ throughout this section.

The κ-stable subregular germ of G_2 should match the stable subregular germ of $PGL(3)$ or of $(SL(2) \times SL(2))/\{\pm 1\}$, for homogeneity reasons (0.1A). The idea is to write down the Igusa expansions of the two orbital integrals and match the $\beta = 2$ terms.

Let $u = \exp(X_\alpha + X_{\alpha+\beta})$. Then the subregular class is $\mathcal{O}_1^{st} = u^{G_2}$. It is Richardson, but not stable as an F-class; see §A.7 for its F-classes. Both the biquadratic and the cubic Cartan subgroups can have non-zero 1-regular κ-germs.

By Table 4.3A there are exactly two branches of $\lambda = 0$ on Z with $\beta = 2$, namely $E_{\{\alpha\}}$ and $E_{\{\beta\}}$, which we write E_α and E_β. By [Ha92a, Ch. VI, §2], all fundamental divisors of G_2 have $\theta = 1$. Explicitly:

E_α is given locally as $\{\lambda = 0\}$ (with multiplicity one) on $Z_\alpha = Z_{\{\alpha\}}$, which is regular with local coordinates $\lambda, x(\alpha), z(\beta), x(\gamma), x(\delta), x(\varepsilon), x(\zeta)$.

E_β is given locally as $\{\lambda = 0\}$ (with multiplicity one) on $Z_\beta = Z_{\{\beta\}}$, which is regular with local coordinates $\lambda, z(\alpha), x(\beta), x(\gamma), x(\delta), x(\varepsilon), x(\zeta)$.

Remark 2026.30. The 1994 text listed a third base divisor with $\beta = 2$, $E_{\emptyset,\alpha,\gamma}$, and eliminated it by [Ha92a, Ch. V, Lemma 1.1]. It is the same divisor as E_α (§4), so no elimination is needed.

As a corollary to [Ha92a, Ch. VI, Lemma 1.1], E_α and E_β are F-rational. The action of the Galois group Γ on T_G and on Z factors through a finite group $\Gamma_T \subset \Omega$. To satisfy (5.1)–(5.4), one may use §2 to average the coordinates of Z_α or Z_β over Γ_T to get F-rational coordinates on these divisors.

If Γ acts on T_G through a subgroup $\langle s_\alpha, s_\zeta \rangle$ of Ω (up to conjugacy), the κ-subregular germ of G_2 should match the stable subregular germ of $H_2 = (SL(2) \times SL(2))/\{\pm 1\}$, and κ is quadratic (§A). If Γ acts on T_G through a cyclic subgroup of Ω of order three, the κ-subregular germ of G_2 should match the stable subregular germ of $H_1 = PGL(3)$, and κ is cubic.

Theorem 6.1 (conditional)**.** *Let T_G be an elliptic Cartan subgroup of G_2 such that Γ acts on T_G through a subgroup $\langle s_\alpha, s_\zeta \rangle$ of Ω (up to conjugacy), or through a cyclic subgroup of order* 3*. Let $\kappa \neq 1$ be a character of $H^1(F, T_G)$, and let $H = H_2$ in the first case and $H = H_1$ in the second. Assume Lemma 5.14, the proviso of Lemma 0.8, and Hypothesis (X_1) of Remark 2026.29 for (T_G, κ). Then Conjecture 0.0A holds for (T_G, κ) and every 1-regular class $\mathcal{O} \subset \mathcal{O}_1^{st}(F)$.*

Remark 2026.31 (the 1994 statement). In 1994, Theorem 6.1 asserted local transfer at the identity for all $f \in C_c^\infty(G(F))$ whose support meets a fixed 1-regular orbit and no unipotent orbit that is not 1-regular. No such f exists: every neighborhood of a unipotent element meets the regular unipotent orbits. The statement has been replaced by a germ identity, which is what the proof gives, and made conditional on (X_1), because the 1994 proof ended with Lemma 5.19, which does not apply (Remark 2026.29). An equivalent formulation with test functions uses f with $\mu_{\mathcal{O}'}(f) = 0$

for every unipotent class $\mathcal{O}' \not\subset \mathcal{O}_1^{st}$; such f exist because the unipotent orbital integrals are linearly independent. The quadratic tori and the tori with Galois image S_3 generated by short reflections also carry non-trivial κ (Table A.6); they are not treated here.

Proof of Theorem 6.1. By the work of Langlands and Shelstad, [LS87], [LS90], transfer at the identity reduces to the matching of germs, Conjecture 0.0A.

We discuss the Igusa expansion of the G_2 side. Since only the $\theta = 1$ term of Formula 0.6(a) contributes [Ha92a, Ch. VI, §2], the $\beta = 2$ term of the Igusa expansion is $|\lambda|\kappa(\mathrm{inv}(\gamma_G^0, \gamma_{G,\lambda}))|D_H(\gamma_{H,\lambda})|^{-1}$ times

$$F_1(1,2,f_\kappa) = PV \int_{U\cap E(F)} A_1(M)h_1|\nu_1| = PV \int_{E(F)\cap U} \pi^*(f)\kappa_1(t_*)|\omega_E|, \tag{6.2}$$

where E is E_α or E_β, depending on T_G, $A_1(M)$, h_1, ν_1 are as in Lemma 5.7, and f_κ includes the κ-factor. The factor $m_\kappa(e)/\kappa_1(\lambda)|_{E(u)}$ is given by [Ha92a, Ch. VI, §2 and Proposition 3.2], provided E is the divisor attached to a line of the Dynkin curve (§2) for the long root of G_2 which is fixed by the Galois action. By Formula 0.6 and Lemma 0.8, this term is equal to a multiple of

$$\sum_{\mathcal{O}\subset\mathcal{O}_1^{st}} \mu_{\mathcal{O}}(f)\Gamma_{\mathcal{O}}^{T_G,\kappa}(\gamma_\lambda).$$

M. Assem has computed the number of F-classes in the stable subregular class when $p \neq 2,3$ [As98]; see §A.7.

By Formula 0.6, (6.2) and Lemma 5.17,

$$PV \int_{U\cap E(F)} A_1(M)h_1|\nu_1| = \int_{\pi(U)\cap\mathcal{O}_1^{st}(F)} f \int_{E_u(F)} \kappa_1(t_*)|\omega_{E(u)}||\omega_{\mathcal{O}_1}|, \tag{6.3}$$

where E is either E_α or E_β. The dependence of (6.3) on the curve $C \subset \exp(\mathcal{O}_F \cdot X)$, where $X \in \mathfrak{t}$ is a tangent vector of C, is through the dependence of $\kappa(t_*)$ and $\omega_{E(u)}$ on X.

Recall [La83, Lemma 2.12] that the measure ω_γ already incorporates the Weyl discriminant $\prod_{\sigma>0}(1-\sigma(\gamma)^{-1})$. From §4,

$$|\omega_{Z_\alpha}| = \frac{|\lambda|}{|z(\beta)|^2} d\lambda\, dx(\alpha)\, dz(\beta)\, dx(\gamma)\, dx(\delta)\, dx(\varepsilon)\, dx(\zeta),$$

$$|\omega_{Z_\beta}| = \frac{|\lambda|}{|z(\alpha)|^2} d\lambda\, dz(\alpha)\, dx(\beta)\, dx(\gamma)\, dx(\delta)\, dx(\varepsilon)\, dx(\zeta).$$

These coordinates are not all F-rational if c_σ is a simple reflection, by (2.16) and (2.17).

On $\mathcal{O}_1$ the roots outside the Levi subgroup of P_α (or of P_β) may be used to parametrize the class; see Lemma 6.4. The following digression is motivated by [Ha92a, Remark V.2.2].

Let $\mathcal{O}$ be the Richardson class of a parabolic subgroup P [St74, §3.9, Theorem 2]. Write $P = MN_P$, so that N_P^G contains $\mathcal{O}$ as a dense subset, $\mathcal{O}\cap N_P$ is dense in N_P, and $\mathcal{O}\cap N_P$ is a single class under P. We have $\dim\mathcal{O} = \dim G - \dim M$. Let $u \in \mathcal{O}\cap N_P$. The P-orbit of u is $\mathcal{O}\cap N_P$, and is dense in N_P, so the coordinates of N_P give coordinates on it. Since $N_P^- P$ is dense in G, the class $\mathcal{O}$ contains the N_P^--orbit of $\mathcal{O}\cap N_P$ as an open subvariety. By Corollary 1.6, for each $\overline{n} \in N_P^-$, $\overline{n} \neq 1$, the conjugate $N_P^{\overline{n}}$ differs from N_P, so $u^{\overline{n}} \notin \mathcal{O}\cap N_P$. It follows that the N_P^--orbit of $\mathcal{O}\cap N_P$ is isomorphic to $N_P^- \times (\mathcal{O}\cap N_P)$.

Lemma 6.4 (Hales)**.** *If $\mathcal{O}$ is the Richardson class of a parabolic P, then the coordinates on $N_P^- \times N_P$ serve as local coordinates on $\mathcal{O}$.*

Remark 2026.32. The 1994 argument used Corollary 1.6 for the statement "$B^{\overline{n}}$ is opposite to B^- and $B^{\overline{n}} \neq B$ unless $\overline{n} = 1$"; the injectivity needed is that of the map $N_P^- \times (\mathscr{O} \cap N_P) \to G$, $(\overline{n}, v) \mapsto v^{\overline{n}}$, which follows from $N_P^- \cap P = 1$ and the fact that P is the normalizer of N_P. The reworded argument above is ours.

Returning to the proof of Theorem 6.1: associated to the roots outside the Levi subgroup of P_α, we have the invariant measure

$$\omega_{\mathscr{O}_1} = \lim_{\lambda \to 0} \frac{dx(\beta) \cdots dx(\zeta)\, dv(\beta) \cdots dv(\zeta)}{\lambda},$$

which is unique up to a constant [Ra72]. On the fiber $E_\beta(u)$, by §4 and [La83, Lemma 2.12],

$$\omega_{E_\beta(u)} = \frac{1 - \alpha(\gamma_\lambda)^{-1}}{\lambda}\Big|_{\lambda=0} dv(\alpha) \frac{dz(\alpha)}{z(\alpha)^2} = \alpha(X)\, dv(\alpha) \frac{dz(\alpha)}{z(\alpha)^2},$$

where $\alpha(X)$ is the value of $d\alpha$ on the tangent vector X of C. Similarly

$$\omega_{E_\alpha(u)} = \beta(X)\, dv(\beta) \frac{dz(\beta)}{z(\beta)^2}.$$

Therefore, the $\beta = 2$ term is

$$|\lambda|\kappa(\mathrm{inv}(\gamma_G^0, \gamma_{G,\lambda}))|D_H(\gamma_{H,\lambda})|^{-1}|\beta(X)| \int_{\mathscr{O}_1} f \int_{E_\alpha(u)\cap U} \kappa(t_*) \Big| dv(\beta) \frac{dz(\beta)}{z(\beta)^2} \Big| |\omega_{\mathscr{O}_1}|, \tag{6.5a}$$

or

$$|\lambda|\kappa(\mathrm{inv}(\gamma_G^0, \gamma_{G,\lambda}))|D_H(\gamma_{H,\lambda})|^{-1}|\alpha(X)| \int_{\mathscr{O}_1} f \int_{E_\beta(u)\cap U} \kappa(t_*) \Big| dv(\alpha) \frac{dz(\alpha)}{z(\alpha)^2} \Big| |\omega_{\mathscr{O}_1}|. \tag{6.5b}$$

The inner integrals, multiplied by $\Delta_0(X)$ and the κ-factor, are the functions $I_E(X,u)$ of Remark 2026.29. By uniqueness of the germ expansion and the linear independence of the $\mu_{\mathscr{O}}$, the coefficient of $\mu_{\mathscr{O}}(f)$ in (6.5a) or (6.5b) is $\Gamma_{\mathscr{O}}^{T_G,\kappa}(\gamma_\lambda)$, up to the normalizing factors in (0.1). Hypothesis (X_1) expresses it as a combination of stable subregular germs of H, which is Conjecture 0.0A for $\mathscr{O}$. □

Remark 2026.33. In 1994 the right side of (6.5b) lacked the prefactor $|\lambda|\kappa(\mathrm{inv})D_H^{-1}$, the primes on dv' were undefined, and the proof ended: "From these expressions and the remarks in §0, one may verify the local translation invariance property of Hales' lemma 5.19." The formulas for $\omega_{E_\alpha(u)}$ and $\omega_{E_\beta(u)}$ are as in 1994 and were not rechecked; in particular the factor $\alpha(X)$ comes from [La83, Lemma 2.12], which we could not consult. The 1994 source also contains, after `\enddocument`, a never-typeset comparison with the stable subregular germs of H_1 and H_2 (from [SS83] and [Re84]). It is not reproduced; it is in the original files. The 1994 text also says that E is E_α or E_β "depending on T_G" without explaining the choice; by [Ha92a, Ch. VI] it should be the branch attached to the line of the Dynkin curve fixed by the Galois action, but this was not checked. A proof of (X_1) would start from that comparison, with $\Sigma_{H_1}^+$ replaced by the short roots (Remark 2026.38).

§ 7 On the 2-regular germ for G_2

Let $G = G_2$ throughout this section.

By Table 4.3A, the only branch of $\lambda = 0$ on Z with $\beta = 3$ is $E_{\emptyset,\alpha,\delta}$. It is the only branch on Z that can contribute to the 2-regular germ, apart from exceptional divisors (Remark 2026.23). It is F-rational, since for every $\sigma \in \Gamma$, $\sigma(E_{\emptyset,\alpha,\delta})$ is again a branch with $\beta = 3$ (observation (I) in §4),

so $\sigma(E_{\emptyset,\alpha,\delta}) = E_{\emptyset,\alpha,\delta}$. To satisfy (5.1)–(5.4), one may use §2 to average the coordinates of $Z_{\emptyset,\alpha,\delta}$ over Γ_T to get F-rational coordinates on this divisor if needed.

The branch $E_{\emptyset,\alpha,\delta}$ is defined by $\lambda = 0$ on the regular open set $Z_{\emptyset,\alpha,\delta} = \{x(\delta) \neq 0, w(\delta) \neq 0, z(\beta) \neq 0\} \subset Z$, with generators $z(\alpha)$, $z(\beta)$, $w(\gamma)$, $w(\delta)$, $x(\delta)$, $x(\varepsilon)$, $x(\zeta)$. The divisor $\lambda = 0$ occurs on this patch as $z(\alpha) = 0$ with multiplicity 2. On $E_{\emptyset,\alpha,\delta}$, $\lambda = x(\alpha) = x(\beta) = z(\alpha) = x(\gamma) = w(\varepsilon) = w(\zeta) = 0$.

By [Ha92a, Ch. VI, §2], only $\theta = 1$ contributes in Formula 0.6(a). Assuming the proviso of Lemma 0.8, the contribution of the 2-regular germ to the κ-orbital integral is $\kappa(\mathrm{inv}(\gamma_G^0, \gamma_{G,\lambda}))|D_H(\gamma_{H,\lambda})|^{-1}$ times

$$|\lambda|^2 PV \int_{E_{\emptyset,\alpha,\delta}(F)} \pi^*(f)\kappa_1(t_*)|\omega_{E_{\emptyset,\alpha,\delta}}| = |\lambda|^2 \int_{\mathscr{O}_2(F)} f \int_{E_{\emptyset,\alpha,\delta}(u)(F)} \kappa_1(t_*)|\omega_{E_{\emptyset,\alpha,\delta}(u)}||\omega_{\mathscr{O}_2}|. \tag{7.1}$$

Theorem 7.2 (conditional)**.** *Let T_G be an elliptic Cartan subgroup of G_2 such that Γ acts on T_G through a subgroup $\langle s_\alpha, s_\zeta \rangle$ of Ω (up to conjugacy), and let $\kappa \neq 1$, so that $H_{T_G,\kappa} = H_2 = (SL(2) \times SL(2))/\{\pm 1\}$. Assume Lemma 5.14, the proviso of Lemma 0.8, and Hypothesis* (X_2) *for (T_G, κ). Then Conjecture 0.0A holds for (T_G, κ) and the 2-regular class $\mathscr{O}_2$: the germ $\Gamma_{\mathscr{O}_2}^{T_G,\kappa}$, normalized by the transfer factor, is a constant multiple of the stable germ of the trivial class of H_2.*

Remark 2026.34. The 1994 statement had the same defects as Theorem 6.1 (Remark 2026.31), and its proof ended with Lemma 5.19. The quadratic tori, whose κ also give H_2 (Table A.6), are not treated.

Proof of Theorem 7.2. By [Sh72] or [Ro80] the stable germ of the trivial class of H_2 is a non-zero constant. So it suffices to show that the G_2 side, as a function of the tangent direction X, is constant; by (7.1) and the argument at the end of §6 that is Hypothesis (X_2). We record what the 1994 manuscript computed toward (X_2).

Let $u = \theta_{\alpha+\beta}(1) \in G_2$. Computing $\mathrm{Ad}(\mathfrak{g}_2)(X_{\alpha+\beta})$ shows that u^{G_2} is 8-dimensional, so $\mathscr{O}_2^{st} = u^{G_2}$ is the class $\tilde{A}_1$. By §A.7 it contains a single F-class, so it is stable. Analogously to Lemma 6.4, the coordinates of $\prod_{\sigma \in \Sigma_G - \Sigma_{H_2}} U_\sigma$ give local coordinates on $\mathscr{O}_2$, where $\Sigma_{H_2}^+ = \{\alpha, \zeta\}$. Put $\Sigma_{\mathscr{O}_2} = \Sigma_G^+ - \Sigma_{H_2}^+ = \{\beta, \gamma, \delta, \varepsilon\}$. Local coordinates on the fiber $E_{\emptyset,\alpha,\delta}(u)$ may be chosen from the remaining coordinates on $E_{\emptyset,\alpha,\delta}$: $z(\beta)$, $x(\zeta)$, $v(\alpha)$, and $v(\zeta)$.

On $E_{\emptyset,\alpha,\delta}$ the residue form, in the normalization of (8.1), is

$$\omega_{E_{\emptyset,\alpha,\delta}} = \frac{\omega_Z}{\lambda^3 \frac{dz(\alpha)}{z(\alpha)}} = \frac{x(\delta)}{w(\delta)^2 z(\beta)^2}\, dz(\beta) \wedge dw(\gamma) \wedge dw(\delta) \wedge dx(\delta) \wedge dx(\varepsilon) \wedge dx(\zeta),$$

times the dv's. We use the form on $\mathscr{O}_2$

$$\begin{aligned} \omega_{\mathscr{O}_2} &:= \lim_{\lambda \to 0} \frac{\left|\bigwedge_{\sigma \in \Sigma_{\mathscr{O}_2}} dx(\sigma) \wedge dv(\sigma)\right|}{|\lambda|^{3/2}} \\ &= \frac{|x(\delta)|^{1/2}}{|w(\delta)|^{3/2}|z(\beta)|^{3/2}}\, dw(\gamma)\, dw(\delta)\, dx(\delta)\, dx(\varepsilon)\, dv(\beta)\, dv(\gamma)\, dv(\delta)\, dv(\varepsilon), \end{aligned} \tag{7.3}$$

where the limit is taken along $E_{\emptyset,\alpha,\delta}$ (the exponent $3/2$ is the order of vanishing of $x(\beta)x(\gamma)$ in λ there). On $E_{\emptyset,\alpha,\delta}(u)$,

$$\omega_{E_{\emptyset,\alpha,\delta}(u)} = \frac{|x(\delta)|^{1/2}}{|w(\delta)|^{1/2}|z(\beta)|^{1/2}}\, dz(\beta)\, dx(\zeta)\, dv(\alpha)\, dv(\zeta). \tag{7.4}$$

One must rewrite this as $\omega_{E_{\emptyset,\alpha,\delta}(u)} = |p_{\mathscr{O}_2,T_G}(X)|\,\omega'_{E_{\emptyset,\alpha,\delta}(u)}$, where $p_{\mathscr{O}_2,T_G}(X)$ is a homogeneous polynomial in X and $\omega'_{E_{\emptyset,\alpha,\delta}(u)}$ does not depend on X. Using (1.22), the 1994 text found

$$p_{\mathscr{O}_2,T_G}(X) = \prod_{\sigma\in\Sigma^+_{H_2}} \sigma(X)^{f_\sigma}, \qquad \sum_\sigma f_\sigma = 2. \tag{7.5}$$

To prove (X_2) one must compute the exponents f_σ, and show that $|p_{\mathscr{O}_2,T_G}(X)|$, $\Delta_0(X)$ and the κ-factors combine to a function of X that is constant near 0. This has not been done. □

Remark 2026.35. In 1994, (7.3) was written with six differentials $dz(\alpha)\,dw(\gamma)\,dw(\delta)$, $dv(\alpha)\,dv(\gamma)\,dv(\delta)$ for an 8-dimensional class, and with "$\Sigma_{\mathscr{O}_3}$" in place of $\Sigma_{\mathscr{O}_2}$. Its coefficient is correct. The coefficient of (7.4) was printed as $|x(\delta)|^{5/2}|w(\delta)|^{7/2}|z(\beta)|^{-1/2}$. The corrected values were computed symbolically (script `g2_patch_jacobians.py`, §7 checks); the normalization of the limit in (7.3) is as in 1994 and has not been justified. The 1994 proof concluded: "The translation invariance of this factor (7.5) follows from the above expression for $|p_{\mathscr{O}_2,T_G}(X)|$ as a product over roots in H_2. By continuity, $\kappa(t_*)$ satisfies the local translation invariance property as well … the theorem follows … from lemma 5.19." A product of roots of H_2 is not translation invariant under any non-zero Y, since those roots span $\mathfrak{t}^*$; and the lemma does not apply (Remark 2026.29).

The transfer factor for the biquadratic tori. The 1994 formula (0.1) uses only two of the factors of the Langlands–Shelstad transfer factor [LS87, §§2–3], $\Delta = \Delta_I\Delta_{II}\Delta_{III_1}\Delta_{III_2}\Delta_{IV}$: namely Δ_{III_1}, which gives $\kappa(\mathrm{inv}(\gamma_G^0,\gamma_G))$, and $\Delta_{IV} = |D_G(\gamma_G)|/|D_H(\gamma_H)|$ (in the notation of §0, where D_G is a product over positive roots). Near the identity the remaining factors make up the term $\Delta_0(X)$ of §0 and of Hypothesis (X_r). We compute it for the tori of Theorem 7.2.

Let T be as in Theorem 7.2, with $\Gamma_T = \{1, s_\alpha, s_\zeta, -1\}$ (note $s_\alpha s_\zeta = -1$, since $\alpha \perp \zeta$). Let $K = F(\sqrt{d_\alpha}, \sqrt{d_\zeta})$ be its splitting field, where $F(\sqrt{d_\alpha})$ is the fixed field of the element acting by s_ζ and $F(\sqrt{d_\zeta})$ that of the element acting by s_α, and let $L = F(\sqrt{d_\alpha d_\zeta})$ be the fixed field of the element acting by -1. For $Y \in \mathfrak{t}(F)$ the Galois group acts on root values by $\sigma(\rho(Y)) = (w_\sigma\rho)(Y)$, so

$$\alpha(Y) = \sqrt{d_\alpha}\,u, \qquad \zeta(Y) = \sqrt{d_\zeta}\,v, \qquad u,v \in F,$$

and every $(u,v) \in F^2$ occurs. Since $\beta = (\zeta - 3\alpha)/2$,

$$\beta(Y) = \tfrac{1}{2}(\zeta-3\alpha)(Y), \quad \gamma(Y) = \tfrac{1}{2}(\zeta-\alpha)(Y), \quad \delta(Y) = \tfrac{1}{2}(\zeta+\alpha)(Y), \quad \varepsilon(Y) = \tfrac{1}{2}(\zeta+3\alpha)(Y),$$

and we put

$$P(Y) := \beta(Y)\gamma(Y)\delta(Y)\varepsilon(Y) = \tfrac{1}{16}(d_\zeta v^2 - d_\alpha u^2)(d_\zeta v^2 - 9d_\alpha u^2) \in F. \tag{7.6}$$

For Y regular, $P(Y) \neq 0$, and $P(cY) = c^4P(Y)$ for $c \in F^\times$.

Proposition 7.6. *Suppose the residue characteristic of F is at least 5. For regular $Y \in \mathfrak{t}(F)$ sufficiently close to 0:*

- *(a) Δ_I is a constant, and $\Delta_{III_2}(\exp Y) = 1$;*
- *(b) $\Delta_{II}(\exp Y) = c\cdot(P(Y), d_\alpha)_F$, where $(\ ,\ )_F$ is the quadratic Hilbert symbol and $c = \pm 1$ depends only on the a-data.*

Hence $\Delta_0(Y) = \Delta_I\Delta_{II}\Delta_{III_2}(\exp Y) = c'\cdot(P(Y),d_\alpha)_F$ with c' independent of Y. It is homogeneous of degree 0.

Proof. (a) Δ_I depends only on T and the a-data. Δ_{III_2} is a continuous character of the compact group $T(F)$, so it is trivial on a neighborhood of 1.

(b) By definition $\Delta_{II}(\gamma) = \prod_\rho \chi_\rho\big((\rho(\gamma)-1)/a_\rho\big)$, the product over representatives of the Γ_T-orbits of roots outside $\Sigma_H = \{\pm\alpha, \pm\zeta\}$. There are two such orbits, $\{\pm\beta, \pm\varepsilon\}$ and $\{\pm\gamma, \pm\delta\}$, since $s_\alpha\beta = \varepsilon$, $s_\zeta\beta = -\varepsilon$, $s_\alpha\gamma = \delta$, $s_\zeta\gamma = -\delta$. In each orbit the stabilizer of a root is trivial and the stabilizer of the pair $\{\rho, -\rho\}$ is $\{1, -1\}$. So $\rho = \beta, \gamma$ is symmetric with $F_\rho = K$ and $F_{\pm\rho} = L$. The a-data satisfy $\sigma_{-1}(a_\rho) = a_{-\rho} = -a_\rho$, and χ_ρ is a character of $K^\times$ whose restriction to $L^\times$ is the character $\omega_{K/L}$ of the quadratic extension K/L. Now $\rho(\exp Y) - 1 = \rho(Y)(1+\eta)$ with $|\eta| \le |\rho(Y)|$, and $\sigma_{-1}(\rho(Y)) = -\rho(Y)$, so $\rho(Y)/a_\rho \in L^\times$. For Y small, $\chi_\rho(1+\eta) = 1$. Hence

$$\Delta_{II}(\exp Y) = \omega_{K/L}\Big(\frac{\beta(Y)\gamma(Y)}{a_\beta a_\gamma}\Big) = \Big(\frac{\beta(Y)\gamma(Y)}{a_\beta a_\gamma}, d_\alpha\Big)_L,$$

since $K = L(\sqrt{d_\alpha})$. As $d_\alpha \in F$, the projection formula for the Hilbert symbol [Ser79, Ch. XIV] gives $(x, d_\alpha)_L = (N_{L/F}x, d_\alpha)_F$. The non-trivial element of $\mathrm{Gal}(L/F)$ acts on root values through s_α, which sends β, γ to ε, δ, so $N_{L/F}(\beta(Y)\gamma(Y)) = P(Y)$. Finally $c = (N_{L/F}(a_\beta a_\gamma), d_\alpha)_F$. □

The root-theoretic steps (the orbits, the stabilizers and (7.6)) are checked by the script `g2_transfer_factor_biquadratic.py`.

Example 7.7. Let F have residue characteristic $p \ge 5$, uniformizer ϖ, and a non-square unit ε.

(a) If one of $F(\sqrt{d_\alpha})$, $F(\sqrt{d_\zeta})$ is unramified over F, then $(P(Y), d_\alpha)_F = 1$ for all regular Y, so Δ_0 is constant. For example, if $d_\alpha = \varepsilon$ and $d_\zeta \in \varpi\mathcal{O}_F^\times$, then in each factor of (7.6) the two terms have valuations of different parity, so $P(Y)$ has even valuation, and $(P(Y), \varepsilon)_F = (-1)^{v(P(Y))} = 1$. If instead $d_\zeta = \varepsilon$, then $P(Y)$ is a square times $1 + \varpi(\cdots)$, hence a square.

(b) If both are ramified (so L/F is unramified), say $d_\alpha = \varpi$ and $d_\zeta = \varepsilon\varpi$, then $(P(Y), d_\alpha)_F = 1$ when $v(u) \ne v(v)$. When $v(u) = v(v)$ it is the Legendre symbol of the residue of $(\varepsilon\bar v^2 - \bar u^2)(\varepsilon\bar v^2 - 9\bar u^2)$, where $\bar u, \bar v$ are the residues of the unit parts of u and v. For $p = 5$ this is -1 for every $\bar u, \bar v$. For $7 \le p \le 23$ it takes both values (script).

So Δ_0 is *not* constant in the direction of approach when both quadratic subfields of K are ramified.

Corollary 7.8. *Let T, κ and $H = H_2$ be as in Theorem 7.2. Assume Lemma 5.14 and the proviso of Lemma 0.8. Then Hypothesis* (X_2) *holds for (T, κ) if and only if, for $u \in \mathcal{O}_2(F)$ and regular $X \in \mathfrak{t}(F)$ near 0,*

$$J(X) := \int_{E_{\emptyset,\alpha,\delta}(u)(F)} \kappa_1(t_*)\,|\omega_{E_{\emptyset,\alpha,\delta}(u)}| = C \cdot |\alpha(X)\zeta(X)| \cdot \big(P(X), d_\alpha\big)_F, \tag{7.7}$$

with C independent of X and of u.

Proof. With the full transfer factor, (X_2) says that the $\mathcal{O}_2$-germ of $\Delta\Phi^{T,\kappa}$ is a constant c_0: the unnormalized stable germ of the trivial class of H is constant. The germ $\Gamma^{T,\kappa}_{\mathcal{O}_2}$ of (0.1), which uses only $\Delta_{III_1}\Delta_{IV}$, is then $c_0\Delta_0(X)^{-1}$. Since $\mathcal{O}_2$ is a single F-class (§A.7), Formula 0.6 and (7.1) give

$$|D_H(\gamma_{\lambda,H})|\, c_0\Delta_0(X)^{-1}\mu_{\mathcal{O}_2}(f) = \kappa(\mathrm{inv}(\gamma_G^0, \gamma_\lambda))\,|\lambda|^2 \int_{\mathcal{O}_2(F)} f(u)J(X)\,|\omega_{\mathcal{O}_2}|$$

for all f. Here $\kappa(\mathrm{inv}(\gamma_G^0, \gamma_\lambda))$ is constant on $T(F)$, and $|D_H(\exp\lambda X)| = |\lambda|^2|\alpha(X)\zeta(X)|$ for λ small. Since $\Delta_0(X)^{-1} = \Delta_0(X)$ is given by Proposition 7.6, (7.7) follows, and conversely. □

So (X_2) splits into two explicit statements about the fiber integral $J(X)$. First, its absolute scaling: by (7.4)–(7.5), the 1994 text asserts $|J(X)| \propto |\alpha(X)|^{f_\alpha}|\zeta(X)|^{f_\zeta}$ with $f_\alpha + f_\zeta = 2$. Corollary 7.8 requires $f_\alpha = f_\zeta = 1$. Second, its sign: $J(X)$ must carry the factor $(P(X), d_\alpha)_F$. This factor involves the roots $\beta, \gamma, \delta, \varepsilon$ *outside* H, and by Example 7.7 it is not constant in general. The form (7.4) is positive, so this sign has to come from $\kappa_1(t_*)$ and from the F-structure of the fiber, which depends on X through (1.22) and the Galois action of §2.

Remark 2026.36 (status of (X_2)). Proposition 7.6, Example 7.7 and Corollary 7.8 are new and are checked at level 1 (by hand, and by the script for the root data and the Hilbert symbols over $\mathbb{Q}_p$); the projection formula and the definitions of the factors $\Delta_I, \dots, \Delta_{IV}$ are standard [Ser79], [LS87]. They show that the 1994 strategy could not have worked even if Lemma 5.19 applied: the X-dependence of Δ_0 is through the roots outside H, and it is not translation invariant. The display of Δ in (0.1) omits $\Delta_I\Delta_{II}\Delta_{II_2}$; for the tori of Theorem 7.2 this matters exactly in the case of Example 7.7(b). To finish (X_2) one still needs three inputs, none of which is available in the 1994 text: (i) the exact coordinate relations (1.22), whose constants $c_1, \dots, c_7$ and omitted terms are not given (they can in principle be computed from Lemma 1.21 in the explicit model of §A.4); (ii) the cocycle t_* on $E_{\emptyset,\alpha,\delta}$, from [Ha92a, Ch. I, Cor. 5.4 and 5.7], which we could not consult; (iii) the Galois action on the coordinates, which §2 gives only schematically (Remark 2026.12). By Corollary 8.8, (7.7) is true given Lemma 5.14 and the proviso of Lemma 0.8. So it predicts $f_\alpha = f_\zeta = 1$ in (7.5), and the sign $(P(X), d_\alpha)_F$ for the κ-weighted integral over the fiber. Either prediction can be tested once (i)–(iii) are available.

§ 8 On the 3-regular germ for G_2

Let $G = G_2$ throughout this section.

By Table 4.3A, the only branch of $\lambda = 0$ on Z with $\beta = 4$ is $E_{\emptyset,\beta,\zeta}$. Apart from exceptional divisors (Remark 2026.23) it is the only branch that can contribute to the 3-regular germ. It is F-rational, since $\sigma(E_{\emptyset,\beta,\zeta})$ is again a branch with $\beta = 4$, by observation (I) of §4 and Proposition 3.4A. To satisfy (5.1)–(5.4), one may use §2 to average the coordinates of $Z_{\emptyset,\beta,\zeta}$ (or suitable $\mathbb{Z}$-linear combinations of them) over Γ_T.

The branch $E_{\emptyset,\beta,\zeta}$ is defined by $\lambda = 0$ on the regular open set $Z_{\emptyset,\beta,\zeta} = \{x(\zeta) \neq 0, w(\zeta) \neq 0, z(\alpha) \neq 0\} \subset Z$. In the local coordinates $z(\alpha)$, $z(\beta)$, $w(\gamma)$, $w(\delta)$, $w(\varepsilon)$, $w(\zeta)$, $x(\zeta)$ on $Z_{\emptyset,\beta,\zeta}$, where $\lambda = x(\zeta)z(\alpha)^3 z(\beta)^2/w(\zeta)$, the form ω_E on $E = E_{\emptyset,\beta,\zeta}$ is

$$\omega_{E_{\emptyset,\beta,\zeta}} = \frac{\omega_Z}{\lambda^4 \frac{dz(\beta)}{z(\beta)}} = \frac{x(\zeta)^2}{w(\zeta)^3 z(\alpha)^2}\, dz(\alpha) \wedge dw(\gamma) \wedge dw(\delta) \wedge dw(\varepsilon) \wedge dw(\zeta) \wedge dx(\zeta), \tag{8.1}$$

times the dv's, using (4.2A). The divisor $\lambda = 0$ occurs on $Z_{\emptyset,\beta,\zeta}$ as $z(\beta) = 0$ with multiplicity 2. On $E_{\emptyset,\beta,\zeta}$ the functions λ, $z(\beta)$, $x(\varepsilon)$, $x(\delta)$, $x(\gamma)$, $x(\beta)$ and $x(\alpha)$ vanish.

By §§4–5 and Formula 0.6, assuming the proviso of Lemma 0.8, the contribution of the 3-regular germ to the κ-orbital integral is

$$\kappa(\mathrm{inv}(\gamma_G^0, \gamma_{G,\lambda}))|D_H(\gamma_{H,\lambda})|^{-1} \cdot |\lambda|^3 \sum_\theta \theta(\lambda) F_1(\theta, 4, \pi^*(f)\kappa_\theta(t_*)),$$

and only $\theta = 1$ contributes [Ha92a, Ch. VI, §2]. By Formula 0.6(b),

$$\int_{E_{\emptyset,\beta,\zeta}(F)} \pi^*(f)\kappa_1(t_*)|\omega_{E_{\emptyset,\beta,\zeta}}| = \int_{\mathscr{O}_3(F)} f \int_{E_{\emptyset,\beta,\zeta}(u)(F)} \kappa_1(t_*)|\omega_{E_{\emptyset,\beta,\zeta}(u)}||\omega_{\mathscr{O}_3}|, \tag{8.2}$$

which is a multiple of the unipotent orbital integral of f over the 3-regular class, which is stable (§A.7).

Theorem 8.3 (conditional)**.** *Let T_G be an elliptic Cartan subgroup of G_2 such that Γ acts on T_G through a cyclic subgroup of Ω of order three, and let $\kappa \neq 1$, so that κ is cubic and $H_{T_G,\kappa} = H_1 = PGL(3)$. Assume Lemma 5.14, the proviso of Lemma 0.8, and Hypothesis (X_3) for (T_G,κ). Then Conjecture 0.0A holds for (T_G,κ) and the 3-regular class $\mathscr{O}_3$: the germ $\Gamma_{\mathscr{O}_3}^{T_G,\kappa}$, normalized by the transfer factor, is a constant multiple of the stable germ of the trivial class of H_1.*

Remark 2026.37. As for Theorems 6.1 and 7.2. The tori with Galois image S_3 generated by short reflections, whose κ also give H_1 (Table A.6), are not treated.

Proof of Theorem 8.3. By [Ro80], the stable germ of the trivial class of H_1, $\Gamma_1^{T_H,st}(\gamma_H)$, is a non-zero constant. So, as in §7, the theorem is Hypothesis (X_3), by (8.2). We record what the 1994 manuscript computed toward (X_3).

Let $u = \theta_\beta(1) \in G_2$. Computing $\mathrm{Ad}(\mathfrak{g}_2)(X_\beta)$ shows that u^{G_2} is 6-dimensional, so $\mathscr{O}_3^{st} = u^{G_2}$ is the class A_1, and by §A.7 it is stable.

To get local coordinates on the fiber $E_{\emptyset,\beta,\zeta}(u)$ over $u \in \mathscr{O}_3$, take the local coordinates on $E_{\emptyset,\beta,\zeta}$ and divide $\omega_{E_{\emptyset,\beta,\zeta}}$ by an invariant measure $\omega_{\mathscr{O}_3}$ on $\mathscr{O}_3$. Analogously to Lemma 6.4, $\prod_{\sigma\in\Sigma_G-\Sigma_L} U_\sigma$ gives local coordinates on $\mathscr{O}_3$, where $\Sigma_L^+ = \{\beta,\varepsilon,\zeta\}$ are the positive roots of the pseudo-Levi subgroup $L = SL(3) \subset G_2$ generated by the long roots. Put $\Sigma_{\mathscr{O}_3} = \Sigma_G^+ - \Sigma_L^+ = \{\alpha,\gamma,\delta\}$. We use the form on $\mathscr{O}_3$

(8.4)
$$|\omega_{\mathscr{O}_3}| := \lim_{\lambda\to 0} \frac{\left|\bigwedge_{\sigma\in\Sigma_{\mathscr{O}_3}} dx(\sigma)\wedge dv(\sigma)\right|}{|\lambda|^2} = \frac{2|x(\zeta)|}{|w(\zeta)||z(\alpha)|^2}\, dz(\alpha)\, dw(\gamma)\, dw(\delta)\, dv(\alpha)\, dv(\gamma)\, dv(\delta).$$

On the generic fiber $E_{\emptyset,\beta,\zeta}(u)$, $u \in \mathscr{O}_3$,

$$|\omega_{E_{\emptyset,\beta,\zeta}(u)}| = \frac{|x(\zeta)|}{2|w(\zeta)|^2}\, dw(\varepsilon)\, dw(\zeta)\, dx(\zeta)\, dv(\beta)\, dv(\varepsilon)\, dv(\zeta). \tag{8.5}$$

One must rewrite this as $\omega_{E_{\emptyset,\beta,\zeta}(u)} = |p_{\mathscr{O}_3,T_G}(X)|\, \omega'_{E_{\emptyset,\beta,\zeta}(u)}$ with ω' independent of X. Then

$$\begin{aligned} |D_H(\gamma_{\lambda,H})| \sum_{\mathscr{O}\subset\mathscr{O}_3^{st}} \mu_{\mathscr{O}}(f)\Gamma_{\mathscr{O}}^{T_G,\kappa}(\gamma_\lambda) &= |\lambda|^3\kappa(\mathrm{inv}(\gamma_G^0,\gamma_\lambda)) \int_{E_{\emptyset,\beta,\zeta}(F)} \pi^*(f)\kappa_1(t_*)|\omega_{E_{\emptyset,\beta,\zeta}}| \\ &= |\lambda|^3\kappa(\mathrm{inv}(\gamma_G^0,\gamma_\lambda))|p_{\mathscr{O}_3,T_G}(X)| \int_{\mathscr{O}_3(F)} f \int_{E_{\emptyset,\beta,\zeta}(u)(F)} \kappa(t_*)|\omega'_{E_{\emptyset,\beta,\zeta}(u)}||\omega_{\mathscr{O}_3}|, \end{aligned} \tag{8.6}$$

where $\omega_{\mathscr{O}_3}$ is normalized consistently with the Langlands–Shelstad transfer factor as in [Ha89]. Using (1.22), the 1994 text found

$$p_{\mathscr{O}_3,T_G}(X) = \prod_{\sigma\in\Sigma_{H_1}^+} \sigma(X)^{e_\sigma}, \qquad \sum_\sigma e_\sigma = 3. \tag{8.7}$$

To prove (X_3) one must compute the e_σ and show that $|p_{\mathscr{O}_3,T_G}(X)|$, $\Delta_0(X)$ and the κ-factors combine to a function of X that is constant near 0. This has not been done. □

Remark 2026.38 (the roots of H_1). The 1994 text took $\Sigma_{H_1}^+ = \{\beta,\varepsilon,\zeta\}$, the long roots. That is the root system of the pseudo-Levi subgroup $L = SL(3)$, which is the right object for coordinates on $\mathscr{O}_3$ (and is used for that above). The endoscopic group $H_1 = PGL(3)$ has, transported to T_G, the roots a of G with $\kappa(a^\vee) = 1$. For the cubic torus these are the short roots: $\Sigma_{H_1}^+ = \{\alpha,\gamma,\delta\}$ (§A, Table A.6).

So (8.7) and Lemma 4.4 use the short roots, as do D_{H_1} and the transfer factor. In 1994, (8.7) was derived with the long roots, so the exponents e_σ have to be recomputed.

Remark 2026.39. Corrections to the 1994 formulas, all checked symbolically (script `g2_patch_jacobians.py`): in (8.1) the coefficient was printed $x(\zeta)^2 w(\zeta)^4 w(\delta)^{-7} z(\alpha)^{-2}$, and the restatement of $|\omega_E|$ before (8.2) had the denominator $|w(\delta)|^7$; in (8.4) the coefficient was $|x(\zeta)|/(|w(\zeta)||z(\alpha)|)$; in (8.5) it was $|x(\zeta)||w(\zeta)|^5/(|z(\alpha)||w(\delta)|^7)$; in (8.6) the power was $|\lambda|^2$, while the $\beta = 4$ term carries $|\lambda|^3$.

Corollary 8.8. *Assume Lemma 5.14 and the proviso of Lemma 0.8. Then Hypotheses* (X_1), (X_2) *and* (X_3) *hold for the pairs* (T_G, κ) *of Theorems 6.1, 7.2 and 8.3.*

Proof. Under these assumptions, the argument at the end of §6 identifies $I_E(X,u)$, for u in a class $\mathscr{O} \subset \mathscr{O}_r^{st}(F)$, with $\Gamma_{\mathscr{O}}^{T_G,\kappa}(\exp X)$ up to normalizing constants independent of X. Conjecture 0.0A is a theorem [Wa97], [Ng10], [GH16] (Remark 2026.1), and expresses this germ as a combination of stable germs of r-regular classes of H. □

Remark 2026.40. Corollary 8.8 shows that the hypotheses (X_r) are true, but it does not give the geometric proof that the 1994 manuscript aimed at. Its proof relies on the identification of normalizations in (0.1), Formula 0.6 and [GH16], which has not been checked in detail (level 3 in the sense of the revision log).

§ 9 A CONJECTURE

We conclude with a conjecture suggested by the argument above. It is restated as a germ identity, with Hales' reduction lemma replaced by an explicit hypothesis of the type (X_r).

Conjecture 9.1. *Let T_G be an elliptic Cartan subgroup of a connected, simply connected, F-quasi-split group G. Let $\mathscr{O}$ be an r-regular unipotent class of $G(F)$, let κ be a character of $H^1(F,T_G)$, and let $H = H_{\kappa,T_G}$ be the corresponding endoscopic group of G* [La79]. *Assume:*

1. *there is an essential, admissible, F-rational $\mathscr{O}$-divisor $E = E_{S_1,S_2,\sigma}$ such that, for each $u \in \mathscr{O}$, the Igusa form on the fiber satisfies $\omega_{E(u)} = p_{\mathscr{O},T_G}(X)\,\omega'_{E(u)}$, where $\omega'_{E(u)}$ is independent of X and $p_{\mathscr{O},T_G}(X) = \prod_{\sigma\in\Sigma_{\mathscr{O}}^+}\sigma(X)$ for some $\Sigma_{\mathscr{O}}^+ \subset \Sigma_G^+$;*
2. *the function $X \mapsto \kappa(\mathrm{inv}(\gamma_G^0, \exp X))\Delta_0(X)|p_{\mathscr{O},T_G}(X)|$ is a constant multiple of the X-dependence of the stable germ of an r-regular class of H (the analogue of (X_r));*
3. *$b(E) \neq 0$ in the F-rational coordinates of $Y''(B_+)$;*
4. *$e(\theta, r+1) \leq 1$, and $e(\theta, r+1) = 0$ unless $\theta = 1$;*
5. *there is only one r-regular unipotent class $\mathscr{O}$ of $G(\overline{F})$, and it is either Richardson or endoscopically Richardson;*
6. *there is only one r-regular unipotent class $\mathscr{O}_H$ of $H(\overline{F})$.*

Then Conjecture 0.0A holds between the r-regular classes of G and of H.

Remark 2026.41. In 1994, hypothesis (2) read "the function $p_{\mathscr{O},T_G}$ in (1) satisfies the hypotheses to Hales' reduction lemma", which cannot hold when H is semisimple of the same rank as G (Remark 2026.29); and the conclusion included a transfer statement for f whose support meets no unipotent class other than those stably conjugate to $\mathscr{O}$, which do not exist (Remark 2026.31). "Endoscopically Richardson" is not defined in the manuscript. Since Conjecture 0.0A is now known (Remark 2026.1), the interest of Conjecture 9.1 lies in its mechanism, not its conclusion. The

variant source `g2new9d.tex` replaces (4) by $e(\theta, r+1) \leq 1$ and (6) by the requirement that $\mathcal{O}$ correspond, under an endoscopic correspondence of unipotent classes attributed to Assem, Hales and Spaltenstein, to the only s-regular class $\mathcal{O}_H$ of H; it cites two works, [A] and [H4], that are not in its bibliography (see the revision log).

There appear to be several low-dimensional cases which satisfy the hypotheses of this conjecture. We mention one: $G = SL(4)$, $\mathcal{O} = \mathcal{O}_{(2,2)}$ the 2-regular class of the partition $4 = 2+2$, $H = GL(2,E)^1$, where E/F is quadratic and the superscript 1 denotes the kernel of $N_{E/F} \circ \det : GL(2,E) \to F^\times$, and $\mathcal{O}_H$ the trivial class of $H(F)$, which is 2-regular in H. The dimensions agree: $\dim \mathcal{O}_{(2,2)} = 8 = \dim SL(4) - \dim H$. The hypotheses have not been checked in this case.

APPENDIX A. BACKGROUND ON G_2

The first four sections of this appendix have been taken from an unpublished joint paper with D. Bump.

The purpose of this appendix is to collect some foundational material on the exceptional group G_2, then to use this material to compute the cuspidal endoscopic groups of G_2, the groups $H^1(\Gamma, T)$, and the F-classes in the stable 2- and 3-regular unipotent orbit.

In the first few sections, we shall be concerned with the phenomenon known as "triality", which yields an embedding of G_2 into the double cover Spin(8) of the split form of SO(8). We shall determine the triality map explicitly at the root level, and also use it to give an explicit Chevalley basis for the Lie algebra $\mathfrak{g}_2$, which should be useful in other calculations.

Acknowledgement: We thank Becci Davies for coding the root diagrams (in $\mathcal{AMS}$-TEX; they are redrawn here).

Background on G_2 and SO(8).

A.1. SO(8). Let J denote the 8×8 matrix with 1's on the skew–diagonal and 0's elsewhere:

$$J := \begin{pmatrix} 0 & & 1 \\ & \iddots & \\ 1 & & 0 \end{pmatrix}.$$

Define the 8×8 **special orthogonal group** by

$$SO(8) := \{A \in GL(8) \mid A \cdot J \cdot {}^tA = J\}^o,$$

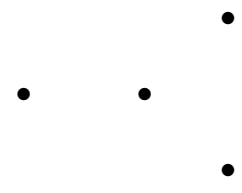
so that $SO(8)$ is semi–simple, connected, neither simply connected nor adjoint, non–compact, and split over $\mathbb{Q}$. Its maximal $\mathbb{Q}$-split torus consists of all matrices of the form

$$\text{(A.1)} \qquad \begin{pmatrix} A & 0 \\ 0 & B \end{pmatrix}, \text{ with } A := \begin{pmatrix} a_1 & & & \\ & a_2 & & \\ & & a_3 & \\ & & & a_4 \end{pmatrix}, \; B := \begin{pmatrix} a_4^{-1} & & & \\ & a_3^{-1} & & \\ & & a_2^{-1} & \\ & & & a_1^{-1} \end{pmatrix}.$$

The compact real form of $SO(8)$ is obtained, of course, by replacing J by the 8×8 identity matrix.

The simply–connected covering group of $SO(8)$ is $SO(8)_{sc} =$ Spin(8), which has center $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$, and since $SO(8)$ has center $\mathbb{Z}/2\mathbb{Z}$, Spin(8) is a 2-fold cover of $SO(8)$ and a 4-fold cover

of $PSO(8)$. The outer automorphisms of Spin(8) are in one–to–one correspondence with the automorphisms of the Dynkin diagram for $SO(8)$:



The Lie algebra

$$so(8) := \{A \in M_8 \mid AJ + J^t A = 0\}$$

may be expressed as the Lie algebra of 8×8 matrices of the form

$$A = \begin{pmatrix} t_1 & a_{12} & a_{13} & a_{14} & a_{15} & a_{16} & a_{17} & 0 \\ a_{21} & t_2 & a_{23} & a_{24} & a_{25} & a_{26} & 0 & -a_{17} \\ a_{31} & a_{32} & t_3 & a_{34} & a_{35} & 0 & -a_{26} & -a_{16} \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & -a_{71} & -a_{61} & -a_{51} & -a_{41} & -a_{31} & -a_{21} & -t_1 \end{pmatrix}. \tag{A.2}$$

If $i \neq j$ then define E_{ij} to be the matrix A as above with $a_{ij} = 1$ and all other entries = 0. Define E_{ii} to be the matrix A as above with $t_i = 1$ and all other entries = 0. Then

$$\begin{aligned} &\{E_{11}, E_{22}, E_{33}, E_{44}, \\ &\quad E_{12}, E_{13}, \ldots, E_{17}, E_{23}, \ldots, E_{26}, E_{34}, E_{35}, \\ &\quad {}^tE_{12}, {}^tE_{13}, \ldots, {}^tE_{17}, {}^tE_{23}, \ldots, {}^tE_{26}, {}^tE_{34}, {}^tE_{35}\} \end{aligned} \tag{A.3}$$

forms a basis for the 28-dimensional vector space $so(8)$.

A.2. The Cayley algebra and G_2. The **split Cayley algebra** is the 2×2 matrix algebra

$$C := \left\{ \begin{pmatrix} \alpha & a \\ b & \beta \end{pmatrix} \mid \alpha, \beta \in \mathbb{Q},\ a, b \in V \right\}, \tag{A.4}$$

where $V \cong \mathbb{Q}^3$ is the cross product algebra

$$\begin{aligned} V := \{xi + yj + zk \mid x, y, z \in \mathbb{Q},\ & i \times i = j \times j = k \times k = 0, \\ & i \times j = k,\ j \times k = i,\ k \times j = -i,\ j \times i = -k, \text{ etc.}\}. \end{aligned}$$

Define

$$G_2(F) := \mathrm{Aut}_F(C \otimes F), \tag{A.5}$$

where F is any field containing $\mathbb{Q}$. Since Der(C) $\cong$ Lie(Aut C), it follows that the Lie algebra $\mathfrak{g}_2$ may be identified with Der(C). For G_2, there are 2 real forms: one being the non-compact $\mathbb{Q}$–split from above, the other being a compact form associated to a division algebra (see[Ja71, pp. 142-147] or [Ti65]). We have

$$\dim_{\mathbb{Q}} C = 8,\ \dim_{\mathbb{Q}} \mathfrak{g}_2 = 14,$$

and there is a trace bilinear form which induces an embedding

$$G_2 = \mathrm{Aut}(C) \hookrightarrow SO(8).$$

Here SO(8) is regarded as the special orthogonal group associated to the trace form $trace(\beta)$ on C, where

$$\beta(x, y) = \frac{1}{2}(x \cdot y^* + y \cdot x^*), \quad \beta(x, x) = N(x),$$

and where N denotes the norm on C. Regarding G_2 as a subgroup of $SO(8)$, we have the following

Triality theorem A.6 ([Ja71], [SV00])**.** *Given any rotation (proper orthogonal map)*

$$A_0 : C \to C,$$

there are orthogonal maps A_1, A_2 such that, for all $x, y \in C$,

$$A_0(xy) = A_1(x)A_2(y).$$

Moreover, the only other orthogonal maps satisfying this property are $-A_1$, $-A_2$, so

$$A_0 \longmapsto (A_1, A_2)$$

is well–defined, up to signs.

Proof. We will verify that our form of triality is the global equivalent of Jacobson's form [Ja71, p. 8]. Let

$$A_0 = 1 + \delta B_0,$$
$$A_1 = 1 + \delta B_1,$$
$$A_2 = 1 + \delta B_2,$$

in $SO(8, \mathbb{Q})$ (regarded as a group of orthogonal maps with respect to the trace form on C), where $\delta^2 = 0$ and $B_i \in so(8, \mathbb{Q})$. Thus A_0, A_1, A_2 may be regarded as lying in the tangent space to SO(8) at the identity (cf. [Bor91, chapter 0]). Then $A_0(xy) = A_1(x)A_2(y)$ is equivalent to

$$B_0(xy) = B_1(x)y + xB_2(y).$$

This is Jacobson's expression. □

Remark 2026.42. In the 1994 text, $k \times j$ was given as i and the form as $\frac{1}{2}(xy^* + y^*x)$, which is not scalar-valued (for $x = 1$, $y = i$ it equals $-i$); both are corrected above. The theorem was stated for arbitrary orthogonal maps; it holds for rotations. The argument above checks only the infinitesimal identity, which does not imply the global statement; for a proof see [SV00, Ch. 3], which we have not checked against this formulation.

Corollary A.7. *The map*

$$SO(8, F) \to PSO(8, F),$$
$$A_0 \longmapsto \quad \pm A_1,$$

defines a projective representation of SO(8), i.e., belongs to PSO(8). In particular, it induces an outer automorphism of degree 3 (the "triality" automorphism) corresponding to a rotation of the Dynkin diagram of $PSO(8, F)$.

Remarks (A.8) If $A_0 \in G_2 = \mathrm{Aut}(C)$ then triality sends $A_0 \longmapsto A_0$. Conversely, if triality sends $A_0 \longmapsto A_0$ then $A_0 \in G_2$, so G_2 is the subgroup fixed by triality. In general, triality sends (up to signs)

$$A_0 \longmapsto A_1,\ A_1 \longmapsto A_2,\ A_2 \longmapsto A_0.$$

This way we can explicitly construct an outer automorphism of $SO(8)$.

(A.9) Triality can be described explicitly on the standard Borel subgroup B of PSO(8): triality cyclically permutes the "root groups" U_α, defined below, in the same way as it permutes the simple roots via its action on the Dynkin diagram; it also permutes the embeddings $PSL(2) \hookrightarrow PSO(8)$ associated to the simple roots. The images of the maximal split torus $T \subset PSL(2)$ embedded in $PSO(8)$ generate the maximal split torus in $PSO(8)$ and, of course, the simple root groups generate the unipotent radical of B. Putting these all together explicitly gives the action of triality on B.

A.3. Root systems for G_2 and $SO(8)$. Since the symmetric group S_3 acts on the Dynkin diagram of $SO(8)$ by triality, it acts on its 24 roots. Of these, 6 roots remain fixed under this action, and 18 are permuted cyclically in groups of three. In particular, the roots corresponding to

$$\{E_{17}\},\ \{E_{16}\},\ \{E_{23}\},$$

are fixed and the triples of roots corresponding to

$$\{E_{14},\, E_{15},\, E_{26}\},\ \{E_{13},\, E_{24},\, E_{25}\},\ \{E_{12},\, E_{34},\, E_{35}\},$$

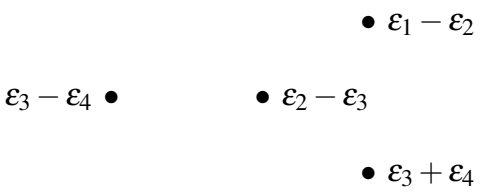
are each permuted by the action of S_3. More explicitly, let us represent a basis of $so(8)$ by

$$\varepsilon_1-\varepsilon_2,\ \varepsilon_2-\varepsilon_3,\ \varepsilon_3-\varepsilon_4,\ \varepsilon_3+\varepsilon_4,$$

where

$$\varepsilon_i : \begin{pmatrix} A & 0 \\ 0 & B \end{pmatrix} \longmapsto a_i, \tag{A.10}$$

in the notation of (A.1), $1 \leq i \leq 4$. Here E_{12} corresponds to $\varepsilon_1-\varepsilon_2$, E_{13} to $\varepsilon_1-\varepsilon_2+\varepsilon_2-\varepsilon_3=\varepsilon_1-\varepsilon_3$, etc. Since $\varepsilon_2-\varepsilon_3$ is connected to the others (since its Killing form inner product with them is non–zero), it corresponds to the center of the Dynkin diagram



Therefore, under the action of S_3, the triality automorphism of so(8) induces a linear map ϕ on the roots such that

$$\phi(\varepsilon_1-\varepsilon_2)=\varepsilon_3-\varepsilon_4,\ \phi(\varepsilon_3-\varepsilon_4)=\varepsilon_3+\varepsilon_4,\ \phi(\varepsilon_3+\varepsilon_4)=\varepsilon_1-\varepsilon_2,\ \phi(\varepsilon_2-\varepsilon_3)=\varepsilon_2-\varepsilon_3. \tag{A.11}$$

Thus

$$\begin{aligned}
\phi(\varepsilon_1) &= \frac{1}{2}(\varepsilon_1+\varepsilon_2+\varepsilon_3-\varepsilon_4),\\
\phi(\varepsilon_2) &= \frac{1}{2}(\varepsilon_1+\varepsilon_2-\varepsilon_3+\varepsilon_4),\\
\phi(\varepsilon_3) &= \frac{1}{2}(\varepsilon_1-\varepsilon_2+\varepsilon_3+\varepsilon_4),\\
\phi(\varepsilon_4) &= \frac{1}{2}(\varepsilon_1-\varepsilon_2-\varepsilon_3-\varepsilon_4).
\end{aligned}$$

Call the induced automorphism on the Lie algebra so(8), ϕ^*. As we've already observed, the Lie subalgebra

$$\{X \in so(8) \mid \phi^*(X) = X\}$$

is the Lie algebra $\mathfrak{g}_2$. Identifying its Lie group with a subgroup of $SO(8)$, we find that its maximal $\mathbb{Q}-$split torus is of the form

$$T' := \left\{ \begin{pmatrix} A & 0 \\ 0 & B \end{pmatrix},\ | \ A := \begin{pmatrix} a_1 & & & \\ & a_2 & & \\ & & a_3 & \\ & & & 1 \end{pmatrix},\ B := \begin{pmatrix} 1 & & & \\ & a_3^{-1} & & \\ & & a_2^{-1} & \\ & & & a_1^{-1} \end{pmatrix} \right. \\ \left. \text{and } a_1 = a_2 a_3 \right\}. \tag{A.12}$$

We will identify T' with the maximal split torus of $SL(3)$ by means of the map

$$\begin{pmatrix} A & 0 \\ 0 & B \end{pmatrix} \in T' \longmapsto \mathrm{diag}(a_2, a_3, a_1^{-1}),$$

with the maximal split torus of $PGL(3)$ by means of the map

$$\begin{pmatrix} A & 0 \\ 0 & B \end{pmatrix} \in T' \longmapsto \mathrm{diag}(a_1a_2, a_1a_3, 1), \tag{A.13}$$

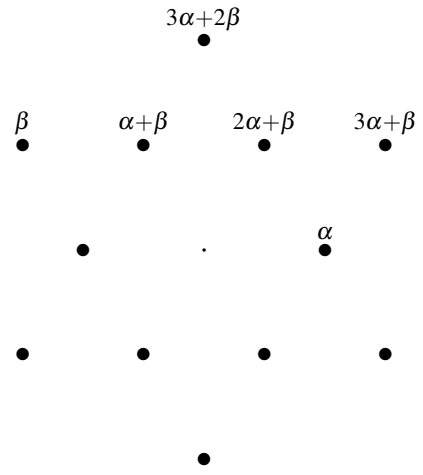
with the maximal split torus of $PGL(2) \times SL(2)$ by means of the map

$$\begin{pmatrix} A & 0 \\ 0 & B \end{pmatrix} \in T' \longmapsto \mathrm{diag}(a_3, 1) \times diag(a_1a_2, 1/a_1a_2), \tag{A.13$'$}$$

and with the maximal split torus of $(SL(2) \times SL(2))/\{\pm 1\}$ by means of the map

$$\begin{pmatrix} A & 0 \\ 0 & B \end{pmatrix} \in T' \longmapsto \mathrm{diag}(a_1, 1/a_1) \times diag(a_2, 1/a_2). \tag{A.13$''$}$$

A.4. The Chevalley basis. The embedding of $\mathfrak{g}_2$ into so(8) gives a convenient framework for doing calculations with explicit matrices. We now write down the Chevalley basis of G_2. The notation for the root system is given by the following diagram:



the root system for G_2

$$X_\alpha := \begin{pmatrix} 0 & 1 & & & & & & \\ & 0 & & & & & & \\ & & 0 & 1 & 1 & & & \\ & & & 0 & 0 & -1 & & \\ & & & & 0 & -1 & & \\ & & & & & 0 & & \\ & & & & & & 0 & -1 \\ & & & & & & & 0 \end{pmatrix}, \quad h_\alpha := \begin{pmatrix} 1 & & & & & & & \\ & -1 & & & & & & \\ & & 2 & & & & & \\ & & & 0 & & & & \\ & & & & 0 & & & \\ & & & & & -2 & & \\ & & & & & & 1 & \\ & & & & & & & -1 \end{pmatrix}.$$

(Note that $X_\alpha = E_{12}+E_{34}+E_{35}$ and $\alpha\begin{pmatrix} A & \\ & B \end{pmatrix} = a_3$, where A, B are as in (A.12).)

$$X_\beta := \begin{pmatrix} 0 & & & & & & & & \\ & 0 & 1 & & & & & & \\ & & 0 & & & & & & \\ & & & 0 & & & & & \\ & & & & 0 & & & & \\ & & & & & 0 & -1 & & \\ & & & & & & 0 & & \\ & & & & & & & 0 & \\ & & & & & & & & 0 \end{pmatrix}, \quad h_\beta := \begin{pmatrix} 0 & & & & & & & & \\ & 1 & & & & & & & \\ & & -1 & & & & & & \\ & & & 0 & & & & & \\ & & & & 0 & & & & \\ & & & & & 1 & & & \\ & & & & & & -1 & & \\ & & & & & & & 0 & \\ & & & & & & & & 0 \end{pmatrix}.$$

(Note that $X_\beta = E_{23}$ and $\beta\begin{pmatrix} A & \\ & B \end{pmatrix} = a_2 a_3^{-1}$, where A, B are as in (A.12).) For $\sigma = \alpha+\beta$:

$$X_\sigma := \begin{pmatrix} 0 & & -1 & & & & & & \\ & 0 & & 1 & 1 & & & & \\ & & 0 & & & & & & \\ & & & 0 & & & -1 & & \\ & & & & 0 & & -1 & & \\ & & & & & 0 & & 1 & \\ & & & & & & 0 & & \\ & & & & & & & 0 & \\ & & & & & & & & 0 \end{pmatrix}, \quad h_\sigma := \begin{pmatrix} 1 & & & & & & & & \\ & 2 & & & & & & & \\ & & -1 & & & & & & \\ & & & 0 & & & & & \\ & & & & 0 & & & & \\ & & & & & 1 & & & \\ & & & & & & -2 & & \\ & & & & & & & -1 & \\ & & & & & & & & -1 \end{pmatrix}.$$

(Note that the entries of $X_{\alpha+\beta}$ above the anti-diagonal are in the positions of E_{13}, E_{24}, E_{25}, with signs -1, 1, 1; the 1994 text wrote $X_{\alpha+\beta} = E_{13}+E_{24}+E_{25}$.) For $\sigma = 2\alpha+\beta$:

$$X_\sigma := \begin{pmatrix} 0 & & & 1 & 1 & & & & \\ & 0 & & & & 1 & & & \\ & & 0 & & & & -1 & & \\ & & & 0 & & & & -1 & \\ & & & & 0 & & & -1 & \\ & & & & & 0 & & & \\ & & & & & & 0 & & \\ & & & & & & & 0 & \\ & & & & & & & & 0 \end{pmatrix}, \quad h_\sigma := \begin{pmatrix} 2 & & & & & & & & \\ & 1 & & & & & & & \\ & & 1 & & & & & & \\ & & & 0 & & & & & \\ & & & & 0 & & & & \\ & & & & & -1 & & & \\ & & & & & & -1 & & \\ & & & & & & & -2 & \\ & & & & & & & & -2 \end{pmatrix}.$$

(Note that $X_{2\alpha+\beta} = E_{14}+E_{15}+E_{26}$.) For $\sigma = 3\alpha+\beta$:

$$X_\sigma := \begin{pmatrix} 0 & & & & 1 & & & \\ & 0 & & & 0 & & & \\ & & 0 & & 0 & 0 & -1 & \\ & & & 0 & & & & \\ & & & & 0 & & & \\ & & & & 0 & & & \\ & & & & & 0 & & \\ & & & & & & 0 & \\ & & & & & & & 0 \end{pmatrix}, \quad h_\sigma := \begin{pmatrix} 1 & & & & & & & \\ & 0 & & & & & & \\ & & 1 & & & & & \\ & & & 0 & & & & \\ & & & & 0 & & & \\ & & & & & -1 & & \\ & & & & & & 0 & \\ & & & & & & & -1 \end{pmatrix}.$$

(Note that $X_{3\alpha+\beta} = E_{16}$.) For $\sigma = 3\alpha+2\beta$:

$$X_\sigma := \begin{pmatrix} 0 & & & & & & 1 & \\ & 0 & & & & & 0 & -1 \\ & & 0 & & & & & \\ & & & 0 & & & & \\ & & & & 0 & & & \\ & & & & & 0 & & \\ & & & & & & 0 & \\ & & & & & & & 0 \end{pmatrix}, \quad h_\sigma := \begin{pmatrix} 1 & & & & & & & \\ & 1 & & & & & & \\ & & 0 & & & & & \\ & & & 0 & & & & \\ & & & & 0 & & & \\ & & & & & 0 & & \\ & & & & & & -1 & \\ & & & & & & & -1 \end{pmatrix}.$$

(Note that $X_{3\alpha+2\beta} = E_{17}$.)

There is a canonical embedding $G_2 \hookrightarrow SO(8)$, given by the natural faithful representation associated to the Cayley algebra. In fact, each one–parameter unipotent subgroup U_σ associated to a short root σ of G_2 is contained in a three dimensional unipotent subgroup of SO(8) generated by three (uniquely determined) roots of D_4 which are permuted by S_3. The possibilities for this embedding of unipotents can be tabulated as follows:

$$\begin{aligned} &U_\alpha \subset G_2 \longleftrightarrow U := \exp[E_{12}F + E_{34}F + E_{35}F], \\ &U_{2\alpha+\beta} \subset G_2 \longleftrightarrow U := \exp[E_{14}F + E_{15}F + E_{26}F], \\ &U_{\alpha+\beta} \subset G_2 \longleftrightarrow U := \exp[E_{13}F + E_{24}F + E_{25}F]. \end{aligned} \tag{A.14}$$

For the long roots of G_2, we have

$$\begin{aligned} &U_\beta \subset G_2 \longleftrightarrow U := \exp[E_{23}F], \\ &U_{3\alpha+2\beta} \subset G_2 \longleftrightarrow U := \exp[E_{17}F], \\ &U_{3\alpha+\beta} \subset G_2 \longleftrightarrow U := \exp[E_{16}F]. \end{aligned} \tag{A.15}$$

In other words, under the action of S_3 on the roots of D_4, the six stable roots correspond to the long roots of G_2 and the other 18, permuted in orbits of 3, correspond to the short roots of G_2.

A.5. Endoscopic groups of G_2. We compute the cuspidal endoscopic groups of G_2, motivated by the description of centralizers of semisimple elements in the dual group [Spa82, Ch. III, §12]. Represent the root α by the vector $(1,0)$ and β by $(-\frac{3}{2}, \frac{\sqrt{3}}{2})$, so that $\gamma = (-\frac{1}{2}, \frac{\sqrt{3}}{2})$, $\delta = (\frac{1}{2}, \frac{\sqrt{3}}{2})$, $\varepsilon = (\frac{3}{2}, \frac{\sqrt{3}}{2})$, $\zeta = (0, \sqrt{3})$, and write $\sigma^\vee = 2\sigma/(\sigma,\sigma)$. Then $\alpha^\vee = (2,0)$ and $\beta^\vee = (-1, \frac{1}{\sqrt{3}})$, and in the basis $(\alpha^\vee, \beta^\vee)$ of X_*,

$$\gamma^\vee = \alpha^\vee + 3\beta^\vee, \quad \delta^\vee = 2\alpha^\vee + 3\beta^\vee, \quad \varepsilon^\vee = \alpha^\vee + \beta^\vee, \quad \zeta^\vee = \alpha^\vee + 2\beta^\vee.$$

The root lattice is $Q = \mathbb{Z}[(1,0), (-\frac{3}{2}, \frac{\sqrt{3}}{2})]$. The weight lattice

$$P = \{x \in \mathbb{R}^2 \mid (x, \Sigma^\vee) \subset \mathbb{Z}\} = \mathbb{Z}\big[(\tfrac{1}{2}, \tfrac{\sqrt{3}}{2}), (0, \sqrt{3})\big]$$

equals Q, since $(\frac{1}{2}, \frac{\sqrt{3}}{2}) = \delta$ and $(0, \sqrt{3}) = \zeta$. So G_2 is both adjoint and simply connected, and its character lattice is $X = Q = P$.

Type $A_1 \times A_1$. Let $\Sigma_{H_2} = \{\pm\alpha, \pm\zeta\}$, a closed subsystem of orthogonal roots; the dual group $\widehat{H}_2$ is the subgroup of $\widehat{G} = G_2(\mathbb{C})$ with roots $\pm\alpha^\vee, \pm\zeta^\vee$. The root lattice of Σ_{H_2} is $Q_1 = \mathbb{Z}[(1,0), (0, \sqrt{3})]$, and its weight lattice is $P_1 = \mathbb{Z}[(1/2, 0), (0, \sqrt{3}/2)]$, with $[P_1 : Q_1] = 4$. The character lattice $X = Q$ of the common torus lies strictly between them: it is the index-2 sublattice of P_1 generated by Q_1 and $\delta = (\alpha+\zeta)/2$. The three lattices between Q_1 and P_1 are $Q_1 + \mathbb{Z}\frac{\alpha}{2}$, $Q_1 + \mathbb{Z}\frac{\zeta}{2}$ and $Q_1 + \mathbb{Z}\frac{\alpha+\zeta}{2}$,

which are the character lattices of $SL(2)\times PGL(2)$, $PGL(2)\times SL(2)$ and $(SL(2)\times SL(2))/\{\pm1\}$. Hence $H_2=(SL(2)\times SL(2))/\{\pm1\}$.

Type A_2. The long roots $\pm\alpha^\vee,\pm\gamma^\vee,\pm\delta^\vee$ of $\widehat{G}$ (the coroots of the short roots of G) form a closed subsystem of type A_2, with simple roots $\alpha^\vee,\gamma^\vee$ and $\alpha^\vee+\gamma^\vee=\delta^\vee$. The corresponding endoscopic group H_1 has roots $\Sigma_{H_1}=\{\pm\alpha,\pm\gamma,\pm\delta\}$, the short roots of G. Its root lattice $\mathbb{Z}[\alpha,\gamma]$ contains $\beta=\gamma-\alpha$, so it equals $X=Q$. Hence H_1 is adjoint: $H_1=PGL(3)$, and $\widehat{H}_1=SL(3)$, in agreement with [SS86, Ch. II, 5.3–5.5].

The short roots of G do not form a closed subsystem ($\alpha+\delta=\varepsilon$), so H_1 is not a subgroup of G_2. The pseudo-Levi subgroup $L=SL(3)\subset G_2$ has the long roots $\pm\beta,\pm\varepsilon,\pm\zeta$; it is used in §8 only for coordinates.

Remark 2026.43. The 1994 text gave the basis $(1/2,0)$, $(0,\sqrt{3}/2)$ for P (but $(1/2,0)\cdot\beta^\vee=-1/2$); took the $A_1\times A_1$ subsystem of $\widehat{G}$ to be generated by $\beta^\vee$ and $\zeta^\vee$, which are not orthogonal; wrote "$Q^\vee$ strictly contains $P^\vee$" for "is strictly contained in $P_1^\vee$"; and used $\Delta_2^\vee=\{\beta^\vee,\varepsilon^\vee\}$ for the A_2 case. These are short roots of $\widehat{G}$ and do not span a closed subsystem, since $\beta^\vee+\varepsilon^\vee=\zeta^\vee$ in the coroot system ($\beta^\vee+(\alpha^\vee+\beta^\vee)=\alpha^\vee+2\beta^\vee$). The conclusions $H_2=(SL(2)\times SL(2))/\{\pm1\}$ and $H_1=PGL(3)$ were right. The argument above is new.

A.6. Tori of G_2. The stable conjugacy classes of Cartan subgroups in a quasi-split group G are in bijection with $H^1(\Gamma,W)$, where $W=\Omega$ is the absolute Weyl group of a fixed Cartan subgroup T of G defined over F [Ro80, §4, Lemma 3]. Since G_2 has no outer automorphisms, the action of Γ on the character lattice $X=X(T)$ of any Cartan subgroup T factors through W, and the Galois image $\Gamma_T\subset W$ determines T up to stable conjugacy [Sa71]. The torus is elliptic if $X^{\Gamma_T}=0$.

By Tate–Nakayama duality [La79, p. 702], for elliptic T,

$$H^1(F,T)\cong X_*/I_\Gamma X_*,\qquad I_\Gamma X_*=\sum_{w\in\Gamma_T}(w-1)X_*,$$

the coinvariants of the cocharacter lattice $X_*=X^\vee$. A character κ of $H^1(F,T)$ is therefore a character of X_* trivial on $I_\Gamma X_*$, and the roots of the endoscopic group $H=H_{T,\kappa}$, transported to T, are the roots a of G with $\kappa(a^\vee)=1$.

In the basis $(\alpha^\vee,\beta^\vee)$ of X_*, the simple reflections and the rotation $w=s_\beta s_\alpha$ of order 6 act by

$$s_\alpha=\begin{pmatrix}-1&1\\0&1\end{pmatrix},\qquad s_\beta=\begin{pmatrix}1&0\\3&-1\end{pmatrix},\qquad s_\beta s_\alpha(m\alpha^\vee+n\beta^\vee)=(n-m)\alpha^\vee+(2n-3m)\beta^\vee,$$

so $w=\left(\begin{smallmatrix}-1&1\\-3&2\end{smallmatrix}\right)$, with determinant 1 and trace 1. For the cyclic group of order 6, $\det(w-1)=1$, so $(w-1)X_*=X_*$ and $H^1(F,T)=0$. For the cyclic group of order 3, generated by $w^2=\left(\begin{smallmatrix}-2&1\\-3&1\end{smallmatrix}\right)$, $(w^2-1)X_*$ is spanned by $(-3,-3)$ and $(1,0)$, that is, by $\alpha^\vee$ and $3\beta^\vee$, of index 3. So $H^1(F,T)\cong\mathbb{Z}/3\mathbb{Z}$, and a non-trivial κ is trivial exactly on the coroots whose $\beta^\vee$-coefficient is divisible by 3: $\alpha^\vee$, $\gamma^\vee$, $\delta^\vee$. For the biquadratic tori, with $\Gamma_T=\langle s_\alpha,s_\zeta\rangle$ (or a conjugate), $(s_\alpha-1)X_*+(s_\zeta-1)X_*=\mathbb{Z}\alpha^\vee+\mathbb{Z}\zeta^\vee$, of index 2, so $H^1(F,T)\cong\mathbb{Z}/2\mathbb{Z}$ and a non-trivial κ is trivial on $\alpha^\vee$ and $\zeta^\vee$. For the quadratic tori, $\Gamma_T=\{\pm1\}$, $I_\Gamma X_*=2X_*$, and the three non-trivial characters of $X_*/2X_*$ are trivial on the coroots $\{\alpha^\vee,\zeta^\vee\}$, $\{\beta^\vee,\delta^\vee\}$ and $\{\gamma^\vee,\varepsilon^\vee\}$, respectively. Running over all elliptic subgroups of W gives the following table (script `g2_tori_endoscopy.py`, which computes $X_*/I_\Gamma X_*$ by Smith normal form).

Remark 2026.44. The 1994 text treated only the biquadratic, order-6 and cubic cases, and asserted that "all other non-split Cartans of G_2 do not contribute". The quadratic tori and the tori with

TABLE A.6. Elliptic Cartan subgroups of G_2 and their endoscopic data.

Galois image $\Gamma_T \subset W$	$H^1(F,T)$	roots of H for $\kappa \neq 1$	H
$\{\pm 1\}$ (quadratic)	$(\mathbb{Z}/2)^2$	$\{\alpha,\zeta\}$, $\{\beta,\delta\}$, $\{\gamma,\varepsilon\}$	H_2
$\langle s_\alpha, s_\zeta\rangle$ and conjugates (biquadratic)	$\mathbb{Z}/2$	$\{\alpha,\zeta\}$ (or a conjugate)	H_2
cyclic of order 3 (cubic)	$\mathbb{Z}/3$	$\{\alpha,\gamma,\delta\}$	H_1
S_3 generated by short reflections	$\mathbb{Z}/3$	$\{\alpha,\gamma,\delta\}$	H_1
cyclic of order 6	0	—	—
S_3 generated by long reflections	0	—	—
W	0	—	—

Galois image S_3 generated by short reflections also carry non-trivial κ, as the table shows. Its conclusions $\mathbb{Z}/2$, 0, $\mathbb{Z}/3$ for the three cases it treated are right, but the computations were not: the displayed formula for $s_\beta s_\alpha$ was $(2m-n)\alpha^\vee + (m-n)\beta^\vee$, of determinant -1; the "cubic" formula had determinant 3; the two coboundary identities were wrong; and the biquadratic case took a quotient $(H^1(\Gamma,T_1)\times H^1(\Gamma,T_2))/\{\pm 1\}$ of cohomology groups, which is not valid. It also called $\langle s_\alpha, s_\zeta\rangle$ a group generated by "simple reflections"; s_ζ is not simple.

A.7. The stability of the 2-regular and 3-regular unipotent orbits for G_2. I thank M. Assem for explaining to me the technique of this section.

Using the appendices above, we find that

$$\mathfrak{m} := cent_{\mathfrak{g}_2}(h_{\alpha+\beta}) = \mathfrak{t} + F\cdot X_{3\alpha+\beta} + F\cdot X_{-3\alpha-\beta},$$

where $\mathfrak{t}$ denotes the Cartan of $\mathfrak{g}_2$. Let $M := Cent_{G_2}(h_{\alpha+\beta})$ and let

$$\mathfrak{g}_2(i) := \{X \in \mathfrak{g}_2 \mid [h_{\alpha+\beta}, X] = iX\}.$$

We compute

$$\mathfrak{g}_2(2) = F\cdot X_{\alpha+\beta}.$$

The map $m \longmapsto Ad(m)X_{\alpha+\beta}$ defines a submersion $M \to \mathfrak{g}_2(2)$ [SS86, ch III, §4.14]. By a lemma of Rao [Ra72], the $G(F)$-orbit of $X_{\alpha+\beta}$ is $Ad(K(V(X_{\alpha+\beta}) + \mathfrak{n}(2)))$, where

$$V(X_{\alpha+\beta}) = \{ad(X)X_{\alpha+\beta} \mid X \in \mathfrak{m}\},$$

K is a maximal compact subgroup of $G_2(F)$, and

$$\mathfrak{n}(2) := \sum_{i>2} \mathfrak{g}_2(i).$$

The fact that $dim\ \mathfrak{g}_2(2) = 1$ gives us that orbit of M in $\mathfrak{g}_2(2)$ is onto, so by Rao's result the 2-regular unipotent orbit is stable.

A similar calculation shows that the 3-regular orbit is stable. (In fact, replacing $h_{\alpha+\beta}$ and $X_{\alpha+\beta}$ by h_β and X_β, we compute

$$\mathfrak{g}_2(2) = F\cdot X_\beta,$$

and the identical argument holds.)

A Lie triple for the subregular orbit is

$$X_+ = X_\beta + X_{2\alpha+\beta}, X_- = X_{-\beta} + X_{-2\alpha-\beta}, H = h_\beta + h_{2\alpha+\beta},$$

and

$$\mathfrak{g}_2(2) = F\cdot X_{\alpha+\beta} + F\cdot X_{2\alpha+\beta} + F\cdot X_{3\alpha+\beta} + F\cdot X_{\beta}.$$

Direct matrix calculations show that the subregular orbit is not stable. The 1994 text reports that M. Assem showed that there are 7 F-classes in the stable class when $p \neq 2,3$; see [As98] and Remark 2026.45.

Remark 2026.45. Two points in this section need attention. First, for $\mathcal{O}_2$ (the class $\tilde{A}_1$) the grading defined by $h_{\alpha+\beta}$ is odd, so the use of Rao's lemma [Ra72] needs justification; the facts $\mathfrak{g}_2(2) = FX_{\alpha+\beta}$ and $\mathfrak{g}_2(2) = FX_{\beta}$ were confirmed. Second, the number of F-classes in the stable subregular class can be computed as follows. The component group of the centralizer of a subregular unipotent element is S_3, and $H^1(F, G_2) = 1$, so the F-classes correspond to $H^1(F, S_3)$, that is, to isomorphism classes of étale cubic F-algebras. For $p \neq 2,3$ there are 8 of these when $q \equiv 1 \pmod 3$ and 6 when $q \equiv 2 \pmod 3$, not 7. We have not been able to compare this with [As98].

SUMMARY OF RESULTS AND OPEN PROBLEMS

What is proved. For $G = G_2$:

(1) The monomial variety Z of §3 is a normal, Cohen–Macaulay affine toric variety (Proposition 3.7), covered up to codimension two by twelve regular patches (Proposition 3.4A).
(2) The divisor $\lambda = 0$ has exactly seven branches on Z, with the Igusa constants of Table 4.3A; the integral values of $\beta - 1$ are $0, 1, 2, 3, 6$, and one branch has $\beta = 11/3$.
(3) The elliptic endoscopic data of G_2 are those of Table A.6, with $H_1 = PGL(3)$ (short roots) and $H_2 = (SL(2) \times SL(2))/\{\pm 1\}$.
(4) The 2- and 3-regular classes are stable (§A.7, with the caveat of Remark 2026.45).

What is reduced to hypotheses. Theorems 6.1, 7.2 and 8.3 reduce the matching of the 1-, 2- and 3-regular κ-germs for the biquadratic and cubic tori to Hypotheses (X_r) on the dependence of explicit fiber integrals on the tangent direction X, together with Lemma 5.14 and the proviso of Lemma 0.8. By Corollary 8.8 the hypotheses (X_r) are in fact true, given those two inputs, because the matching is now known [GH16].

Open problems.

(1) Compute $\Delta_0(X)$ and the fiber integrals $I_E(X, u)$ of Remark 2026.29 for G_2, and so prove (X_r) directly, as Hales does for $GSp(4)$ [Ha89, §6]. This would give explicit formulas for the κ-germs and the matching constants, which [GH16] does not.
(2) Choose a toric resolution of Z, determine the exceptional divisors with integral β (Remark 2026.23), their images under π and their F-points, and so settle the proviso of Lemma 0.8. Treat the spurious divisors over $Y_1 \setminus Y''$.
(3) Prove that the contribution of the branch with $\beta = 11/3$ vanishes.
(4) Give a complete proof of Lemma 5.14, or replace it.
(5) Treat the quadratic tori and the tori with Galois image S_3 generated by short reflections (Table A.6).
(6) Prove Conjecture 3.7A (normality of Z in general), and decide Conjecture 3.4.
(7) Reconcile the count of F-classes in the subregular class with [As98] (Remark 2026.45).

GLOSSARY

κ-orbital integral:
The combination $\Phi^{T,\kappa}(\gamma,f)=\sum\kappa(\mathrm{inv}(\gamma^0,\gamma'))\Phi(\gamma',f)$ over the F-classes γ' in the stable class of a regular semisimple γ.

Shalika germ:
The function $\Gamma^T_{\mathscr{O}}$ (or $\Gamma^{T,\kappa}_{\mathscr{O}}$) giving the coefficient of the unipotent orbital integral $\mu_{\mathscr{O}}(f)$ in the expansion of an orbital integral near the identity [Sh72].

r-regular class:
A unipotent class $\mathscr{O}$ with $\dim\mathrm{Cent}_G(u)=\mathrm{rk}\,G+2r$ for $u\in\mathscr{O}$. For G_2: regular ($r=0$), subregular (1), $\tilde{A}_1$ (2), A_1 (3), trivial (6).

Endoscopic group:
A quasi-split group H attached to a pair (T,κ), whose dual group is the centralizer of κ in $\widehat{G}$ [La79], [LS87].

Transfer factor:
The function $\Delta_{G/H}$ of Langlands and Shelstad [LS87] that normalizes the comparison of κ-orbital integrals on G with stable orbital integrals on H.

Variety of stars:
Langlands' variety S of configurations of Borel subgroups indexed by the chambers of the Weyl group (§1).

Igusa constants:
For a branch E of $\lambda=0$ on a smooth model, $a(E)=\mathrm{ord}_E\lambda$, $b(E)=1+\mathrm{ord}_E\omega$, and $\beta(E)=b(E)/a(E)$; the branch contributes terms $\theta(\lambda)|\lambda|^{\beta-1}$ to the Igusa expansion.

Branch, base divisor:
An irreducible component of $\phi^{-1}(0)$ (or of $\lambda=0$); a base divisor is a branch of Y'', as opposed to an exceptional or spurious divisor of the resolution.

Toric variety:
A normal variety with a dense torus orbit; here $Z=\mathrm{Spec}\,k[S]$ for a finitely generated saturated semigroup S [Ke72].

Principal value integral:
The regularized integral over a branch that gives the coefficient $F_1(\theta,\beta,f)$ [LS83], [La83].

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REFERENCES

[As98] M. Assem, *On stability and endoscopic transfer of unipotent orbital integrals on p-adic symplectic groups*, Mem. Amer. Math. Soc. **134** (1998), no. 635.

[As00] M. Assem, "Remarks on the transfer factors for unipotent orbital integrals in p-adic classical split groups", Compositio Math. **120** (2000), 227–290.

[Bor91] A. Borel, *Linear Algebraic Groups*, 2nd ed., Springer-Verlag, 1991.

[Ca89] R. Carter, *Simple Groups of Lie Type*, Wiley-Interscience, 1989.

[De02] S. DeBacker, "Homogeneity results for invariant distributions of a reductive p-adic group", Ann. Sci. École Norm. Sup. (4) **35** (2002), 391–422.

[DG80] M. Demazure and P. Gabriel, *Introduction to Algebraic Geometry and Algebraic Groups*, North-Holland Math. Studies **39**, 1980.

[GH16] J. Gordon and T. Hales, "Endoscopic transfer of orbital integrals in large residual characteristic", Amer. J. Math. **138** (2016), 109–148.

[Ha89] T. Hales, "Shalika germs on $GSp(4)$", in *Orbites unipotentes et représentations*, Astérisque **171–172** (1989), 195–256.

[Ha92a] T. Hales, *The Subregular Germ of Orbital Integrals*, Mem. Amer. Math. Soc. **99** (1992), no. 476.

[Ha92b] T. Hales, "Orbital integrals on $U(3)$", in *The Zeta Functions of Picard Modular Surfaces*, ed. R. Langlands and D. Ramakrishnan, Centre de Recherches Mathématiques, Montréal, 1992, 303–333.

[Ha94] T. Hales, "Hyperelliptic curves and harmonic analysis (why harmonic analysis on reductive p-adic groups is not elementary)", in *Representation Theory and Analysis on Homogeneous Spaces*, Contemp. Math. **177**, Amer. Math. Soc., 1994, 137–169.

[Har77] R. Hartshorne, *Algebraic Geometry*, Springer-Verlag, 1977.

[Ho72] M. Hochster, "Rings of invariants of tori, Cohen–Macaulay rings generated by monomials, and polytopes", Ann. of Math. **96** (1972), 318–337.

[Ig78] J.-I. Igusa, *Lectures on Forms of Higher Degree*, Tata Institute, Bombay, 1978.

[Ja71] N. Jacobson, *Exceptional Lie Algebras*, Marcel Dekker, 1971.

[Ke72] G. Kempf, F. Knudsen, D. Mumford and B. Saint-Donat, *Toroidal Embeddings I*, Lecture Notes in Math. **339**, Springer, 1973.

[KP79] H. Kraft and C. Procesi, "Closures of conjugacy classes of matrices are normal", Invent. Math. **53** (1979), 227–247.

[La79] R. Langlands, "Stable conjugacy: definitions and lemmas", Canad. J. Math. **31** (1979), 700–725.

[La83] R. Langlands, "Orbital integrals on forms of $SL(3)$, I", Amer. J. Math. **105** (1983), 465–506.

[LS83] R. Langlands and D. Shelstad, "On principal values on p-adic manifolds", in *Lie Group Representations II*, Lecture Notes in Math. **1041**, Springer, 1984, 250–279.

[LS87] R. Langlands and D. Shelstad, "On the definition of transfer factors", Math. Ann. **278** (1987), 219–271.

[LS89] R. Langlands and D. Shelstad, "Orbital integrals on forms of $SL(3)$, II", Canad. J. Math. **41** (1989), 480–507.

[LS90] R. Langlands and D. Shelstad, "Descent for transfer factors", in *The Grothendieck Festschrift*, vol. II, Birkhäuser, 1990, 485–563.

[Ng10] B. C. Ngô, "Le lemme fondamental pour les algèbres de Lie", Publ. Math. IHÉS **111** (2010), 1–169.

[Ra72] R. Ranga Rao, "Orbital integrals in reductive groups", Ann. of Math. **96** (1972), 505–510.

[Re84] J. Repka, "Shalika's germs for p-adic $GL(n)$, II: the subregular term", Pacific J. Math. **113** (1984), 173–182.

[Re85] J. Repka, "Germs associated to regular unipotent classes in p-adic $SL(n)$", Canad. Math. Bull. **28** (1985), 257–266.

[Ro80] J. Rogawski, "An application of the building to orbital integrals", Compositio Math. **42** (1980), 417–423.

[Sa71] I. Satake, *Classification Theory of Semi-simple Algebraic Groups*, Marcel Dekker, 1971.

[Ser79] J.-P. Serre, *Local Fields*, Graduate Texts in Math. **67**, Springer, 1979.

[Sh72] J. Shalika, "A theorem on semi-simple p-adic groups", Ann. of Math. **95** (1972), 226–242.

[She82] D. Shelstad, "L-indistinguishability for real groups", Math. Ann. **259** (1982), 385–430.

[She89] D. Shelstad, "A formula for regular unipotent germs", in *Orbites unipotentes et représentations*, Astérisque **171–172** (1989), 275–277.

[Sl80] P. Slodowy, *Simple Singularities and Simple Algebraic Groups*, Lecture Notes in Math. **815**, Springer, 1980.

[Spa82] N. Spaltenstein, *Classes unipotentes et sous-groupes de Borel*, Lecture Notes in Math. **946**, Springer, 1982.

[Sp79] T. A. Springer, "Reductive groups", in *Automorphic Forms, Representations, and L-functions*, Part I, Proc. Sympos. Pure Math. **33**, Amer. Math. Soc., 1979, 3–27.

[SS83] P. Sally and J. Shalika, "The Fourier transform of orbital integrals on $SL(2)$ over a p-adic field", in *Lie Group Representations II*, Lecture Notes in Math. **1041**, Springer, 1984, 303–340.

[SS86] T. A. Springer and R. Steinberg, "Conjugacy classes", in *Seminar on Algebraic Groups and Related Finite Groups*, Lecture Notes in Math. **131**, Springer, 1970 (2nd printing 1986), 167–266.

[St74] R. Steinberg, *Conjugacy Classes in Algebraic Groups*, Lecture Notes in Math. **366**, Springer, 1974.

[SV00] T. A. Springer and F. D. Veldkamp, *Octonions, Jordan Algebras and Exceptional Groups*, Springer Monographs in Math., Springer, 2000.

[Ti65] J. Tits, "Classification of algebraic semisimple groups", in *Algebraic Groups and Discontinuous Subgroups*, Proc. Sympos. Pure Math. **9**, Amer. Math. Soc., 1966, 33–62.

[Ts15] C.-C. Tsai, "Inductive structure of Shalika germs and affine Springer fibers", preprint, arXiv:1512.00445, 2015.

[Wa97] J.-L. Waldspurger, "Le lemme fondamental implique le transfert", Compositio Math. **105** (1997), 153–236.

[Wa01] J.-L. Waldspurger, *Intégrales orbitales nilpotentes et endoscopie pour les groupes classiques non ramifiés*, Astérisque **269** (2001).

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